From Discounted Value to Conjugate Risk - CAPM Phase Geometry, the Financial Meaning of Q, and the R → −Q → −R Measurement Cycle

CAPM Phase Geometry, the Financial Meaning of Q, and the R → −Q → −R Measurement Cycle

How a Mature Discounted-Cash-Flow Model Generates a Complex Valuation Plane, a Conjugate Risk Exposure, and a Closed Financial Measurement Structure

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Source Note

Earlier work introduced a complex completion of mature financial valuation:

$Z = R + iQ.$ (0.1)

Here R is admitted value, Q is the orthogonal pressure coordinate implied by a declared financial filter, A is the pre-filter value amplitude, and θ is the angle generated by the relation between A and R:

$A^2 = R^2 + Q^2.$ (0.2)

$R = A \cos \theta.$ (0.3)

$Q = A \sin \theta.$ (0.4)

$Z = A \exp(i\theta).$ (0.5)

The CAPM implementation defines A from a declared baseline discount rate and R from the ordinary CAPM required return. Q is then derived from the same valuation relation rather than introduced as an independent risk score. Earlier articles developed this geometry into phase dynamics, commitment gates, ledger time, relative valuation frames, derivative composite states, and observer-bounded financial worlds.

One central question nevertheless remained unresolved.

What exactly is Q as a financial measurement?

Calling Q “retained pressure” identifies its broad role but does not yet establish its precise mathematical and financial identity. Q is not the scalar discount haircut A − R. It is not automatically expected loss, Value at Risk, volatility, opportunity cost, or a second asset price. Nor is multiplication by i adequately explained by saying that hidden risk simply becomes visible loss.

The present article completes that missing step.

Its principal result is:

$\partial R/\partial \theta = -Q.$ (0.6)

Q is therefore the magnitude of the first-order dollar exposure of admitted value to movement in the declared valuation phase.

This leads to a closed measurement structure:

$$R \rightarrow -Q \rightarrow -R \rightarrow Q \rightarrow R. \quad (0.7)$$

The first quarter-turn changes the financial readout from mark to conjugate phase exposure. The second quarter-turn reverses the signed valuation orientation. Exposure becomes economic profit or loss only when the valuation phase actually moves. That economic consequence becomes financial history only when it passes a recognition or settlement gate and enters a ledger.

The article is divided into two major parts.

Part I develops the construction entirely within familiar finance, calculus, matrix algebra, and sensitivity analysis. It requires no knowledge of quantum mechanics, tensor calculus, Hilbert spaces, gauge theory, or differential geometry.

Part II asks what deeper structures become visible after the finance-first result has been established: measurement rotation, quadrature relations, relative valuation frames, derivative composite states, gate-and-ledger commitment, residual structure, and observer-bounded valuation worlds.

The resulting framework is a conceptual and mathematical research programme. It is not investment advice.

Abstract

Modern finance converts future economic claims into scalar present values. Under CAPM-based discounted-cash-flow valuation, beta and the equity risk premium determine a required return, and that required return determines an admitted value R. The scalar result is operationally useful, but it does not preserve the complete geometry implied by comparing the CAPM-discounted value with a declared baseline valuation.

For a future cash flow CF_t, define the baseline-discounted amplitude A_t and the ordinary CAPM value R_t by:

$$A_t = CF_t/(1 + r_{base})^t. \quad (0.8)$$

$$R_t = CF_t/(1 + r_{CAPM})^t. \quad (0.9)$$

$$r_{CAPM} = r_{base} + \beta ERP. \quad (0.10)$$

The CAPM valuation phase is defined by:

$$\cos \theta_t = R_t/A_t. \quad (0.11)$$

Therefore:

$$\cos \theta_t = [(1 + r_{base})/(1 + r_{CAPM})]^t. \quad (0.12)$$

The orthogonal coordinate is:

$$Q_t = \sqrt{(A_t^2 - R_t^2)}. \quad (0.13)$$

The completed valuation state is:

$$Z_t = R_t + iQ_t = A_t \exp(i\theta_t). \quad (0.14)$$

The first principal result is that Q is not merely a geometric remainder. Along a fixed-amplitude valuation orbit:

$$\partial R/\partial\theta = -Q. \quad (0.15)$$

Thus Q is the magnitude of the first-order dollar sensitivity of admitted value to valuation-phase movement. Define the signed CAPM Phase Delta by:

$$\Delta_\theta \equiv \partial R/\partial\theta. \quad (0.16)$$

Then:

$$\Delta_\theta = -Q. \quad (0.17)$$

Ordinary required-return sensitivity and phase sensitivity are exactly equivalent:

$$dR = -[tR/(1 + r)]dr. \quad (0.18)$$

$$dR = -Qd\theta. \quad (0.19)$$

Hence:

$$Qd\theta = [tR/(1 + r)]dr. \quad (0.20)$$

The phase representation does not replace or alter ordinary CAPM sensitivity. It expresses the same local value change in a different risk coordinate.

The article then defines the complex-structure operator $\mathcal{J}$ on the real two-dimensional valuation state:

$$\mathcal{J}[R,Q]^T = [-Q,R]^T. \quad (0.21)$$

It follows that:

$$\mathcal{J}^2 = -I. \quad (0.22)$$

$$\mathcal{J}^4 = I. \quad (0.23)$$

A family of rotated financial measurements is defined by:

$$M_\varphi(Z) = \mathrm{Re}[\exp(i\varphi)Z]. \quad (0.24)$$

Therefore:

$$M_\varphi(Z) = R\cos\varphi - Q\sin\varphi. \quad (0.25)$$

The special readouts are:

$$M_0(Z) = R. \quad (0.26)$$

$$M_{\pi/2}(Z) = -Q. \quad (0.27)$$

$$M_\pi(Z) = -R. \quad (0.28)$$

$$M_{3\pi/2}(Z) = Q. \quad (0.29)$$

$$M_{2\pi}(Z) = R. \quad (0.30)$$

This produces the closed financial measurement cycle:

$R \to -Q \to -R \to Q \to R.$ (0.31)

The first quarter-turn converts mark into conjugate phase exposure. The second reverses the signed valuation orientation. In a linear long–short position space, the four readouts correspond to the long mark, long phase exposure, short mark, and short phase exposure.

The construction does not imply that −Q is automatically a realized loss. Q is an exposure coefficient. For a finite phase movement $\Delta\theta$:

$R_{new} = R\cos\Delta\theta - Q\sin\Delta\theta.$ (0.32)

Therefore:

$\Delta R = R(\cos\Delta\theta - 1) - Q\sin\Delta\theta.$ (0.33)

For small $\Delta\theta$:

$\Delta R = -Q\Delta\theta - (R/2)(\Delta\theta)^2 + (Q/6)(\Delta\theta)^3 + O((\Delta\theta)^4).$ (0.34)

The article also derives the exact relationship between Q and the scalar CAPM haircut:

$H(\theta) = A - R(\theta).$ (0.35)

$dH/d\theta = Q.$ (0.36)

$H(\theta) = \int_0^\theta Q(\varphi)d\varphi.$ (0.37)

Q is therefore the marginal growth rate of the CAPM filter haircut with respect to valuation phase. It becomes a marginal opportunity-cost measure only under the additional condition that A represents the best foregone alternative.

Part II develops the broader implications. It distinguishes state evolution, measurement rotation, and ledger commitment; explains why $i^2 = -1$ represents measurement-orientation reversal rather than double economic loss; introduces quadrature and relative-frame interpretations; and situates the one-state CAPM construction inside derivative composite systems and observer-bounded valuation worlds.

The broader comparison with quantum theory remains methodological. Finance can reproduce complex coordinates, phase, conjugate measurement, contextual readout, commitment gates, trace, residual, and backreaction without thereby becoming a literal quantum system.

Keywords

CAPM; discounted cash flow; complex valuation; conjugate risk; phase delta; valuation phase; sensitivity analysis; financial measurement; risk exposure; complex operator; long–short duality; opportunity cost; ledger; residual; observer-bounded valuation.

Part I — The Complex Valuation Plane for Finance

0. Reader Contract: A Finance-First Construction

0.1 What this article proves

This article begins from ordinary CAPM-based discounted-cash-flow valuation.

It does not modify the CAPM required-return formula:

r_CAPM = r_base + βERP. (0.38)

It does not modify the ordinary discounted present value:

R_t = CF_t/(1 + r_CAPM)^t. (0.39)

It introduces a declared baseline valuation:

A_t = CF_t/(1 + r_base)^t. (0.40)

The ratio R_t/A_t is then represented as the cosine of a valuation angle:

cos θ_t = R_t/A_t. (0.41)

A norm-preserving completion defines:

Q_t = √(A_t² − R_t²). (0.42)

The complex state is:

Z_t = R_t + iQ_t. (0.43)

The article proves five principal results.

First, ordinary CAPM DCF is exactly the real projection of the declared baseline amplitude:

R_t = A_t cos θ_t. (0.44)

Second, Q is the conjugate phase exposure of R:

∂R/∂θ = −Q. (0.45)

Third, ordinary required-return sensitivity and valuation-phase sensitivity are the same local P&L expressed in different coordinates:

−[tR/(1 + r)]dr = −Qdθ. (0.46)

Fourth, the complex-structure operator closes after four applications:

$\mathcal{J}^2 = -I$. (0.47)

$\mathcal{J}^4 = I$. (0.48)

Fifth, the rotated measurement family produces:

R → −Q → −R → Q → R. (0.49)

These are mathematical and financial measurement results. They do not depend on quantum mechanics.

0.2 What this article does not claim

The article does not claim that Q is already a standard CAPM variable.

It does not claim that Q is an independent empirical observation in the one-period static model.

Once A and R are declared:

$$Q = \sqrt{(A^2 - R^2)}. \quad (0.50)$$

Therefore Q is algebraically determined by A and R.

The article does not claim:

$$Q = A - R. \quad (0.51)$$

It does not claim that Q is automatically:

- expected loss;
- realized loss;
- beta;
- volatility;
- Value at Risk;
- Expected Shortfall;
- duration;
- convexity;
- option premium;
- market price;
- opportunity cost.

The article does not claim that multiplication by i causes an asset to lose value.

It does not claim that:

$$R \to -Q$$

describes a chronological fall in the market price.

It does not claim that applying the operator twice causes two economic losses.

The identity:

$$\mathcal{J}^2 R = -R \quad (0.52)$$

is first a statement about measurement orientation in the completed valuation plane.

The article does not claim that a company is the negative position of its shareholders.

In a signed tradable-position space, a short position may have value $-R$ when the corresponding long position has value R. That clean long–short relation does not automatically transfer to every corporate, accounting, regulatory, or institutional relationship.

The article does not claim that finance is a literal quantum system.

Complex coordinates, phase, conjugate variables, contextual measurement, gates, trace, and residuals can appear in classical and institutional systems. Their appearance does not prove quantum ontology.

0.3 Required mathematical background

Part I assumes familiarity with:

- present-value discounting;
- CAPM;
- first and second derivatives;
- the chain rule;
- Taylor expansion;
- two-dimensional vectors;
- elementary matrix multiplication;
- sine and cosine;
- the complex number i.

No knowledge of advanced physics is required.

The most unfamiliar object may be the operator $\mathcal{J}$. In Part I it will be introduced simply as the matrix:

$\mathcal{J} = [0\ -1;\ 1\ 0]$. (0.53)

Applied to a two-dimensional vector, it performs a 90° rotation:

$\mathcal{J}[R,Q]^T = [-Q,R]^T$. (0.54)

Everything else follows from ordinary algebra and calculus.

0.4 Five levels of statement

The article distinguishes five levels of claim.

Level 1 — Mathematical identity

Examples include:

$A^2 = R^2 + Q^2$. (0.55)

$\partial R/\partial\theta = -Q$. (0.56)

$\mathcal{J}^2 = -I$. (0.57)

These results follow from the declared representation.

Level 2 — Financial construction

Examples include:

- A is the value of the same cash flow under the declared baseline discount protocol;
- R is the value admitted under the CAPM required return;
- Q is the orthogonal coordinate implied by that valuation relation;
- −Q is the signed phase exposure of R.

These statements depend on how A, R, and θ are constructed.

Level 3 — Measurement interpretation

Examples include:

- M0 reads the mark;
- M_π/2 reads the conjugate phase exposure;
- M_π reverses the signed valuation orientation;
- the sequence R → −Q → −R → Q is a measurement cycle.

These statements depend on the declared sign and orientation convention.

Level 4 — Empirical hypothesis

Examples include:

- Q may improve cross-asset pressure comparison;
- phase exposure may support multi-horizon risk attribution;
- market-implied Q may reveal residual disagreement with model-implied Q;
- phase dynamics may improve regime diagnosis.

These claims require empirical testing.

Level 5 — Foundational implication

Examples include:

- measurement orientation may be as important as the measured state;
- a visible scalar result may not exhaust the effective state;
- gate and trace may be required to turn movement into financial history;
- some structures associated with quantum observation may belong to a more general observer-bound grammar.

These are research hypotheses, not established physical conclusions.

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1. The Scalar Achievement of CAPM DCF

1.1 CAPM converts market exposure into a required return

The standard CAPM relation is:

$$r_{CAPM} = r_f + \beta(E[r_m] - r_f). \quad (1.1)$$

Let:

$$ERP = E[r_m] - r_f. \quad (1.2)$$

Then:

$$r_{CAPM} = r_f + \beta ERP. \quad (1.3)$$

For the more general construction used in this article:

$$r = r_{base} + \beta ERP. \quad (1.4)$$

The baseline r_{base} may be the risk-free rate, but it may also be another explicitly declared reference rate.

CAPM performs an important compression.

A security's relation to the market is summarized by beta. Beta and the equity risk premium then produce a required return. That required return can be used to discount future cash flows.

The basic chain is:

$$\text{Market Exposure} \rightarrow \beta \rightarrow \text{Required Return} \rightarrow \text{Present Value}. \quad (1.5)$$

Nothing in the present article rejects this chain.

1.2 CAPM inside discounted-cash-flow valuation

For a single future cash flow CF_t:

$$R_t = CF_t/(1 + r)^t. \quad (1.6)$$

For multiple cash flows:

$$R = \Sigma_t CF_t/(1 + r_t)^t. \quad (1.7)$$

The single-period construction will be used first because it makes the geometry transparent.

The quantity R_t has a familiar operational meaning.

It is a present value under the declared CAPM required return. It can support:

- investment comparison;
- project appraisal;
- equity valuation;

- capital allocation;
- transaction negotiation;
- risk-adjusted hurdle rates;
- portfolio analysis.

The scalar output is not a defect.

Contracts, ledgers, decisions, and prices require scalar commitments.

1.3 Scalar admission necessarily suppresses structure

Suppose two assets have the same CAPM present value:

$R_1 = R_2$. (1.8)

The scalar valuation treats them as equal on the admitted-value axis.

But they may not have the same baseline valuation.

Suppose:

$A_1 \neq A_2$. (1.9)

Then the same admitted value may arise from different combinations of baseline amplitude and CAPM filtering.

One asset may have:

- a large baseline cash-flow value;
- a large required-return adjustment.

Another may have:

- a smaller baseline cash-flow value;
- a milder required-return adjustment.

Both may produce the same R.

The scalar result is still correct under its protocol. But it no longer displays the structural difference.

The question is therefore not:

> Is CAPM wrong because it gives one number?

The better question is:

> Can the same CAPM relation be represented in a way that preserves both the admitted value and the pressure implied by the filter?

1.4 The complex extension begins from CAPM's success

The proposed extension does not calculate R differently.

It introduces a declared reference valuation A and asks how the ordinary CAPM value R is positioned relative to that reference.

The scalar construction is:

$A \rightarrow \text{CAPM Filter} \rightarrow R.$ (1.10)

The complex completion is:

$A \rightarrow \theta \rightarrow R + iQ.$ (1.11)

The CAPM value remains:

$R = CF_t/(1 + r)^t.$ (1.12)

The extension adds:

- the valuation ratio R/A;
- the implied phase θ;
- the orthogonal coordinate Q;
- the measurement relation between R and Q.

The purpose is not to produce a different present value.

It is to determine whether the valuation relation already implies a richer risk-measurement structure.

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2. Declaring the Baseline Valuation

2.1 The same cash flow under two discount protocols

Consider the same future cash flow CF_t.

Under the declared baseline rate r_{base}:

$A_t = CF_t/(1 + r_{base})^t.$ (2.1)

Under the CAPM required return r:

$R_t = CF_t/(1 + r)^t.$ (2.2)

Where:

$r = r_{base} + \beta ERP.$ (2.3)

A_t and R_t are not values of two different cash flows.

They are two present-value readings of the same declared future cash flow under two discount protocols.

This distinction is essential.

2.2 What A means

A_t is the baseline-discounted cash-flow amplitude.

It is not necessarily:

- the market price;
- the intrinsic value;
- the socially correct value;
- the maximum possible value;
- the best investment alternative;
- the value that would exist in a perfect world.

It is:

> the value of the declared cash flow under the declared baseline discount rule.

If:

r_base = r_f, (2.4)

then A_t is the risk-free discounted value of the expected cash flow.

If another baseline is chosen, A_t changes.

Therefore:

No declared baseline → no meaningful A. (2.5)

2.3 What R means

R_t is the admitted value under the CAPM discount protocol:

R_t = CF_t/(1 + r_base + βERP)^t. (2.6)

It is “admitted” in the limited sense that it is the value surviving the declared CAPM filter.

The word admitted does not mean objectively true under every possible financial framework.

A certainty-equivalent model, stochastic discount factor, credit model, liquidity model, or option-pricing model may produce a different admitted value.

Every financial value is protocol-relative in this limited operational sense.

2.4 Dimensional consistency

The construction uses the following units:

Symbol	Meaning	Unit

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CF_t	future cash flow	currency
A_t	baseline-discounted amplitude	currency
R_t	CAPM-admitted present value	currency
Q_t	orthogonal valuation coordinate	currency
r_base	baseline discount rate	dimensionless rate
ERP	equity risk premium	dimensionless rate
β	market-risk coefficient	dimensionless
t	horizon	time-period count
θ_t	valuation phase	dimensionless angle
Z_t	completed valuation state	currency-valued complex coordinate

Q has the same monetary unit as A and R.

This does not make Q a second cash balance or a separately tradable price.

A derivative sensitivity can also be measured in currency without being a standalone asset price.

The unit tells us what the quantity measures, not what kind of economic object it is.

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3. From the CAPM Ratio to a Valuation Phase

3.1 The admitted-value ratio

Define:

$c_t = R_t/A_t$. (3.1)

Substitute:

$R_t = CF_t/(1 + r)^t$. (3.2)

$A_t = CF_t/(1 + r_{base})^t$. (3.3)

Then:

$c_t = [(1 + r_{base})/(1 + r)]^t$. (3.4)

Using:

$r = r_{base} + \beta ERP$, (3.5)

we obtain:

$c_t = [(1 + r_{base})/(1 + r_{base} + \beta ERP)]^t$. (3.6)

The ratio c_t summarizes the amount of baseline amplitude remaining on the admitted-value axis after the CAPM discount adjustment.

3.2 Definition of the CAPM valuation phase

Define:

$\theta_t = \arccos(c_t)$. (3.7)

Therefore:

$\cos \theta_t = c_t$. (3.8)

Hence:

$\cos \theta_t = R_t/A_t$. (3.9)

And:

$\cos \theta_t = [(1 + r_{base})/(1 + r)]^t$. (3.10)

The phase θ_t is not imported from physics.

It is a change of coordinates applied to a mature financial ratio.

3.3 The ordinary positive-claim domain

Assume:

$CF_t > 0$. (3.11)

$t > 0$. (3.12)

$1 + r > 0$. (3.13)

$r \geq r_{base}$. (3.14)

Then:

$0 < R_t/A_t \leq 1$. (3.15)

Therefore:

$0 \leq \theta_t < \pi/2$. (3.16)

This has an important consequence.

The ordinary positive-cash-flow CAPM construction directly occupies only the first quadrant of the completed valuation plane.

Ordinary increases in beta, ERP, or the CAPM required return reduce R toward zero but do not naturally make R negative.

The wider cycle:

$R \rightarrow -Q \rightarrow -R \rightarrow Q \rightarrow R$

is therefore not generated merely by increasing the ordinary positive discount rate.

It belongs to the completed measurement structure, not to the original positive-claim CAPM domain alone.

3.4 Phase is protocol-relative

θ depends on:

- the selected cash flow;
- the selected horizon;
- the baseline rate;
- beta;
- the equity risk premium;
- the norm convention;
- the sign convention.

A different baseline produces a different A.

A different A produces a different ratio R/A.

A different ratio produces a different θ.

Therefore θ is not a universal property of the asset independently of valuation protocol.

The correct statement is:

> θ is the angular coordinate canonically implied by the declared valuation geometry.

It is canonical within the protocol, not absolute outside it.

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4. The CAPM Projection Theorem

4.1 Statement

Theorem 1 — CAPM Projection Theorem

Given:

$A_t = CF_t/(1 + r_{base})^t$, (4.1)

$r = r_{base} + \beta ERP$, (4.2)

and:

$\cos \theta_t = [(1 + r_{base})/(1 + r)]^t$, (4.3)

then:

$A_t \cos \theta_t = CF_t/(1 + r)^t$. (4.4)

Therefore:

$R_t = A_t \cos \theta_t$. (4.5)

Ordinary CAPM DCF is exactly the real-axis projection of the declared baseline-discounted amplitude.

4.2 Proof

Begin with:

$A_t \cos \theta_t$. (4.6)

Substitute the definition of A_t:

$A_t \cos \theta_t = [CF_t/(1 + r_{base})^t] \cos \theta_t$. (4.7)

Substitute the definition of $\cos \theta_t$:

$A_t \cos \theta_t = [CF_t/(1 + r_{base})^t][(1 + r_{base})/(1 + r)]^t$. (4.8)

Cancel the baseline factors:

$A_t \cos \theta_t = CF_t/(1 + r)^t$. (4.9)

But:

$R_t = CF_t/(1 + r)^t$. (4.10)

Therefore:

$R_t = A_t \cos \theta_t$. (4.11)

Q.E.D.

4.3 Interpretation

The theorem does not alter the CAPM present value.

It changes the representation of how the value is formed.

The scalar description is:

$R_t = CF_t/(1 + r)^t$. (4.12)

The geometric description is:

$R_t = A_t \cos \theta_t$. (4.13)

These are numerically identical.

The geometric form adds a declared magnitude and orientation.

That additional structure allows the orthogonal coordinate Q to be defined without introducing a second independent risk assumption.

4.4 Why this is not an alternative pricing model

The construction does not say:

CAPM gives the wrong R, and the complex model gives a better R. (4.14)

It says:

CAPM gives R, and the complex model embeds that same R inside a larger coordinate system. (4.15)

The new model therefore succeeds or fails on a different question:

> Does the completed coordinate system improve sensitivity analysis, risk attribution, scenario interpretation, cross-frame comparison, or empirical diagnosis?

Elegance alone is not sufficient.

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5. Completing the Valuation State

5.1 Definition of Q

Once:

$R_t = A_t \cos \theta_t$, (5.1)

define:

$Q_t = A_t \sin \theta_t$. (5.2)

Then:

$R_t^2 + Q_t^2 = A_t^2\cos^2\theta_t + A_t^2\sin^2\theta_t$. (5.3)

Using:

$\cos^2\theta_t + \sin^2\theta_t = 1$, (5.4)

we obtain:

$A_t^2 = R_t^2 + Q_t^2$. (5.5)

Therefore:

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (5.6)

The positive root is used in the ordinary first-quadrant construction.

5.2 Explicit CAPM formula for Q

Using:

$A_t = CF_t/(1 + r_{base})^t$, (5.7)

and:

$\cos \theta_t = [(1 + r_{base})/(1 + r)]^t$, (5.8)

we have:

$\sin \theta_t = \sqrt{\{1 - [(1 + r_{base})/(1 + r)]^{(2t)}\}}$. (5.9)

Therefore:

$Q_t = [CF_t/(1 + r_{base})^t]\sqrt{1 - [(1 + r_{base})/(1 + r)]^{(2t)}}$. (5.10)

Using:

$r = r_{base} + \beta ERP$, (5.11)

the fully expanded form is:

$Q_t = [CF_t/(1 + r_{base})^t]\sqrt{1 - [(1 + r_{base})/(1 + r_{base} + \beta ERP)]^{(2t)}}$. (5.12)

Thus Q is determined by:

$Q_t = Q_t(CF_t, r_{base}, \beta, ERP, t)$. (5.13)

Q is derived, not appended.

5.3 The completed complex state

Define:

$Z_t = R_t + iQ_t$. (5.14)

Substitute:

$R_t = A_t \cos \theta_t$. (5.15)

$Q_t = A_t \sin \theta_t$. (5.16)

Then:

$Z_t = A_t \cos \theta_t + iA_t \sin \theta_t$. (5.17)

Factor out A_t:

$Z_t = A_t(\cos \theta_t + i \sin \theta_t)$. (5.18)

Using Euler's identity:

$\exp(i\theta_t) = \cos \theta_t + i \sin \theta_t$, (5.19)

we obtain:

$Z_t = A_t \exp(i\theta_t)$. (5.20)

The complex state preserves:

- the magnitude A_t;
- the admitted coordinate R_t;
- the orthogonal coordinate Q_t;
- the valuation orientation θ_t.

5.4 The unresolved question

At this stage, Q has a valid mathematical definition.

It is the orthogonal complement required by:

$A^2 = R^2 + Q^2$. (5.21)

But a financial scholar may still object:

> A Pythagorean complement is not yet a financial quantity.

That objection is correct.

The next task is not to attach a verbal metaphor to Q.

The next task is to derive its exact role under financial measurement and sensitivity analysis.

That derivation begins by first establishing what Q is not.

6. What Q Is Not

6.1 Q is not the scalar CAPM haircut

The most immediate temptation is to interpret Q as the amount of value removed by the CAPM risk adjustment.

Define the scalar haircut:

$H_t = A_t - R_t$. (6.1)

Using:

$R_t = A_t \cos \theta_t$, (6.2)

we obtain:

$H_t = A_t(1 - \cos \theta_t)$. (6.3)

But Q_t is:

$Q_t = A_t \sin \theta_t$. (6.4)

Therefore:

$Q_t \neq H_t$. (6.5)

Equivalently:

$Q_t \neq A_t - R_t$. (6.6)

The two quantities have different mathematical roles.

H_t is a same-axis difference between two scalar valuations.

Q_t is an orthogonal coordinate in the completed valuation plane.

A same-axis subtraction asks:

> How much lower is the CAPM value than the baseline value?

The orthogonal completion asks:

> What conjugate coordinate is implied when the baseline magnitude A_t is preserved while R_t is treated as its real projection?

These are not the same question.

6.2 Why Q can be much larger than A − R

For small θ:

$\cos \theta \approx 1 - \theta^2/2$. (6.7)

$\sin \theta \approx \theta$. (6.8)

Therefore:

$H = A(1 - \cos \theta) \approx A\theta^2/2$. (6.9)

And:

$Q = A \sin \theta \approx A\theta$. (6.10)

Eliminating θ gives:

$H \approx Q^2/(2A)$. (6.11)

Thus the scalar haircut is second order in the phase displacement, while Q is first order.

This explains why Q may be numerically much larger than A − R when the phase angle is small.

That numerical difference does not mean the framework has created additional economic loss.

It means that Q and H measure different structures:

H = accumulated real-axis displacement. (6.12)

Q = conjugate first-order phase scale. (6.13)

6.3 Q is not expected loss

Expected loss usually has a structure such as:

Expected Loss = Probability of Default × Loss Given Default × Exposure. (6.14)

Or more generally:

$EL = \Sigma_s p_s L_s$. (6.15)

Q does not have this definition.

The CAPM-derived Q can be calculated without specifying:

- default probability;
- recovery rate;
- loss distribution;
- tail quantile;
- scenario frequency.

Therefore Q is not expected loss unless an additional model is introduced that separately proves such an equivalence.

The geometric identity alone does not provide that proof.

6.4 Q is not realized loss

A realized loss requires an actual change in value and a recognition event.

If the valuation phase changes by $d\theta$ while A remains fixed:

$$dR = -Qd\theta. \quad (6.16)$$

The loss or gain is dR.

Q is the coefficient multiplying the phase movement.

If:

$$d\theta = 0, \quad (6.17)$$

then:

$$dR = 0, \quad (6.18)$$

even when:

$$Q > 0. \quad (6.19)$$

Therefore Q is an exposure without being an event.

The existence of risk sensitivity does not imply that the risk has been realized.

6.5 Q is not volatility

Volatility measures dispersion over a declared return process.

A standard volatility measure may be written:

$$\sigma = \sqrt{Var(r)}. \quad (6.20)$$

Q is instead derived from the valuation ratio:

$$Q = \sqrt{(A^2 - R^2)}. \quad (6.21)$$

Volatility may influence beta, expected return, option prices, discount rates, or market-implied

valuation.

But Q is not mathematically identical to σ.

Two assets may have the same volatility and different Q values because they have different:

- cash flows;
- horizons;
- base rates;
- betas;
- admitted values.

Conversely, two assets may have the same Q and different volatilities.

6.6 Q is not beta

Beta is a dimensionless market-exposure coefficient:

$\beta = \mathrm{Cov}(r_i, r_m)/\mathrm{Var}(r_m)$. (6.22)

Q is measured in currency.

The relation between them is mediated by the complete CAPM DCF construction:

$\beta \rightarrow r_{CAPM} \rightarrow R \rightarrow \theta \rightarrow Q$. (6.23)

Beta contributes to Q, but beta is not Q.

The distinction matters because the same beta can produce different Q values for different:

- cash-flow magnitudes;
- maturities;
- baseline rates;
- equity risk premia.

6.7 Q is not Value at Risk or Expected Shortfall

Value at Risk at confidence level α may be written schematically as:

$VaR_\alpha = -\mathrm{Quantile}_\alpha(\Delta V)$. (6.24)

Expected Shortfall is:

$ES_\alpha = -E[\Delta V \mid \Delta V \leq -VaR_\alpha]$. (6.25)

Both depend on a distribution of future value changes.

Q does not require such a distribution.

It is a local coordinate derived from the declared valuation geometry.

Q may later be used inside a scenario or distributional model, but that does not make it identical to VaR or Expected Shortfall.

6.8 Q is not automatically duration

For the single cash flow:

R = CF_t/(1 + r)^t. (6.26)

The required-return sensitivity is:

∂R/∂r = −tR/(1 + r). (6.27)

A duration-like measure therefore arises naturally.

But Q is not equal to:

tR/(1 + r). (6.28)

Instead, Q becomes equal to the derivative of R only after the risk coordinate is changed from r to θ:

∂R/∂θ = −Q. (6.29)

Thus Q is not ordinary duration.

It is the exposure coefficient associated with a different declared risk coordinate.

6.9 Q is not automatically opportunity cost

Opportunity cost normally refers to the value of the best foregone alternative.

If V_best is the best unchosen alternative and V_chosen is the chosen alternative, then:

OC = V_best − V_chosen. (6.30)

The CAPM baseline A is not automatically V_best.

A is the value of the same cash flow under the baseline discount protocol.

Therefore:

A − R (6.31)

is not automatically opportunity cost.

It is first a scalar CAPM filter haircut.

Only when an additional decision model establishes:

A = V_best, (6.32)

and:

R = V_chosen, (6.33)

may:

A − R = OC. (6.34)

Under that additional condition, Q acquires a marginal opportunity-cost interpretation. That result will be derived later.

6.10 Q is not a second market price

R may be compared with a quoted market price.

Q generally cannot.

Q is not ordinarily:

- paid by a buyer;
- received by a seller;
- posted as a standalone quote;
- recorded as a separate cash balance;
- traded independently of the underlying valuation construction.

The fact that Q is measured in currency does not make it a price.

Financial sensitivities such as dollar duration and option delta exposure may also be currency-valued without being separate prices.

The correct initial classification is:

Q = conjugate valuation coordinate. (6.35)

The stronger financial identity will be derived from differentiation.

---

7. Static Information and the Null Hypothesis

7.1 Q is algebraically determined by A and R

For the one-period construction:

$Q = \sqrt{(A^2 - R^2)}$. (7.1)

Therefore, once A and R are known, Q is known.

No additional market observation is required.

This leads to an important limitation.

Proposition 1 — Static Non-Independence

In a one-period static CAPM completion, Q contains no algebraically independent information beyond the declared pair (A,R). (7.2)

This proposition should not be hidden.

It is central to the empirical discipline of the framework.

7.2 The complex completion does not manufacture information

A coordinate transformation cannot create new empirical facts from nothing.

The following representations contain the same static information:

(A,R). (7.3)

(A,θ). (7.4)

(R,Q). (7.5)

(A,θ,Q). (7.6)

Provided the relevant sign and domain conventions are known, each representation can be recovered from the others.

For example:

$\theta = \arccos(R/A)$. (7.7)

$Q = \sqrt{(A^2 - R^2)}$. (7.8)

$A = \sqrt{(R^2 + Q^2)}$. (7.9)

$\theta = \arctan(Q/R)$. (7.10)

The complex representation therefore does not reveal an independent hidden variable merely by changing notation.

7.3 What the completion actually preserves

Although it creates no new static information, it preserves and reorganizes information that scalar reporting may suppress.

The scalar result R alone does not preserve A.

If only R is reported, the observer cannot recover:

- the declared baseline amplitude;
- the filter ratio;
- the implied angle;
- the orthogonal coordinate.

Two valuations can have the same R but different A:

$R_1 = R_2$. (7.11)

$A_1 \neq A_2$. (7.12)

Then:

$Q_1 \neq Q_2$. (7.13)

Thus the additional information enters through the declared baseline A, not through an unexplained independent Q.

The complete comparison is:

Scalar report = R. (7.14)

Completed report = (A,R,Q,θ,P). (7.15)

Here P denotes the valuation protocol.

7.4 The strongest null hypothesis

The framework should be tested against the following null hypothesis:

> In a one-period static setting, Q is only an elegant reparameterization of already available valuation information and provides no incremental practical benefit. (7.16)

This null hypothesis is strong but appropriate.

To justify Q operationally, the framework must improve at least one of the following:

- sensitivity interpretation;
- cross-asset comparison;
- multi-horizon aggregation;
- scenario analysis;
- model reconciliation;
- residual diagnosis;
- communication of risk structure;
- intervention design.

If it improves none of them, Q remains optional notation.

7.5 Where genuine novelty may arise

The static one-period identity is only the beginning.

Potential empirical or diagnostic novelty may arise when:

1. A and R vary through time;
2. different assets share the same R but not the same A;
3. different valuation protocols generate different Q values;
4. model-implied Q differs from market-implied Q;
5. multiple cash-flow phases interact;
6. a common phase shock acts across a portfolio;
7. Q predicts nonlinear repricing or gate pressure;
8. residuals remain after the declared phase model has been applied.

The key transition is:

Static Coordinate → Dynamic Exposure. (7.17)

The next theorem supplies that transition.

---

8. The Conjugate Risk Theorem

8.1 Holding the amplitude fixed

Consider the completed valuation state:

$Z = A \exp(i\theta)$. (8.1)

Its real coordinate is:

$R(\theta) = A \cos \theta$. (8.2)

Its orthogonal coordinate is:

$Q(\theta) = A \sin \theta$. (8.3)

For the moment, hold A fixed.

This means that the underlying baseline-discounted cash-flow amplitude is unchanged while the valuation orientation moves.

The movement is therefore angular rather than radial.

8.2 Differentiate admitted value with respect to phase

Differentiate:

$R(\theta) = A \cos \theta$. (8.4)

Because A is fixed:

$\partial R/\partial \theta = -A \sin \theta$. (8.5)

But:

$Q = A \sin \theta$. (8.6)

Therefore:

$\partial R/\partial \theta = -Q$. (8.7)

This is the principal identity of the article.

8.3 The Conjugate Risk Theorem

Theorem 2 — Conjugate Risk Theorem

For a fixed-amplitude valuation state:

$Z = R + iQ = A \exp(i\theta)$, (8.8)

the admitted value and orthogonal coordinate satisfy:

$\partial R/\partial \theta = -Q$. (8.9)

And:

$\partial Q/\partial \theta = R$. (8.10)

Therefore R and Q form a conjugate rotational pair.

8.4 Proof of the second relation

Differentiate:

$Q(\theta) = A \sin \theta$. (8.11)

Holding A fixed:

$\partial Q/\partial \theta = A \cos \theta$. (8.12)

But:

$R = A \cos \theta$. (8.13)

Therefore:

$\partial Q/\partial \theta = R$. (8.14)

Q.E.D.

8.5 Definition of CAPM Phase Delta

Define the signed valuation-phase sensitivity:

$\Delta_\theta \equiv \partial R/\partial \theta$. (8.15)

Then:

$\Delta_\theta = -Q.$ (8.16)

The quantity Q is therefore the magnitude of CAPM Phase Delta in the ordinary first-quadrant construction:

$Q = |\Delta_\theta|.$ (8.17)

Because:

$Q \geq 0,$ (8.18)

and:

$\partial R/\partial\theta \leq 0$ (8.19)

for:

$0 \leq \theta < \pi/2.$ (8.20)

8.6 Financial interpretation

The ordinary CAPM value R answers:

> What is the admitted present value under the declared required return?

The phase derivative answers:

> How rapidly does that admitted value change when the valuation orientation moves?

The answer is:

$-Q$ currency units per unit phase. (8.21)

Thus Q is not another price.

It is an exposure coefficient.

The finance-first interpretation is:

> Q is the magnitude of the first-order dollar sensitivity of admitted value to movement in the declared CAPM valuation phase. (8.22)

8.7 Units

θ is dimensionless.

Therefore:

$\partial R/\partial\theta$ (8.23)

has the same monetary unit as R.

Hence Q has the correct unit for a dollar exposure to a dimensionless risk coordinate.

This is analogous to other financial sensitivities.

For example, an option delta multiplied by a unit change in the underlying produces a value change.

Here:

Phase Delta × Phase Movement = Value Change. (8.24)

Specifically:

$-Q \times d\theta = dR$. (8.25)

8.8 Sign convention

Under the declared orientation:

$R = A \cos \theta$. (8.26)

$Q = A \sin \theta$. (8.27)

Increasing θ rotates value away from the positive real axis.

Therefore:

$\partial R/\partial \theta = -Q$. (8.28)

A positive phase movement produces a negative first-order change in R:

$d\theta > 0 \Rightarrow dR < 0$. (8.29)

A negative phase movement produces a positive first-order change:

$d\theta < 0 \Rightarrow dR > 0$. (8.30)

The negative sign is therefore economically interpretable.

It expresses the chosen orientation in which increasing CAPM phase means stronger valuation filtering.

8.9 Q as a tangent quantity

The point:

(R,Q) (8.31)

lies on the circle:

$R^2 + Q^2 = A^2$. (8.32)

The tangent vector to the valuation orbit is:

$d/d\theta \, [R,Q]^T = [-Q,R]^T$. (8.33)

Thus Q appears not only as the vertical coordinate.

It also determines the real component of the tangent direction.

This is a deeper result.

The same number Q has two linked roles:

1. orthogonal coordinate of the valuation state;
2. magnitude of the real-axis response to angular movement.

The geometry therefore does not merely store Q.

It turns Q into an operational sensitivity.

---

9. Is θ Merely an Arbitrary Reparameterization?

9.1 The objection

A mathematically trained reader may object:

> Any smooth monotonic variable can be reparameterized. If θ is chosen after observing R, then $\partial R/\partial\theta = -Q$ may be a manufactured identity rather than a substantive financial result.

This objection must be taken seriously.

A derivative depends on the selected coordinate.

For example, if:

$u = f(\theta)$, (9.1)

then:

$\partial R/\partial u = (\partial R/\partial\theta)(\partial\theta/\partial u)$. (9.2)

A different risk coordinate produces a different exposure coefficient.

Why, then, should θ be privileged?

9.2 θ is fixed by three declarations

θ is not selected arbitrarily after the derivative is calculated.

It is fixed by three prior declarations:

1. the baseline amplitude A;
2. the norm:

$A^2 = R^2 + Q^2$; (9.3)

3. the orientation convention:

$R = A\cos\theta$, (9.4)

$Q = A \sin \theta$. (9.5)

Once these are fixed:

$\theta = \arccos(R/A)$. (9.6)

The phase coordinate is therefore the canonical angular coordinate of that declared Euclidean valuation geometry.

9.3 Canonical does not mean universal

The word canonical must be used carefully.

θ is canonical relative to:

- a declared baseline;
- a declared norm;
- a declared orientation;
- a declared valuation protocol.

It is not canonical independently of all such choices.

Change the baseline A, and θ changes.

Change the metric, and Q changes.

Change the orientation convention, and the sign of Q changes.

Therefore:

Canonical within protocol ≠ universal outside protocol. (9.7)

9.4 The role of the Euclidean norm

The model adopts:

$A^2 = R^2 + Q^2$. (9.8)

This is a substantive modelling choice.

A more general completion could use:

$A^2 = g_{RR} R^2 + 2g_{RQ} RQ + g_{QQ} Q^2$. (9.9)

Under such a metric, the orthogonal coordinate and phase relation would change.

The present article therefore does not claim that Euclidean geometry is the only possible financial geometry.

It claims that, once the Euclidean norm is declared, the resulting phase and quarter-turn operator are mathematically determined.

9.5 Why retain the Euclidean completion?

The Euclidean construction has several advantages.

First, it preserves magnitude:

$|Z| = A.$ (9.10)

Second, it gives a unique orthogonal complement up to sign:

$Q = \pm\sqrt{(A^2 - R^2)}.$ (9.11)

Third, it supports a standard rotation group:

$U(\varphi) = \exp(\varphi \mathcal{J}).$ (9.12)

Fourth, it yields the closed derivative relations:

$\partial R/\partial\theta = -Q.$ (9.13)

$\partial Q/\partial\theta = R.$ (9.14)

Fifth, it allows the same structure to be compared with familiar quadrature methods in engineering and signal analysis.

These advantages justify examining the Euclidean model seriously.

They do not remove the need for empirical comparison with alternative metrics.

9.6 The financial content is not the angle alone

The financial content of the construction lies in the entire declared chain:

$CF_t \to A_t \to R_t/A_t \to \theta_t \to Q_t.$ (9.15)

If θ were chosen independently of CAPM, the construction would be arbitrary.

But θ is tied to the mature valuation ratio:

$\cos \theta_t = [(1 + r_{base})/(1 + r_{CAPM})]^t.$ (9.16)

Therefore changes in:

- beta;
- ERP;
- required return;
- horizon;

produce determinate changes in θ.

The next section translates the phase sensitivity back into these ordinary CAPM variables.

---

10. Translating Phase Exposure Back into CAPM Variables

10.1 The ordinary required-return sensitivity

For one future cash flow:

$R = CF_t/(1 + r)^t$. (10.1)

Differentiate with respect to r:

$\partial R/\partial r = -tCF_t/(1 + r)^{(t+1)}$. (10.2)

Because:

$R = CF_t/(1 + r)^t$, (10.3)

we may write:

$\partial R/\partial r = -tR/(1 + r)$. (10.4)

This is the familiar negative sensitivity of present value to the required return.

Define the positive dollar exposure coefficient:

$D_r \equiv -\partial R/\partial r$. (10.5)

Therefore:

$D_r = tR/(1 + r)$. (10.6)

10.2 Phase sensitivity to the required return

Recall:

$\cos \theta = R/A$. (10.7)

Hold A fixed with respect to the CAPM risk increment under examination.

Differentiate both sides with respect to r:

$-\sin \theta \cdot \partial\theta/\partial r = (1/A)\partial R/\partial r$. (10.8)

Because:

$\sin \theta = Q/A$, (10.9)

we have:

$-(Q/A)\partial\theta/\partial r = (1/A)\partial R/\partial r$. (10.10)

Multiply by A:

$-Q\partial\theta/\partial r = \partial R/\partial r$. (10.11)

Substitute:

$\partial R/\partial r = -tR/(1 + r)$. (10.12)

Then:

$Q\partial\theta/\partial r = tR/(1 + r)$. (10.13)

Therefore:

$\partial\theta/\partial r = tR/[(1 + r)Q]$. (10.14)

10.3 The exact bridge equation

By the chain rule:

$\partial R/\partial\theta = (\partial R/\partial r)/(\partial\theta/\partial r)$. (10.15)

Substitute:

$\partial R/\partial r = -tR/(1 + r)$, (10.16)

and:

$\partial\theta/\partial r = tR/[(1 + r)Q]$. (10.17)

Then:

$\partial R/\partial\theta = [-tR/(1 + r)]/[tR/((1 + r)Q)]$. (10.18)

All common factors cancel:

$\partial R/\partial\theta = -Q$. (10.19)

The phase result is therefore exactly consistent with ordinary CAPM discount-rate sensitivity.

10.4 Differential form

The ordinary required-return representation is:

$dR = -[tR/(1 + r)]dr$. (10.20)

The phase representation is:

$dR = -Qd\theta$. (10.21)

Therefore:

$Qd\theta = [tR/(1 + r)]dr$. (10.22)

This is the central bridge equation between familiar finance and the complex valuation plane.

It shows that:

Dollar exposure to phase × phase movement
= Dollar exposure to required return × required-return movement. (10.23)

10.5 Q as a change-of-risk-coordinate exposure

Ordinary sensitivity analysis depends on the selected risk coordinate.

Under the required-return coordinate r, the exposure coefficient is:

$D_r = tR/(1 + r)$. (10.24)

Under the valuation-phase coordinate θ, the exposure coefficient is:

$D_\theta = Q$. (10.25)

The two are linked by:

$D_\theta \, d\theta = D_r \, dr$. (10.26)

Thus Q is not a replacement for duration-like sensitivity.

It is the exposure coefficient obtained after changing from the rate coordinate to the phase coordinate.

The correct description is:

> Q is the phase-normalized dollar exposure of the admitted CAPM value. (10.27)

The term phase-normalized does not mean that Q is already a standard duration measure.

It means that the same local P&L has been expressed per unit movement in θ rather than per unit movement in r.

10.6 Why the phase coordinate may still be useful

A change of coordinates is not automatically useful.

Its value depends on whether the new coordinate improves:

- comparison;
- aggregation;
- interpretation;
- invariance;
- diagnosis.

The phase coordinate may be useful because it converts the bounded ratio:

$0 < R/A \leq 1$ (10.28)

into an angular orientation:

$0 \leq \theta < \pi/2$. (10.29)

This may allow:

- assets with different cash-flow scales to be compared by orientation;

- multiple filters to be represented as rotations;
- common phase shocks to be aggregated;
- mark and exposure to form a closed operator structure.

These possibilities must be tested rather than assumed.

10.7 Required-return form of the operator

Because:

$\partial/\partial\theta = (\partial\theta/\partial r)^{-1}\ \partial/\partial r$, (10.30)

and:

$\partial\theta/\partial r = tR/[(1 + r)Q]$, (10.31)

we obtain:

$\partial/\partial\theta = [(1 + r)Q/(tR)]\partial/\partial r$. (10.32)

Define:

$\mathcal{J}_r \equiv [(1 + r)Q/(tR)]\partial/\partial r$. (10.33)

Then:

$\mathcal{J}_r\ R = -Q$. (10.34)

The next task is to determine what happens when the same operator is applied again.

11. Beta, ERP, and Base-Rate Bumps

11.1 Required return is a composite risk coordinate

The CAPM required return is:

$r = r_{base} + \beta ERP$. (11.1)

Therefore a small movement in r may arise from several sources:

$dr = dr_{base} + ERP \cdot d\beta + \beta \cdot dERP$. (11.2)

The same change in admitted value can therefore be attributed to:

- a movement in the baseline rate;
- a movement in beta;
- a movement in the equity risk premium;
- a joint movement in more than one component.

The phase representation must preserve this attribution structure.

11.2 Beta sensitivity of admitted value

Holding r_base and ERP fixed:

$\partial r/\partial \beta = ERP.$ (11.3)

Using:

$\partial R/\partial r = -tR/(1 + r)$, (11.4)

the chain rule gives:

$\partial R/\partial \beta = (\partial R/\partial r)(\partial r/\partial \beta).$ (11.5)

Therefore:

$\partial R/\partial \beta = -tERP \cdot R/(1 + r).$ (11.6)

This is the ordinary dollar sensitivity of the discounted value to beta.

11.3 Beta sensitivity of valuation phase

From:

$\partial \theta/\partial r = tR/[(1 + r)Q]$, (11.7)

and:

$\partial r/\partial \beta = ERP$, (11.8)

we obtain:

$\partial \theta/\partial \beta = tERP \cdot R/[(1 + r)Q].$ (11.9)

Therefore:

$\partial R/\partial \theta = (\partial R/\partial \beta)/(\partial \theta/\partial \beta).$ (11.10)

Substituting the two derivatives:

$\partial R/\partial \theta = [-tERP \cdot R/(1 + r)]/[tERP \cdot R/((1 + r)Q)].$ (11.11)

Hence:

$\partial R/\partial \theta = -Q.$ (11.12)

The result is unchanged.

The phase exposure can therefore be recovered by a beta bump, provided the same bump is translated into the corresponding movement in θ.

11.4 ERP sensitivity

Holding r_base and β fixed:

$\partial r/\partial ERP = \beta$. (11.13)

Therefore:

$\partial R/\partial ERP = -t\beta R/(1 + r)$. (11.14)

Similarly:

$\partial \theta/\partial ERP = t\beta R/[(1 + r)Q]$. (11.15)

Hence:

$\partial R/\partial \theta = (\partial R/\partial ERP)/(\partial \theta/\partial ERP) = -Q$. (11.16)

Thus rate, beta, and ERP bumps all recover the same phase exposure when they move the valuation state along the same declared CAPM orbit.

11.5 The beta-space and ERP-space operators

Define the beta-space phase operator:

$\mathcal{J}_\beta \equiv [(1 + r)Q/(tERP \cdot R)]\partial/\partial\beta$. (11.17)

Then:

$\mathcal{J}_\beta R = -Q$. (11.18)

Similarly, define the ERP-space phase operator:

$\mathcal{J}_{ERP} \equiv [(1 + r)Q/(t\beta R)]\partial/\partial ERP$. (11.19)

Then:

$\mathcal{J}_{ERPR} = -Q$. (11.20)

These are not different underlying operators.

They are different coordinate representations of the same vector field:

$\mathcal{J} = \partial/\partial\theta$. (11.21)

11.6 Base-rate movements require additional care

A movement in r_base is more complicated.

The baseline rate appears in both:

A = CF_t/(1 + r_base)^t, (11.22)

and:

r = r_base + βERP. (11.23)

Therefore a movement in r_base generally changes both:

- the amplitude A;

- the valuation orientation θ.

The resulting movement is not purely angular.

The total differential of R is:

$$dR = (\partial R/\partial A)dA + (\partial R/\partial \theta)d\theta. \quad (11.24)$$

Because:

$$R = A \cos \theta, \quad (11.25)$$

we have:

$$\partial R/\partial A = \cos \theta = R/A. \quad (11.26)$$

And:

$$\partial R/\partial \theta = -Q. \quad (11.27)$$

Therefore:

$$dR = (R/A)dA - Qd\theta. \quad (11.28)$$

This decomposition separates:

- radial change in the baseline economic amplitude;
- angular change in the valuation filter.

A base-rate shock cannot automatically be treated as a pure phase shock unless A is held fixed by protocol.

11.7 Radial and angular attribution

Define the proportional radial change:

$$g_A \equiv dA/A. \quad (11.29)$$

Then:

$$dA = Ag_A. \quad (11.30)$$

Substituting into:

$$dR = (R/A)dA - Qd\theta, \quad (11.31)$$

gives:

$$dR = Rg_A - Qd\theta. \quad (11.32)$$

Similarly, since:

$$Q = A \sin \theta, \quad (11.33)$$

we obtain:

$$dQ = Qg_A + Rd\theta. \quad (11.34)$$

These equations distinguish two different sources of movement:

Radial economic change:

$dR_{radial} = Rg_A$. (11.35)

Angular valuation change:

$dR_{angular} = -Qd\theta$. (11.36)

The distinction is essential.

A fall in R may arise because the expected cash flow amplitude has weakened, because the valuation filter has tightened, or because both have occurred.

11.8 Joint CAPM shocks

For a general CAPM shock:

$dr = dr_{base} + ERP \cdot d\beta + \beta \cdot dERP$, (11.37)

the angular component is:

$d\theta = (\partial\theta/\partial r)dr$ + additional baseline-amplitude correction. (11.38)

If r_{base} is fixed:

$d\theta = [tR/((1 + r)Q)](ERP \cdot d\beta + \beta \cdot dERP)$. (11.39)

Then:

$dR = -Qd\theta$. (11.40)

If r_{base} also moves, the full decomposition must retain dA:

$dR = (R/A)dA - Qd\theta$. (11.41)

This is the correct local attribution equation.

11.9 Finite-difference implementation

The derivatives need not be calculated symbolically.

Suppose a small beta bump ε is applied.

Define:

$\beta_+ = \beta + \varepsilon$. (11.42)

$\beta_- = \beta - \varepsilon$. (11.43)

Revalue:

$R_+ = R(\beta_+)$. (11.44)

$R_- = R(\beta_-)$. (11.45)

And calculate:

$\theta_+ = \theta(\beta_+)$. (11.46)

$\theta_- = \theta(\beta_-)$. (11.47)

Then:

$\partial R/\partial\theta \approx (R_+ - R_-)/(\theta_+ - \theta_-)$. (11.48)

For sufficiently small ε:

$(R_+ - R_-)/(\theta_+ - \theta_-) \approx -Q$. (11.49)

The same procedure may be applied to:

- required return;
- ERP;
- another declared filter parameter.

This gives a practical bump-and-revalue interpretation of Q.

---

12. The Complex-Structure Operator

12.1 State-vector representation

Represent the valuation state as:

$x(\theta) = [R(\theta),Q(\theta)]^T$. (12.1)

Using:

$R(\theta) = A\cos\theta$, (12.2)

$Q(\theta) = A\sin\theta$, (12.3)

we have:

$x(\theta) = A[\cos\theta,\sin\theta]^T$. (12.4)

Differentiate with respect to θ:

$dx/d\theta = A[-\sin\theta,\cos\theta]^T$. (12.5)

Therefore:

$dx/d\theta = [-Q,R]^T$. (12.6)

12.2 Definition of the quarter-turn operator

Define:

$\mathcal{J} = [0\ -1;\ 1\ 0]$. (12.7)

Then:

$\mathcal{J}[R,Q]^T = [-Q,R]^T$. (12.8)

Therefore:

$dx/d\theta = \mathcal{J}x$. (12.9)

On the declared valuation orbit:

$\mathcal{J} = d/d\theta$. (12.10)

This equality should be interpreted as an equality of action on the valuation state and functions generated by its phase orbit.

It is not a claim that every derivative operator in finance is multiplication by i.

12.3 First application

Apply $\mathcal{J}$ to x:

$\mathcal{J}x = [-Q,R]^T$. (12.11)

The real component changes from:

R (12.12)

to:

$-Q$. (12.13)

The imaginary-coordinate component changes from:

Q (12.14)

to:

R. (12.15)

This is the real two-dimensional representation of multiplication by i:

$i(R + iQ) = -Q + iR$. (12.16)

12.4 Second application

Apply $\mathcal{J}$ again:

$\mathcal{J}^2x = \mathcal{J}[-Q,R]^T$. (12.17)

Therefore:

$\mathcal{J}^2x = [-R,-Q]^T$. (12.18)

In matrix form:

$\mathcal{J}^2 = [-1\ 0;\ 0\ -1]$. (12.19)

Hence:

$\mathcal{J}^2 = -I$. (12.20)

The second quarter-turn reverses the entire signed orientation of the valuation state.

12.5 Third and fourth applications

A third application gives:

$\mathcal{J}^3 x = [Q,-R]^T$. (12.21)

A fourth gives:

$\mathcal{J}^4 x = [R,Q]^T$. (12.22)

Therefore:

$\mathcal{J}^4 = I$. (12.23)

The complete state cycle is:

$$[R,Q] \\ \rightarrow [-Q,R] \\ \rightarrow [-R,-Q] \\ \rightarrow [Q,-R] \\ \rightarrow [R,Q]. \quad (12.24)$$

The corresponding real-axis readout cycle is:

$R \rightarrow -Q \rightarrow -R \rightarrow Q \rightarrow R$. (12.25)

12.6 Operator Closure Theorem

Theorem 3 — Operator Closure Theorem

For the completed Euclidean valuation state:

$x = [R,Q]^T$, (12.26)

define:

$\mathcal{J} = [0\ -1;\ 1\ 0]$. (12.27)

Then:

$\mathcal{J}^2 = -I$. (12.28)

$\mathcal{J}^4 = I$. (12.29)

And:

$\mathcal{J} x = dx/d\theta$. (12.30)

Therefore the mark and its conjugate phase exposure form a four-step closed measurement

structure.

12.7 The operator in required-return coordinates

Previously:

$\mathcal{J}_r \equiv [(1 + r)Q/(tR)]\partial/\partial r.$ (12.31)

Because:

$\mathcal{J}_r = \partial/\partial\theta,$ (12.32)

we obtain:

$\mathcal{J}_rR = -Q.$ (12.33)

To verify the second application, first differentiate Q with respect to r.

From:

$Q = \sqrt{(A^2 - R^2)},$ (12.34)

with A fixed:

$\partial Q/\partial r = -(R/Q)\partial R/\partial r.$ (12.35)

Using:

$\partial R/\partial r = -tR/(1 + r),$ (12.36)

we obtain:

$\partial Q/\partial r = tR^2/[(1 + r)Q].$ (12.37)

Therefore:

$\mathcal{J}_rQ = [(1 + r)Q/(tR)][tR^2/((1 + r)Q)].$ (12.38)

Hence:

$\mathcal{J}_rQ = R.$ (12.39)

It follows that:

$\mathcal{J}_r^2R = \mathcal{J}_r(-Q) = -R.$ (12.40)

The operator cycle is therefore fully recoverable from ordinary CAPM parameter differentiation.

12.8 Why the coefficient inside $\mathcal{J}_r$ does not create a contradiction

The required-return representation:

$\mathcal{J}_r = [(1 + r)Q/(tR)]\partial/\partial r$ (12.41)

contains state-dependent coefficients.

This does not invalidate the construction.

It means that θ is a nonlinear coordinate on the CAPM valuation curve.

The operator $\mathcal{J}_r$ is the vector field that moves one unit in θ, expressed in the coordinate r.

The fundamental object is:

$\mathcal{J} = \partial/\partial\theta$. (12.42)

The expression $\mathcal{J}_r$ is its representation after a change of coordinates.

---

13. Why the Quarter-Turn Operator Is Not Arbitrary

13.1 Required properties

Suppose a linear operator K is intended to represent the transition from the value axis to its conjugate measurement axis.

Require that K:

1. preserves vector length;
2. maps every nonzero vector to an orthogonal vector;
3. preserves orientation consistently;
4. performs a quarter-turn;
5. returns the negative state after two applications.

These conditions imply:

$K^2 = -I$. (13.1)

13.2 Length preservation

The Euclidean norm is:

$\|x\|^2 = x^T x = R^2 + Q^2$. (13.2)

A length-preserving operator satisfies:

$\|Kx\|^2 = \|x\|^2$. (13.3)

Equivalently:

$K^T K = I$. (13.4)

Thus K must be orthogonal.

13.3 Orthogonality of the rotated state

A quarter-turn must also satisfy:

$x^T K x = 0$ (13.5)

for every x.

This requires K to be skew-symmetric:

$K^T = -K$. (13.6)

In two real dimensions, every skew-symmetric matrix has the form:

$K = [0\ -a;\ a\ 0]$. (13.7)

Length preservation requires:

$a^2 = 1$. (13.8)

Therefore:

$a = +1$ (13.9)

or:

$a = -1$. (13.10)

Hence:

$K = [0\ -1;\ 1\ 0]$ (13.11)

or:

$K = [0\ 1;\ -1\ 0]$. (13.12)

These are the two possible quarter-turn orientations.

13.4 Uniqueness up to orientation

Proposition 2 — Quarter-Turn Uniqueness

On a declared two-dimensional Euclidean valuation plane, a linear operator that preserves length, maps every vector to an orthogonal vector, and satisfies $K^2 = -I$ is unique up to orientation. (13.13)

The two possibilities are:

$\mathcal{J}_+ = [0\ -1;\ 1\ 0]$. (13.14)

$\mathcal{J}_- = -\mathcal{J}_+$. (13.15)

The first represents counterclockwise rotation.

The second represents clockwise rotation.

The article adopts $\mathcal{J}_+$.

13.5 The sign of Q depends on orientation

Under the adopted convention:

$\mathcal{J}[R,Q]^T = [-Q,R]^T$. (13.16)

Under the opposite convention:

$-\mathcal{J}[R,Q]^T = [Q,-R]^T$. (13.17)

Thus the sign of the conjugate readout depends on the declared orientation.

This does not make the framework arbitrary.

Orientation conventions also appear in:

- coordinate geometry;
- electrical engineering;
- Fourier analysis;
- rotational mechanics.

What matters is that the convention is declared and applied consistently.

13.6 Metric dependence

Uniqueness holds only after the Euclidean metric has been declared.

If the valuation plane uses a non-Euclidean metric G:

$\|x\|_G^2 = x^T G x$, (13.18)

then the compatible complex structure must satisfy:

$\mathcal{J}^T G \mathcal{J} = G$. (13.19)

And:

$\mathcal{J}^2 = -I$. (13.20)

The resulting conjugate coordinate may differ from the Euclidean Q.

The present construction is therefore:

- exact under its declared metric;
- not a proof that all financial filters must use that metric.

---

14. Generator and Finite Rotation

14.1 Infinitesimal generator

The local evolution equation is:

$dx/d\theta = \mathcal{J}x$. (14.1)

This says that an infinitesimal phase movement $d\theta$ changes the state by:

$dx = \mathcal{J}x\, d\theta$. (14.2)

In components:

$dR = -Q d\theta$. (14.3)

$dQ = R d\theta$. (14.4)

These are local sensitivity equations.

14.2 Finite rotation

A finite rotation by φ is generated by exponentiating $\mathcal{J}$:

$U(\varphi) = \exp(\varphi \mathcal{J})$. (14.5)

Using the matrix exponential:

$\exp(\varphi \mathcal{J}) = I + \varphi \mathcal{J} + \varphi^2 \mathcal{J}^2/2! + \varphi^3 \mathcal{J}^3/3! + \cdots$. (14.6)

Since:

$\mathcal{J}^2 = -I$, (14.7)

and:

$\mathcal{J}^3 = -\mathcal{J}$, (14.8)

the even and odd terms separate:

$U(\varphi) = I \cos\varphi + \mathcal{J} \sin\varphi$. (14.9)

Therefore:

$U(\varphi) = [\cos\varphi\ -\sin\varphi;\ \sin\varphi\ \cos\varphi]$. (14.10)

This is the ordinary rotation matrix.

14.3 Action on the valuation state

Apply $U(\varphi)$ to:

$x = [R,Q]^{T}$. (14.11)

Then:

$x' = U(\varphi)x$. (14.12)

Therefore:

$R' = R\cos\varphi - Q\sin\varphi.$ (14.13)

$Q' = R\sin\varphi + Q\cos\varphi.$ (14.14)

The magnitude is preserved:

$R'^2 + Q'^2 = R^2 + Q^2 = A^2.$ (14.15)

14.4 Quarter-turn

For:

$\varphi = \pi/2,$ (14.16)

we have:

$\cos(\pi/2) = 0.$ (14.17)

$\sin(\pi/2) = 1.$ (14.18)

Therefore:

$U(\pi/2) = \mathcal{J}.$ (14.19)

And:

$[R,Q]^T \to [-Q,R]^T.$ (14.20)

Thus multiplication by i is the finite quarter-turn generated by $\mathcal{J}$.

14.5 Half-turn

For:

$\varphi = \pi,$ (14.21)

we have:

$U(\pi) = -I.$ (14.22)

Therefore:

$[R,Q]^T \to [-R,-Q]^T.$ (14.23)

The valuation orientation is reversed.

This is the geometric meaning of:

$i^2 = -1.$ (14.24)

It does not yet specify a market event.

It specifies a half-turn in measurement orientation.

14.6 Full turn

For:

$\varphi = 2\pi$, (14.25)

we obtain:

$U(2\pi) = I$. (14.26)

Therefore:

$[R,Q]^T \rightarrow [R,Q]^T$. (14.27)

The state returns after one complete ordinary rotation.

14.7 Generator and quarter-turn are related but distinct

The statements:

$\mathcal{J} = \partial/\partial\theta$ (14.28)

and:

$U(\pi/2) = \mathcal{J}$ (14.29)

can appear to identify differentiation directly with multiplication by i.

The precise relation is:

- $\mathcal{J}$ is the infinitesimal generator;
- $\exp(\varphi\mathcal{J})$ is the finite rotation;
- at $\varphi = \pi/2$, the finite rotation matrix equals $\mathcal{J}$ because $\mathcal{J}$ itself is the quarter-turn matrix.

This special equality follows from the simple two-dimensional complex structure.

It should not be generalized carelessly to arbitrary differential operators.

---

15. A Family of Financial Measurements

15.1 Measurement is not the same as state transformation

A rotation matrix may be applied in two conceptually different ways.

Active interpretation

The state is rotated while the measurement axis remains fixed.

Passive interpretation

The state remains fixed while the measurement axis is rotated.

Both produce related coordinate formulas.

For financial interpretation, the passive version is often more appropriate.

The asset state need not change merely because the analyst changes from:

- mark measurement;
- risk measurement;
- opposite-position measurement.

15.2 Definition of the measurement family

Let:

$Z = R + iQ.$ (15.1)

Define the phase-indexed real readout:

$M_{\varphi}(Z) = \mathrm{Re}[\exp(i\varphi)Z].$ (15.2)

Expanding:

$\exp(i\varphi)Z = (\cos\varphi + i\sin\varphi)(R + iQ).$ (15.3)

Therefore:

$\exp(i\varphi)Z = R\cos\varphi - Q\sin\varphi + i(R\sin\varphi + Q\cos\varphi).$ (15.4)

Hence:

$M_{\varphi}(Z) = R\cos\varphi - Q\sin\varphi.$ (15.5)

15.3 Mark measurement

At:

$\varphi = 0,$ (15.6)

we obtain:

$M_0(Z) = R.$ (15.7)

This is the ordinary admitted-value readout.

It asks:

> What value appears on the declared real valuation axis?

15.4 Conjugate-exposure measurement

At:

$$\varphi = \pi/2, \quad (15.8)$$

we obtain:

$$M_{\pi/2}(Z) = -Q. \quad (15.9)$$

This is the signed CAPM phase exposure.

It asks:

> What is the first-order real-value response to positive movement in the declared valuation phase?

The answer is:

$$-Q. \quad (15.10)$$

15.5 Opposite-mark measurement

At:

$$\varphi = \pi, \quad (15.11)$$

we obtain:

$$M_{\pi}(Z) = -R. \quad (15.12)$$

This reverses the signed valuation orientation.

In a linear position space, $-R$ may represent the mark of the opposite position.

It does not automatically represent:

- the company's value;
- a creditor's loss;
- a regulator's capital charge.

Those interpretations require additional mappings.

15.6 Opposite-exposure measurement

At:

$$\varphi = 3\pi/2, \quad (15.13)$$

we obtain:

$$M_{3\pi/2}(Z) = Q. \quad (15.14)$$

This is the signed phase exposure associated with the opposite valuation orientation.

15.7 Return to the original mark

At:

φ = 2π, (15.15)

we obtain:

M_2π(Z) = R. (15.16)

The measurement cycle closes.

15.8 Measurement-Cycle Theorem

Theorem 4 — Measurement-Cycle Theorem

For:

M_φ(Z) = Re[exp(iφ)Z], (15.17)

the quarter-turn readouts satisfy:

M0(Z) = R. (15.18)

M_π/2(Z) = −Q. (15.19)

M_π(Z) = −R. (15.20)

M_3π/2(Z) = Q. (15.21)

M_2π(Z) = R. (15.22)

Therefore:

R → −Q → −R → Q → R. (15.23)

---

16. The Financial Meaning of the Measurement Cycle

16.1 The first quarter-turn changes the measurement type

The transition:

R → −Q (16.1)

should not initially be read as:

The asset loses R + Q. (16.2)

Nor should it be read as:

The market price becomes −Q. (16.3)

It means:

> The readout has changed from admitted mark to signed phase exposure.

The measured state is the same Z.

The measurement orientation has changed.

16.2 The second quarter-turn reverses the signed position orientation

The transition:

−Q → −R (16.4)

means that another quarter-turn has been applied.

After two quarter-turns:

U(π) = −I. (16.5)

The entire state has reversed orientation:

Z → −Z. (16.6)

In a signed position space, this may correspond to moving from a long orientation to the opposite short orientation.

16.3 The third quarter-turn reads the opposite exposure

The transition:

−R → Q (16.7)

moves from the opposite mark to the opposite phase exposure.

If the long mark has exposure:

−Q, (16.8)

then the opposite position has exposure:

Q. (16.9)

This is consistent with linear risk attribution.

16.4 The fourth quarter-turn closes the system

The transition:

Q → R (16.10)

returns to the original mark.

Therefore the cycle does not represent an irreversible chronology.

It is a closed family of readouts.

Irreversible financial history requires an additional gate-and-ledger process that will be introduced later.

16.5 Mark and risk form a conjugate pair

The cycle may be summarized as:

Mark
→ Conjugate Exposure
→ Opposite Mark
→ Opposite Exposure
→ Mark. (16.11)

In symbols:

R
→ −Q
→ −R
→ Q
→ R. (16.12)

This is the finance-first meaning of multiplication by i in the completed CAPM plane.

16.6 Why $i^2 = -1$ is not a double-loss rule

The identity:

$i^2 = -1$ (16.13)

does not mean:

Risk occurs twice and becomes a negative asset. (16.14)

It means:

Two quarter-turns reverse orientation. (16.15)

The first quarter-turn changes the readout from mark to conjugate exposure.

The second quarter-turn changes the signed orientation of the mark.

Therefore the correct interpretation is:

$i^2 = -1$
= measurement-orientation reversal. (16.16)

It is not:

$i^2 = -1$
= two realized losses. (16.17)

17. Long–Short Duality

17.1 Signed position space

The interpretation of −R becomes financially precise when the valuation state belongs to a linear signed position space.

Let n denote the number of units held in a claim.

If the value functional is linear in position size:

V(n) = nR. (17.1)

Then:

V(−n) = −V(n). (17.2)

A positive position is long.

The corresponding negative position is short.

This relation is natural for:

- ordinary long–short security positions;
- clean bilateral derivative claims;
- signed portfolio exposures;
- positions for which valuation is approximately linear in notional.

It is not automatically valid for every corporate or institutional relationship.

17.2 Complex state of the long position

Let the completed state of the unit long position be:

Z_long = R + iQ. (17.3)

The ordinary mark measurement gives:

M0(Z_long) = R. (17.4)

The conjugate-exposure measurement gives:

M_π/2(Z_long) = −Q. (17.5)

Therefore the long position has:

Long Mark = R. (17.6)

Long Phase Delta = −Q. (17.7)

The negative sign means that a positive movement in valuation phase reduces the admitted value to first order.

17.3 Complex state of the short position

The opposite position is:

Z_short = −Z_long. (17.8)

Therefore:

Z_short = −R − iQ. (17.9)

Its ordinary mark is:

M₀(Z_short) = −R. (17.10)

Its phase exposure is:

M_π/2(Z_short) = Q. (17.11)

Thus:

Short Mark = −R. (17.12)

Short Phase Delta = Q. (17.13)

The value and exposure signs both reverse.

17.4 Long–Short Duality Proposition

Proposition 3 — Long–Short Duality

For a completed valuation state in a linear signed position space:

Z_long = R + iQ, (17.14)

and:

Z_short = −Z_long, (17.15)

the corresponding mark and phase-exposure pairs are:

Long = (R,−Q). (17.16)

Short = (−R,Q). (17.17)

Therefore the quarter-turn measurement cycle may be read as:

Long Mark
→ Long Phase Exposure
→ Short Mark
→ Short Phase Exposure
→ Long Mark. (17.18)

In symbols:

$$R \to -Q \to -R \to Q \to R. \quad (17.19)$$

17.5 Why −Q is not the counterparty's mark

The table makes an important distinction visible:

Measurement	Long position	Short position
Mark	R	−R
Phase Delta	−Q	Q

The opposite party's mark is −R, not −Q.

The quantity −Q is the long position's sensitivity to positive phase movement.

Likewise, Q is the short position's sensitivity.

Therefore:

$$-Q \neq \text{Counterparty Mark}. \quad (17.20)$$

Instead:

$$-Q = \text{Long Phase Delta}. \quad (17.21)$$

And:

$$Q = \text{Short Phase Delta}. \quad (17.22)$$

This resolves the earlier temptation to interpret R and −Q as the simultaneous values of two different stakeholders.

They are different measurement types.

The actual long–short value pair is:

$$R \leftrightarrow -R. \quad (17.23)$$

The actual long–short exposure pair is:

$$-Q \leftrightarrow Q. \quad (17.24)$$

17.6 Clean bilateral derivatives

The long–short interpretation is especially transparent for a clean bilateral derivative.

Let one party hold a derivative with value:

$$V_A = R. \quad (17.25)$$

Ignoring counterparty credit, funding asymmetry, collateral friction, tax differences, and transaction costs, the other party holds:

$$V_B = -R. \quad (17.26)$$

Therefore:

$V_A + V_B = 0$. (17.27)

If both positions are represented within the same phase geometry:

$\Delta_{\theta,A} = -Q$. (17.28)

$\Delta_{\theta,B} = Q$. (17.29)

Hence:

$\Delta_{\theta,A} + \Delta_{\theta,B} = 0$. (17.30)

Both mark and first-order phase exposure clear across the two sides.

This is a genuine financial closure.

17.7 Why the company is not automatically the short side of its shareholders

For ordinary equity, a shareholder may hold a positive market value R.

It does not follow that the issuing company holds a liability with market value −R under the same measurement protocol.

The company's accounting equity, treasury-stock treatment, cost of capital, financing obligations, and enterprise value are governed by different maps.

Therefore:

Company Value ≠ −Shareholder Market Value (17.31)

as a general identity.

Likewise:

Cost of Equity ≠ Q. (17.32)

The cost of equity is a rate.

Q is a currency-valued phase exposure.

A relationship between them exists through the CAPM discount process, but they are not the same financial object.

17.8 Where the duality applies

The long–short interpretation is most defensible when all of the following hold:

1. the claim is defined in a signed linear position space;
2. the same valuation protocol is used for both sides;
3. the same cash flows and horizon are being valued;

4. the same baseline amplitude is used;

5. counterparty and funding asymmetries are ignored or separately modelled;

6. the position reversal is represented by multiplication by −1.

Under these conditions:

$$Z_{short} = -Z_{long} \quad (17.33)$$

is financially meaningful.

Without them, −Z may remain a mathematical orientation reversal without being the value of a specific stakeholder.

---

18. Exposure Is Not Economic Loss

18.1 The distinction between exposure and movement

The identity:

$$\partial R/\partial \theta = -Q \quad (18.1)$$

does not mean that the asset has already lost Q.

It means that Q is the coefficient governing the first-order response of R to a phase movement.

The economic value change is:

$$dR = -Q d\theta. \quad (18.2)$$

If:

$$d\theta = 0, \quad (18.3)$$

then:

$$dR = 0. \quad (18.4)$$

even when Q is large.

A large exposure without movement produces no immediate P&L.

A small exposure with a sufficiently large movement may produce a material P&L.

Therefore exposure and event must remain separate.

18.2 Exact finite rotation

Suppose the state begins at:

$$Z = R + iQ. \quad (18.5)$$

A finite positive phase movement $\Delta\theta$ gives:

$Z_{new} = \exp(i\Delta\theta)Z$. (18.6)

The new real coordinate is:

$R_{new} = R\cos\Delta\theta - Q\sin\Delta\theta$. (18.7)

The new imaginary coordinate is:

$Q_{new} = R\sin\Delta\theta + Q\cos\Delta\theta$. (18.8)

Therefore the exact change in admitted value is:

$\Delta R = R_{new} - R$. (18.9)

Hence:

$\Delta R = R(\cos\Delta\theta - 1) - Q\sin\Delta\theta$. (18.10)

This is the exact finite-rotation P&L at fixed amplitude.

18.3 First-order approximation

For small $\Delta\theta$:

$\sin\Delta\theta \approx \Delta\theta$. (18.11)

$\cos\Delta\theta \approx 1 - (\Delta\theta)^2/2$. (18.12)

Retaining only the first-order term:

$\Delta R \approx -Q\Delta\theta$. (18.13)

Thus:

Phase Delta $= -Q$. (18.14)

Phase Movement $= \Delta\theta$. (18.15)

First-Order P&L $= -Q\Delta\theta$. (18.16)

This is structurally analogous to ordinary sensitivity analysis:

First-Order P&L = Exposure × Risk-Factor Movement. (18.17)

18.4 Higher-order expansion

Using the Taylor series:

$\sin\Delta\theta = \Delta\theta - (\Delta\theta)^3/6 + (\Delta\theta)^5/120 - \cdots$, (18.18)

and:

$\cos\Delta\theta = 1 - (\Delta\theta)^2/2 + (\Delta\theta)^4/24 - \cdots$, (18.19)

substitute into the exact P&L formula:

$\Delta R = R(\cos \Delta\theta - 1) - Q \sin \Delta\theta$. (18.20)

Then:

$\Delta R = -Q\Delta\theta - (R/2)(\Delta\theta)^2 + (Q/6)(\Delta\theta)^3 + (R/24)(\Delta\theta)^4 - (Q/120)(\Delta\theta)^5 + O((\Delta\theta)^6)$. (18.21)

The first two terms are:

$\Delta R \approx -Q\Delta\theta - (R/2)(\Delta\theta)^2$. (18.22)

The first term is directional exposure.

The second term is phase curvature.

18.5 The unusual sign of phase curvature

The second derivative is:

$\partial^2 R/\partial\theta^2 = -R$. (18.23)

Therefore the phase curvature is negative whenever:

$R > 0$. (18.24)

This differs from the positive convexity often associated with ordinary fixed-income price–yield relationships.

The difference is not a contradiction.

It arises because θ is not the yield coordinate.

Under the angular coordinate:

$R = A \cos \theta$ (18.25)

is locally concave in the first quadrant.

Under the required-return coordinate:

$R = CF_t/(1 + r)^t$ (18.26)

has a different second derivative.

Coordinate change alters the form of higher-order sensitivity.

Therefore Phase Gamma must not be confused with ordinary rate convexity.

18.6 Amplitude movement

The preceding finite-rotation formula assumes A is fixed.

If A also changes:

$Z_{new} = (A + \Delta A)\exp[i(\theta + \Delta\theta)]$. (18.27)

To first order:

$$dR = (R/A)dA - Qd\theta. \quad (18.28)$$

Or:

$$dR = Rg_A - Qd\theta, \quad (18.29)$$

where:

$$g_A = dA/A. \quad (18.30)$$

Thus value change has at least two components:

$$\text{Radial P\&L} = Rg_A. \quad (18.31)$$

$$\text{Angular P\&L} = -Qd\theta. \quad (18.32)$$

A complete attribution must not misclassify cash-flow deterioration as phase rotation or phase rotation as cash-flow deterioration.

18.7 Dynamic residual

A real valuation process may not be fully explained by radial and angular movements.

Introduce a residual term ε_R:

$$dR = Rg_A - Qd\theta + \varepsilon_R. \quad (18.33)$$

Similarly:

$$dQ = Qg_A + Rd\theta + \varepsilon_Q. \quad (18.34)$$

The complex residual is:

$$\varepsilon = \varepsilon_R + i\varepsilon_Q. \quad (18.35)$$

A large residual may indicate:

- an omitted risk factor;
- changing liquidity;
- a model break;
- non-CAPM repricing;
- path dependence;
- a change of protocol;
- data error;
- institutional intervention.

The residual prevents the complex geometry from claiming more explanatory power than it has earned.

18.8 Exposure is not recognition

Even when an actual ΔR occurs, it may not immediately enter every financial ledger.

Examples include:

- an unrealized market loss;
- an unrecognized impairment;
- a contingent legal burden;
- an unexercised option payoff;
- an intraday price movement before official close;
- an economic loss excluded under an accounting rule.

Therefore three distinctions are required:

Exposure ≠ Movement. (18.36)

Movement ≠ Recognized P&L. (18.37)

Recognized P&L ≠ Final Economic Resolution. (18.38)

These distinctions will become central in the gate-and-ledger analysis.

---

19. The Periodic Phase-Sensitivity Hierarchy

19.1 Zeroth-order value

Begin with:

$V^{(0)}(\theta) = R(\theta)$. (19.1)

Where:

$R(\theta) = A \cos \theta$. (19.2)

This is the admitted mark.

19.2 First derivative

Differentiate once:

$V^{(1)}(\theta) = \partial R/\partial \theta$. (19.3)

Therefore:

$V^{(1)}(\theta) = -Q$. (19.4)

This is the signed CAPM Phase Delta.

19.3 Second derivative

Differentiate again:

V^(2)(θ) = ∂²R/∂θ². (19.5)

Since:

∂Q/∂θ = R, (19.6)

we obtain:

V^(2)(θ) = −R. (19.7)

The second derivative returns the opposite signed mark.

19.4 Third derivative

Differentiate once more:

V^(3)(θ) = ∂³R/∂θ³. (19.8)

Therefore:

V^(3)(θ) = Q. (19.9)

The third derivative is the opposite signed phase exposure.

19.5 Fourth derivative

A fourth derivative gives:

V^(4)(θ) = ∂⁴R/∂θ⁴. (19.10)

Therefore:

V^(4)(θ) = R. (19.11)

The derivative hierarchy closes:

R → −Q → −R → Q → R. (19.12)

19.6 General derivative formula

The nth derivative may be written:

$\partial^n R/\partial\theta^n = A\cos(\theta + n\pi/2)$. (19.13)

Equivalently:

$\partial^n R/\partial\theta^n = M_n\pi/2(Z)$. (19.14)

Thus differentiation and quarter-turn measurement are two expressions of the same phase

structure.

19.7 Periodicity

Because:

$\cos(\theta + 2\pi) = \cos\theta$, (19.15)

the hierarchy satisfies:

$V^{(n+4)}(\theta) = V^{(n)}(\theta)$. (19.16)

The period is four derivative orders.

This is not typical of an arbitrary valuation function.

It follows specifically from the circular fixed-amplitude geometry.

19.8 Naming discipline

It may be tempting to assign new Greek-style names to every derivative:

- Phase Delta = −Q;
- Phase Gamma = −R;
- Phase Speed = Q;
- Phase Closure = R.

This article should resist unnecessary terminology.

Only the first derivative has an immediate risk-management role:

$\Delta_\theta = -Q$. (19.17)

The higher derivatives are best described directly as:

- second phase derivative;
- third phase derivative;
- fourth phase derivative.

Their periodicity is mathematically important even if no permanent financial names are adopted.

19.9 Comparison with option Greeks

For an option value $V(S,\sigma,t,\ldots)$, ordinary Greeks include:

$\text{Delta} = \partial V/\partial S$. (19.18)

$\text{Gamma} = \partial^2 V/\partial S^2$. (19.19)

Vega = ∂V/∂σ. (19.20)

The CAPM phase hierarchy instead differentiates with respect to θ:

Phase Delta = ∂R/∂θ = −Q. (19.21)

The methodology is Greek-like because it measures local sensitivity.

But Q is not an option Greek unless the underlying valued claim is itself an option and θ is declared as one of its risk coordinates.

19.10 Why the hierarchy matters

The hierarchy shows that Q is not merely appended to R.

R and Q are dynamically linked.

Knowing the operator action on either coordinate determines the other:

$\mathcal{J}$R = −Q. (19.22)

$\mathcal{J}$Q = R. (19.23)

This is stronger than maintaining two independent spreadsheet columns.

The complex structure imposes a lawful transition relation between them.

---

20. Haircut, Marginal Pressure, and Opportunity Cost

20.1 Scalar CAPM haircut

Define the scalar valuation haircut:

H(θ) = A − R(θ). (20.1)

Because:

R(θ) = A cos θ, (20.2)

we have:

H(θ) = A(1 − cos θ). (20.3)

At:

θ = 0, (20.4)

the CAPM filter equals the baseline:

R = A. (20.5)

Therefore:

H(0) = 0. (20.6)

As θ increases, the admitted value falls and the scalar haircut grows.

20.2 Marginal haircut theorem

Differentiate H with respect to θ:

$dH/d\theta = -dR/d\theta$. (20.7)

Since:

$dR/d\theta = -Q$, (20.8)

we obtain:

Proposition 4 — Marginal Haircut Relation

$dH/d\theta = Q$. (20.9)

Thus Q is the marginal growth rate of the scalar CAPM haircut per unit valuation phase.

This is an exact identity.

20.3 Integral relation

Since:

$H(0) = 0$, (20.10)

integrating gives:

$H(\theta) = \int_0^\theta Q(\varphi)d\varphi$. (20.11)

Because:

$Q(\varphi) = A \sin \varphi$, (20.12)

we have:

$H(\theta) = \int_0^\theta A \sin \varphi \, d\varphi$. (20.13)

Therefore:

$H(\theta) = A(1 - \cos \theta)$. (20.14)

The scalar haircut is the accumulated phase-pressure integral.

20.4 Q is a marginal quantity, not the accumulated haircut

The distinction is:

$Q(\theta)$ = Instantaneous marginal pressure at phase θ. (20.15)

$H(\theta)$ = Accumulated real-axis haircut from 0 to θ. (20.16)

Therefore:

$Q \neq H$. (20.17)

But:

$Q = dH/d\theta$. (20.18)

This is a much more precise relation than saying that Q is “similar to” a hidden cost.

20.5 Small-angle relation

For small θ:

$Q \approx A\theta$. (20.19)

And:

$H \approx A\theta^2/2$. (20.20)

Therefore:

$H \approx Q^2/(2A)$. (20.21)

The marginal pressure Q grows linearly in θ.

The accumulated haircut grows quadratically.

This explains why a substantial phase sensitivity may coexist with a relatively small scalar haircut.

20.6 Exact relation between H and Q

Because:

$A = \sqrt{(R^2 + Q^2)}$, (20.22)

the scalar haircut may be written:

$H = \sqrt{(R^2 + Q^2)} - R$. (20.23)

Rationalizing:

$H = Q^2/[\sqrt{(R^2 + Q^2)} + R]$. (20.24)

Since:

$\sqrt{(R^2 + Q^2)} = A$, (20.25)

we obtain:

$H = Q^2/(A + R)$. (20.26)

This relation is exact.

For small Q relative to A:

$A + R \approx 2A$, (20.27)

and therefore:

$H \approx Q^2/(2A)$. (20.28)

20.7 Opportunity-cost condition

Traditional opportunity cost requires an alternative-choice structure.

Let:

V_{best} = value of the best foregone alternative. (20.29)

Let:

V_{chosen} = value of the chosen alternative. (20.30)

Then:

$OC = V_{best} - V_{chosen}$. (20.31)

To identify the CAPM haircut with opportunity cost, one must additionally establish:

$A = V_{best}$. (20.32)

$R = V_{chosen}$. (20.33)

Only then:

$OC = A - R = H$. (20.34)

And:

$dOC/d\theta = Q$. (20.35)

Under these additional conditions, Q is:

> the marginal opportunity cost per unit valuation phase.

Without those conditions, the correct term remains:

> marginal CAPM filter haircut.

20.8 Opportunity cost as a choice-dependent interpretation

The same numerical A may play different roles under different protocols.

Valuation protocol

A = baseline-discounted value of the same cash flow. (20.36)

Then:

A – R = CAPM filter haircut. (20.37)

Decision protocol

A = best foregone alternative value. (20.38)

Then:

A – R = opportunity cost. (20.39)

The geometry alone cannot decide which interpretation applies.

The protocol must declare it.

This is an example of a broader rule:

Same Formula ≠ Same Financial Meaning without the Same Construction. (20.40)

20.9 Shadow-cost interpretation

Even when A is not a best alternative, Q may still be described as a shadow sensitivity:

Q = marginal value removed from the real-axis admission rate per unit phase. (20.41)

This language is safer than calling Q an actual cost.

It emphasizes that Q measures the pressure gradient of the filter, not a settled payment.

---

21. Measurement, Movement, Gate, and Ledger

21.1 Four distinct stages

A complete financial runtime requires four distinct operations:

1. measurement;
2. movement;
3. gate;
4. ledger.

They must not be collapsed into one another.

The basic sequence is:

Measurement → Exposure → State Movement → Economic P&L → Gate → Ledger + Residual. (21.1)

21.2 Measurement identifies exposure

The ordinary measurement reads:

$M_0(Z) = R.$ (21.2)

The conjugate measurement reads:

$M_{\pi/2}(Z) = -Q.$ (21.3)

This identifies the signed phase exposure.

No P&L has yet been generated merely by reading $-Q$.

Measurement answers:

> What would be the first-order response under a unit positive phase movement?

21.3 Movement generates economic consequence

Suppose the phase changes by $\Delta\theta$.

The exact economic value change is:

$\Delta R = R(\cos \Delta\theta - 1) - Q \sin \Delta\theta.$ (21.4)

For a small movement:

$\Delta R \approx -Q\Delta\theta.$ (21.5)

Thus:

Exposure × Movement → Economic P&L. (21.6)

The exposure does not create the movement.

The movement does not arise because the analyst changed measurement equipment.

The risk factor must actually change.

21.4 Gate determines recognition

A financial gate is a declared rule deciding whether an economic change becomes an operative event.

Examples include:

- official market close;
- option exercise;
- margin call;
- covenant breach;
- default threshold;
- impairment test;

- settlement;
- collateral revaluation;
- regulatory capital rule;
- accounting recognition test.

Represent a gate schematically as:

$$G_P(\Delta R, X, L) \in \{\text{Admit}, \text{Defer}, \text{Reject}\}. \quad (21.7)$$

Here:

$$P = \text{declared protocol}; \quad (21.8)$$

$$X = \text{current financial state}; \quad (21.9)$$

$$L = \text{existing ledger}. \quad (21.10)$$

The gate does not calculate Q.

It determines what becomes committed.

21.5 Ledger records trace

If the gate admits the event:

$$G_P(\Delta R, X, L) = \text{Admit}, \quad (21.11)$$

the ledger updates:

$$L_k \to L_{k+1}. \quad (21.12)$$

The new entry may represent:

- realized P&L;
- recognized impairment;
- collateral posting;
- exercise payoff;
- margin requirement;
- regulatory capital consumption;
- contractual settlement.

The event now becomes historical trace.

21.6 Deferred consequence

If the gate defers recognition:

G_P(ΔR,X,L) = Defer, (21.13)

the economic pressure may remain active without becoming a committed ledger entry.

Examples include:

- unrealized mark-to-market movement;
- unrecognized expected loss;
- contingent liability;
- unexercised protection;
- incomplete trade confirmation;
- pending regulatory decision.

This remainder is not necessarily zero.

It becomes part of the residual state.

21.7 Residual after commitment

Even when a gate admits part of the movement, not all pressure may disappear.

Let the total economic change be:

ΔR_total. (21.14)

Let the admitted ledger consequence be:

ΔR_ledger. (21.15)

Define the residual:

ε_gate = ΔR_total − ΔR_ledger. (21.16)

The residual may represent:

- unrecognized economic damage;
- remaining tail exposure;
- pending cash flow;
- unresolved legal risk;
- liquidity pressure;
- model disagreement;
- incomplete settlement.

Thus:

Commitment ≠ Exhaustion. (21.17)

A gate may create trace while leaving residual pressure.

21.8 Why multiplication by i is not the gate

Multiplication by i produces:

$iZ = -Q + iR.$ (21.18)

This changes the coordinate orientation or measurement type.

It does not decide whether a financial consequence is recognized.

Therefore:

i = Measurement Quarter-Turn. (21.19)

Gate = Commitment Rule. (21.20)

Ledger = Historical Record. (21.21)

These are different structures.

The earlier phrase “risk rotates into realized loss” is acceptable only as compressed intuition.

The rigorous process is:

Risk Exposure
→ Actual Phase Movement
→ Economic Value Change
→ Recognition Gate
→ Ledgered Loss or Gain. (21.22)

21.9 Active and passive rotation

This distinction also clarifies two forms of rotation.

Passive rotation

The same state Z is read under another measurement orientation.

No economic P&L is generated.

Active rotation

The state itself moves from θ to $\theta + \Delta\theta$.

R changes.

Economic P&L is generated.

The same rotation mathematics can describe both.

The financial interpretation depends on whether the state or the measurement basis is changing.

21.10 Backreaction

A ledgered event may change future valuation conditions.

For example:

- a margin call triggers asset sales;
- a downgrade raises funding costs;
- an impairment reduces distributable capital;
- an exercise event changes hedge demand;
- a covenant breach changes control rights;
- a reported loss changes investor behaviour.

Represent the backreaction map as:

$X_{k+1} = \mathscr{B}(X_k, L_{k+1})$. (21.23)

The ledger does not merely describe the financial world.

It may alter what happens next.

21.11 Complete runtime

The complete local runtime may therefore be written:

X_k
→ Declare P_k
→ Construct Z_k
→ Measure $(R_k, -Q_k)$
→ Move by $\Delta\theta_k$ and ΔA_k
→ Generate ΔR_k
→ Apply $Gate_k$
→ Update L_{k+1}
→ Retain ε_k
→ Backreact into X_{k+1}. (21.24)

This is the point at which a valuation model begins to become world-like.

It no longer reports only a number.

It organizes possible transitions, commitments, traces, and consequences.

---

22. Numerical Verification

22.1 Declared parameters

Consider a single cash flow with:

$CF_t = 100.00$. (22.1)

t = 5. (22.2)

r_base = 0.03. (22.3)

β = 1.20. (22.4)

ERP = 0.05. (22.5)

The CAPM required return is:

r = r_base + βERP. (22.6)

Therefore:

r = 0.03 + 1.20(0.05). (22.7)

Hence:

r = 0.09. (22.8)

22.2 Baseline amplitude

The baseline-discounted amplitude is:

A = 100/(1.03)^5. (22.9)

Therefore:

A ≈ 86.2609. (22.10)

22.3 CAPM-admitted value

The CAPM present value is:

R = 100/(1.09)^5. (22.11)

Therefore:

R ≈ 64.9931. (22.12)

22.4 Valuation ratio and phase

The admitted-value ratio is:

c = R/A. (22.13)

Therefore:

c ≈ 64.9931/86.2609. (22.14)

Hence:

c ≈ 0.7534. (22.15)

The phase is:

θ = arccos(c). (22.16)

Therefore:

θ ≈ 0.7170 radians. (22.17)

Equivalently:

θ ≈ 41.08°. (22.18)

22.5 Orthogonal coordinate

The orthogonal coordinate is:

Q = √(A² − R²). (22.19)

Therefore:

Q ≈ √[(86.2609)² − (64.9931)²]. (22.20)

Hence:

Q ≈ 56.7158. (22.21)

The completed state is:

Z ≈ 64.9931 + i56.7158. (22.22)

22.6 Scalar haircut

The scalar CAPM haircut is:

H = A − R. (22.23)

Therefore:

H ≈ 86.2609 − 64.9931. (22.24)

Hence:

H ≈ 21.2678. (22.25)

Observe:

Q ≈ 56.7158. (22.26)

H ≈ 21.2678. (22.27)

Therefore:

Q ≠ H. (22.28)

The exact relationship is:

$$H = Q^2/(A + R). \quad (22.29)$$

Substituting:

$$H \approx (56.7158)^2/(86.2609 + 64.9931). \quad (22.30)$$

Hence:

$$H \approx 21.2678. \quad (22.31)$$

22.7 Required-return sensitivity

The ordinary sensitivity is:

$$\partial R/\partial r = -tR/(1 + r). \quad (22.32)$$

Substitute:

$$\partial R/\partial r = -5(64.9931)/1.09. \quad (22.33)$$

Therefore:

$$\partial R/\partial r \approx -298.1335. \quad (22.34)$$

A one-percentage-point increase in required return corresponds locally to:

$$\Delta r = 0.01. \quad (22.35)$$

Therefore:

$$\Delta R \approx -298.1335(0.01). \quad (22.36)$$

Hence:

$$\Delta R \approx -2.9813. \quad (22.37)$$

22.8 Phase sensitivity to required return

The phase sensitivity is:

$$\partial\theta/\partial r = tR/[(1 + r)Q]. \quad (22.38)$$

Substitute:

$$\partial\theta/\partial r = 5(64.9931)/[1.09(56.7158)]. \quad (22.39)$$

Therefore:

$$\partial\theta/\partial r \approx 5.2568. \quad (22.40)$$

For:

$$\Delta r = 0.01, \quad (22.41)$$

the corresponding local phase movement is:

$\Delta\theta \approx 5.2568(0.01)$. (22.42)

Hence:

$\Delta\theta \approx 0.052568$ radians. (22.43)

22.9 Phase P&L

The phase representation gives:

$\Delta R \approx -Q\Delta\theta$. (22.44)

Substitute:

$\Delta R \approx -56.7158(0.052568)$. (22.45)

Therefore:

$\Delta R \approx -2.9813$. (22.46)

This matches the ordinary required-return calculation.

Thus:

$-[tR/(1 + r)]\Delta r = -Q\Delta\theta$. (22.47)

The two risk coordinates produce the same first-order P&L.

22.10 The measurement cycle

The four principal readouts are:

$M_0(Z) = R \approx 64.9931$. (22.48)

$M_{\pi/2}(Z) = -Q \approx -56.7158$. (22.49)

$M_{\pi}(Z) = -R \approx -64.9931$. (22.50)

$M_{3\pi/2}(Z) = Q \approx 56.7158$. (22.51)

$M_{2\pi}(Z) = R \approx 64.9931$. (22.52)

Therefore:

64.9931
→ −56.7158
→ −64.9931
→ 56.7158
→ 64.9931. (22.53)

These are measurement readouts.

They are not a chronological price history.

22.11 Long–short comparison

For the long position:

Long Mark ≈ 64.9931. (22.54)

Long Phase Delta ≈ −56.7158. (22.55)

For the short position:

Short Mark ≈ −64.9931. (22.56)

Short Phase Delta ≈ 56.7158. (22.57)

The mark and exposure both clear:

64.9931 + (−64.9931) = 0. (22.58)

−56.7158 + 56.7158 = 0. (22.59)

This is the cleanest financial realization of the four-step cycle.

23. Multi-Period Cash Flows

23.1 Each cash flow has its own valuation phase

A realistic asset normally contains more than one future cash flow.

For each horizon t, define:

$A_t = CF_t/(1 + r_{base,t})^t$. (23.1)

$R_t = CF_t/(1 + r_t)^t$. (23.2)

$\cos \theta_t = R_t/A_t$. (23.3)

$Q_t = A_t \sin \theta_t$. (23.4)

The completed cash-flow state is:

$Z_t = R_t + iQ_t$. (23.5)

Equivalently:

$Z_t = A_t \exp(i\theta_t)$. (23.6)

The phase θ_t may differ across horizons because the effects of discounting, beta, term premia, funding conditions, and risk exposure accumulate differently through time.

Therefore a multi-period asset does not generally possess one primitive phase shared by all cash flows.

Its first representation is a collection:

$\{Z_1,Z_2,\ldots,Z_T\}$. (23.7)

23.2 Complex aggregation

Define the aggregate complex valuation state:

$$Z = \Sigma_t Z_t. \quad (23.8)$$

Therefore:

$$Z = \Sigma_t A_t \exp(i\theta_t). \quad (23.9)$$

Its real coordinate is:

$$R = \mathrm{Re}(Z). \quad (23.10)$$

Hence:

$$R = \Sigma_t A_t \cos \theta_t. \quad (23.11)$$

Its imaginary coordinate is:

$$Q = \mathrm{Im}(Z). \quad (23.12)$$

Hence:

$$Q = \Sigma_t A_t \sin \theta_t. \quad (23.13)$$

The aggregate state is:

$$Z = R + iQ. \quad (23.14)$$

This preserves the ordinary additive structure of discounted cash flows on the real axis:

$$R = \Sigma_t R_t. \quad (23.15)$$

At the same time, it aggregates the phase-exposure coordinates:

$$Q = \Sigma_t Q_t. \quad (23.16)$$

The last equality assumes that all Q_t terms use the same orientation convention.

23.3 Aggregate magnitude is not the sum of individual amplitudes

The magnitude of the aggregate complex state is:

$$A_Z = |Z|. \quad (23.17)$$

Therefore:

$$A_Z^2 = R^2 + Q^2. \quad (23.18)$$

But in general:

$$A_Z \neq \Sigma_t A_t. \quad (23.19)$$

The triangle inequality gives:

$$|Z| \le \Sigma_t |Z_t|. \quad (23.20)$$

Therefore:

$$A_Z \le \Sigma_t A_t. \quad (23.21)$$

Equality holds only when all cash-flow vectors point in the same phase direction:

$$\theta_1 = \theta_2 = \cdots = \theta_T. \quad (23.22)$$

If phases differ, the complex cash-flow vectors partially offset one another.

This is ordinary vector addition.

It does not require physical wave interference.

Nevertheless, it reveals a structure that scalar aggregation does not display:

> Cash flows may add fully in monetary amount while failing to align fully in valuation orientation.

23.4 Aggregate Q remains a conjugate exposure under a common phase rotation

Suppose the entire multi-period state experiences a common measurement or valuation rotation φ:

$$Z(\varphi) = \exp(i\varphi)Z. \quad (23.23)$$

Then:

$$Z(\varphi) = \exp(i\varphi)(R + iQ). \quad (23.24)$$

The real coordinate becomes:

$$R(\varphi) = R\cos\varphi - Q\sin\varphi. \quad (23.25)$$

Differentiate:

$$dR(\varphi)/d\varphi = -R\sin\varphi - Q\cos\varphi. \quad (23.26)$$

At:

$$\varphi = 0, \quad (23.27)$$

we obtain:

$$dR(\varphi)/d\varphi|_{(\varphi=0)} = -Q. \quad (23.28)$$

Thus:

Theorem 5 — Common-Phase Aggregation Theorem

For any aggregate complex valuation state:

$$Z = \Sigma_t A_t \exp(i\theta_t) = R + iQ, \quad (23.29)$$

the aggregate imaginary coordinate Q is the magnitude of the first-order real-value exposure to a common rotation of the entire state:

$$d\mathrm{Re}[\exp(i\varphi)Z]/d\varphi|_{(\varphi=0)} = -Q. \quad (23.30)$$

The conjugate risk theorem therefore survives aggregation under a common phase shock.

23.5 Second-order closure also survives aggregation

Differentiate again:

$$d^2R(\varphi)/d\varphi^2 = -R\cos\varphi + Q\sin\varphi. \quad (23.31)$$

At:

$$\varphi = 0, \quad (23.32)$$

we obtain:

$$d^2R(\varphi)/d\varphi^2|_{(\varphi=0)} = -R. \quad (23.33)$$

The aggregate state therefore preserves the same hierarchy:

$$R \to -Q \to -R \to Q \to R. \quad (23.34)$$

The closure does not require every θ_t to be equal.

It requires only that the applied rotation φ be common to the aggregate state.

23.6 Term-specific phase shocks

A common phase rotation is not the only possible risk movement.

Each horizon may experience its own phase change:

$$\theta_t \to \theta_t + d\theta_t. \quad (23.35)$$

The real value is:

$$R = \Sigma_t A_t \cos\theta_t. \quad (23.36)$$

Holding each A_t fixed:

$$dR = -\Sigma_t A_t \sin\theta_t\, d\theta_t. \quad (23.37)$$

Since:

$$Q_t = A_t \sin\theta_t, \quad (23.38)$$

we obtain:

$$dR = -\Sigma_t Q_t\, d\theta_t. \quad (23.39)$$

Define the phase-exposure vector:

$$\mathbf{Q} = (Q_1, Q_2, \ldots, Q_T). \quad (23.40)$$

Define the phase-movement vector:

$d\boldsymbol{\theta} = (d\theta_1, d\theta_2, \ldots, d\theta_T)$. (23.41)

Then:

$dR = -\mathbf{Q} \cdot d\boldsymbol{\theta}$. (23.42)

This is the multi-period generalization of:

$dR = -Q d\theta$. (23.43)

The scalar Q is sufficient for a common phase shock.

The vector **Q** is required for horizon-specific movements.

23.7 Phase-exposure term structure

The sequence:

$Q_1, Q_2, \ldots, Q_T$ (23.44)

forms a phase-exposure term structure.

It shows where along the cash-flow horizon the asset carries the greatest sensitivity to valuation-phase movement.

Possible summary measures include the total oriented exposure:

$Q_{total} = \Sigma_t Q_t$. (23.45)

And the Q-weighted horizon:

$T_Q = [\Sigma_t tQ_t]/[\Sigma_t Q_t]$. (23.46)

The quantity T_Q is not ordinary duration.

It is the average horizon of the phase-exposure coordinates under the declared construction.

It may be useful for comparing assets whose ordinary present values are similar but whose retained phase exposures are concentrated at different maturities.

23.8 A common rate shock is not generally a common phase shock

Suppose all cash flows are discounted using one required return r.

For each horizon:

$\partial\theta_t/\partial r = tR_t/[(1 + r)Q_t]$. (23.47)

Because t, R_t, and Q_t differ across horizons:

$\partial\theta_t/\partial r$ (23.48)

generally differs across t.

Therefore a common rate shock dr produces:

$d\theta_t = [tR_t/((1 + r)Q_t)]dr.$ (23.49)

The resulting P&L is:

$dR = -\Sigma_t Q_t d\theta_t.$ (23.50)

Substitute $d\theta_t$:

$dR = -\Sigma_t [tR_t/(1 + r)]dr.$ (23.51)

This is exactly the ordinary multi-period required-return sensitivity.

But it also demonstrates:

Common Rate Shock ≠ Common Phase Rotation (23.52)

in general.

A common rate shock creates a maturity-dependent phase movement.

A common phase shock is a different declared scenario.

23.9 Parallel phase shocks and parallel rate shocks

The two scenarios should be distinguished.

Parallel rate shock

$dr_t = dr$ (23.53)

for all horizons.

This produces different $d\theta_t$ values.

Parallel phase shock

$d\theta_t = d\varphi$ (23.54)

for all horizons.

This produces:

$dR = -Q_{total} d\varphi.$ (23.55)

Neither scenario is more fundamental by definition.

They represent different stress protocols.

A financial model must declare which movement is being tested.

23.10 Multi-period radial and angular decomposition

Allow both A_t and θ_t to vary.

For each cash flow:

$$dR_t = (R_t/A_t)dA_t - Q_t\, d\theta_t. \quad (23.56)$$

Summing:

$$dR = \Sigma_t\, (R_t/A_t)dA_t - \Sigma_t\, Q_t\, d\theta_t. \quad (23.57)$$

Define:

$$g_{A,t} = dA_t/A_t. \quad (23.58)$$

Then:

$$dR = \Sigma_t\, R_t g_{A,t} - \Sigma_t\, Q_t\, d\theta_t. \quad (23.59)$$

This separates:

- cash-flow or baseline-amplitude change;
- valuation-phase change;
- maturity location.

A residual term may be added:

$$dR = \Sigma_t\, R_t g_{A,t} - \Sigma_t\, Q_t\, d\theta_t + \varepsilon_R. \quad (23.60)$$

The residual captures movement not explained by the declared multi-period geometry.

23.11 Portfolio aggregation

The same method extends across assets.

Let asset i have:

$$Z_i = R_i + iQ_i. \quad (23.61)$$

Let w_i be its signed portfolio weight.

The portfolio state is:

$$Z_P = \Sigma_i\, w_i Z_i. \quad (23.62)$$

Therefore:

$$R_P = \Sigma_i\, w_i R_i. \quad (23.63)$$

$$Q_P = \Sigma_i\, w_i Q_i. \quad (23.64)$$

Under a common portfolio phase rotation:

$$dR_P = -Q_P d\varphi. \quad (23.65)$$

Under asset-specific phase movements:

$$dR_P = -\Sigma_i w_i Q_i d\theta_i. \quad (23.66)$$

Thus Q can aggregate like a signed exposure when the protocol and orientation are common.

If different assets use incompatible baselines or metrics, their Q values cannot be added without first reconciling those protocols.

23.12 Classical cancellation and diversification

Suppose two positions have:

$$Q_1 > 0, \quad (23.67)$$

and:

$$w_2 Q_2 < 0. \quad (23.68)$$

Then their aggregate phase exposure may partially cancel:

$$Q_P = w_1 Q_1 + w_2 Q_2. \quad (23.69)$$

This is a classical diversification or hedging relation.

It does not imply quantum interference.

The complex notation simply makes directional exposure cancellation explicit.

---

24. Empirical Discipline and Falsification

24.1 The mathematical identity is not yet an empirical discovery

The relation:

$$Q = \sqrt{(A^2 - R^2)} \quad (24.1)$$

is true by construction.

Likewise:

$$\partial R/\partial \theta = -Q \quad (24.2)$$

follows from:

$$R = A \cos \theta. \quad (24.3)$$

An empirical study cannot claim success merely by confirming identities that were used to define the variables.

The framework becomes empirically meaningful only when it is tested against observations not mechanically guaranteed by the construction.

24.2 The tautology problem

Suppose θ is reconstructed after observing R:

$\theta = \arccos(R/A)$. (24.4)

Then Q is calculated:

$Q = A \sin \theta$. (24.5)

If one subsequently verifies:

$dR/d\theta = -Q$, (24.6)

using the same reconstructed data, the result may be mathematically correct but empirically tautological.

A meaningful test must use at least one independent element, such as:

- an ex ante phase shock;
- an independently specified baseline;
- an out-of-sample market movement;
- a competing valuation protocol;
- market-implied prices;
- a gate event not used in constructing Q;
- a future residual outcome.

24.3 The static null hypothesis

The strongest static null hypothesis is:

> Q is a deterministic re-expression of A and R and therefore adds no incremental information, predictive power, or decision value. (24.7)

This null hypothesis should be accepted unless evidence shows that the completed geometry improves at least one practical task.

24.4 Diagnostic tests

A first empirical programme may test whether Q improves diagnosis rather than prediction.

For example, compare assets with similar admitted values:

$R_i \approx R_j$. (24.8)

But different completed states:

$A_i \neq A_j$. (24.9)

$$Q_i \neq Q_j. \quad (24.10)$$

The question is whether the Q difference corresponds to meaningful differences in:

- future repricing sensitivity;
- funding pressure;
- liquidity demand;
- drawdown behaviour;
- capital consumption;
- regime dependence;
- model disagreement.

If Q merely restates the difference in A without improving interpretation, its diagnostic value is limited.

24.5 Dynamic attribution test

A local attribution model may be written:

$$\Delta R_{i,t+1} = R_{i,t} g_{A,i,t+1} - Q_{i,t} \Delta\theta_{i,t+1} + \varepsilon_{i,t+1}. \quad (24.11)$$

The research question is not whether the identity can fit historical ΔR after all terms are reconstructed from the same outcome.

The proper question is whether ex ante estimates of:

$$g_A, \quad (24.12)$$

$$Q, \quad (24.13)$$

and:

$$\Delta\theta \quad (24.14)$$

produce useful out-of-sample attribution or forecasting.

24.6 Benchmark against ordinary rate sensitivity

The phase model must be compared with the simpler benchmark:

$$\Delta R \approx -D_r \Delta r. \quad (24.15)$$

Where:

$$D_r = tR/(1 + r). \quad (24.16)$$

The phase form is:

$$\Delta R \approx -Q\Delta\theta. \quad (24.17)$$

Since the two are locally equivalent:

$$Q\Delta\theta = D_r\Delta r, \quad (24.18)$$

the phase representation cannot claim superior one-factor predictive accuracy merely from this identity.

It must demonstrate another advantage, such as:

- more stable cross-asset normalization;
- clearer multi-filter decomposition;
- better common-shock aggregation;
- better residual detection;
- improved communication of nonlinearity.

24.7 Comparison with duration and convexity

The relevant conventional benchmarks include:

$$\text{Dollar Duration} = -\partial R/\partial r. \quad (24.19)$$

$$\text{Rate Convexity} = \partial^2 R/\partial r^2. \quad (24.20)$$

The phase measures are:

$$\text{Phase Delta} = \partial R/\partial\theta = -Q. \quad (24.21)$$

$$\text{Second Phase Derivative} = \partial^2 R/\partial\theta^2 = -R. \quad (24.22)$$

Because higher derivatives change under coordinate transformation, phase curvature must not be presented as a replacement for ordinary convexity without empirical justification.

A useful test is whether phase coordinates make nonlinear scenario behaviour more regular or more comparable across claims.

24.8 Comparison with option Greeks

For derivative claims, the phase exposure should be compared with:

- delta;
- gamma;
- vega;
- theta decay;
- rho;
- cross-Greeks.

A general derivative value may be:

$V = V(S,\sigma,r,t,K,\ldots)$. (24.23)

The phase coordinate may be added as a composite function:

$\theta = \theta(S,\sigma,r,t,K,\ldots)$. (24.24)

Then:

$\partial V/\partial\theta$ (24.25)

must be compared with the existing chain of Greeks.

If Phase Delta is only a nonlinear combination of established Greeks with no added stability or interpretive benefit, it should not be treated as an independent risk dimension.

24.9 Model-implied and market-implied Q

Suppose A is declared independently and the observed market price is P.

If:

$0 \le P \le A$, (24.26)

define the market-implied phase:

$\theta_market = \arccos(P/A)$. (24.27)

And:

$Q_market = \sqrt{(A^2 - P^2)}$. (24.28)

The CAPM model gives:

$Q_CAPM = \sqrt{(A^2 - R_CAPM^2)}$. (24.29)

Define the residual pressure:

$\Delta Q = Q_market - Q_CAPM$. (24.30)

This residual may reflect:

- non-CAPM factor pricing;
- liquidity;
- credit;
- sentiment;
- option value;
- model error;
- baseline misspecification.

But it should not automatically be labelled alpha or mispricing.

24.10 Domain failure as information

If:

$P > A$, (24.31)

then:

$A^2 - P^2 < 0$. (24.32)

The ordinary first-quadrant Euclidean Q is no longer real.

This should not be hidden by forcing a value into the formula.

Possible interpretations include:

1. the baseline A is unsuitable;
2. the market prices growth or optionality absent from the declared cash flow;
3. the positive-claim geometry is too small;
4. another metric is required;
5. the market state lies outside the model's admissible domain.

Model-domain failure can itself be diagnostically useful.

24.11 Cross-sectional testing

A cross-sectional study may test whether Q explains future outcomes after controlling for established measures.

A schematic regression is:

$Y_{i,t+1} = \alpha + b_Q Q_{i,t} + b_R R_{i,t} + b_\beta \beta_{i,t} + b_D D_{i,t} + b_\sigma \sigma_{i,t} + \text{controls} + \varepsilon_{i,t+1}$. (24.33)

Possible dependent variables Y include:

- future drawdown;
- spread widening;
- liquidity deterioration;
- forecast error;
- capital stress;
- gate-event probability;
- residual volatility.

The coefficient b_Q has meaning only if Q is not included alongside mechanically equivalent variables in a way that creates unstable multicollinearity.

24.12 Time-series testing

A time-series model may examine whether changes in Q precede or accompany regime transitions.

For example:

GateEvent_(t+1) = F(Q_t,dQ_t/dt,Δθ_t,ε_t,controls). (24.34)

The hypothesis is not that high Q must cause a gate event.

The weaker test is whether Q improves classification after controlling for conventional risk measures.

24.13 Protocol-robustness test

Because Q depends on the declared baseline, empirical work should repeat the construction under multiple admissible baselines:

A^(1),A^(2),...,A^(m). (24.35)

This produces:

Q^(1),Q^(2),...,Q^(m). (24.36)

A useful result should not depend entirely on one arbitrary baseline choice.

Researchers should report:

- baseline sensitivity;
- sign stability;
- rank stability;
- predictive stability;
- residual stability.

24.14 Out-of-sample discipline

The valuation protocol must be fixed before the evaluation period.

The following should be declared in advance:

- cash-flow model;
- baseline rate;
- CAPM inputs;
- horizon;
- metric;

- orientation;
- gate rule;
- benchmark models;
- evaluation criterion.

Otherwise, the complex geometry may be tuned after the fact to fit the observed path.

24.15 Conditions for rejection

The framework should be reduced or rejected if:

1. Q adds no diagnostic benefit beyond A and R;
2. phase attribution is less stable than ordinary rate attribution;
3. results depend excessively on arbitrary baseline selection;
4. multi-period aggregation creates misleading cancellation;
5. market-implied Q is dominated by model error;
6. no measurable intervention or decision improves;
7. the complex notation obscures rather than clarifies established finance.

The theory earns its place only through explanatory or operational gain.

24.16 Part I conclusion

Part I has established a complete finance-first result.

The CAPM complex completion begins from:

$A = CF_t/(1 + r_{base})^t$. (24.37)

$R = CF_t/(1 + r_{CAPM})^t$. (24.38)

The valuation phase is:

$\theta = \arccos(R/A)$. (24.39)

The orthogonal coordinate is:

$Q = \sqrt{(A^2 - R^2)}$. (24.40)

The principal identity is:

$\partial R/\partial\theta = -Q$. (24.41)

The quarter-turn operator satisfies:

$\mathcal{J}^2 = -I$. (24.42)

The measurement cycle is:

$$R \to -Q \to -R \to Q \to R. \quad (24.43)$$

The exact local P&L equivalence is:

$$-Q d\theta = -[tR/(1 + r)]dr. \quad (24.44)$$

And the scalar haircut satisfies:

$$A - R = \int_0^{\theta} Q(\varphi)d\varphi. \quad (24.45)$$

The finance-first conclusion is therefore:

> Q is the conjugate coordinate and first-order phase exposure generated by the same declared valuation geometry that produces R.

Nothing in this conclusion requires quantum mechanics.

Part II now asks what broader conceptual structure becomes available once mark and conjugate exposure are treated as one completed financial state.

Part II — From Conjugate Risk to Observer-Bound Valuation Worlds

25. Three Operations That Must Never Be Confused

25.1 Why the distinction is necessary

The same rotation equations can describe several different operations.

Without a careful distinction, one may incorrectly conclude that:

- changing measurement equipment changes the asset;
- reading $-Q$ realizes a loss;
- applying i is the same as crossing a settlement gate;
- a half-turn is a chronological transition from a long position to a short position.

To prevent these errors, Part II distinguishes:

1. state evolution;
2. measurement rotation;
3. commitment.

25.2 State evolution

The financial state evolves when its economic or valuation inputs change.

The original state is:

$Z = A \exp(i\theta)$. (25.1)

A changed state is:

$Z' = (A + \Delta A)\exp[i(\theta + \Delta\theta)]$. (25.2)

This may result from:

- revised cash-flow expectations;
- beta movement;
- ERP movement;
- funding changes;
- liquidity changes;
- contractual events;
- market repricing.

State evolution can generate economic P&L.

25.3 Measurement rotation

Measurement rotation changes which component of the same state is read.

For fixed Z:

$M_{\varphi}(Z) = \mathrm{Re}[\exp(i\varphi)Z]$. (25.3)

Changing φ changes the readout:

$M_0(Z) = R$. (25.4)

$M_{\pi/2}(Z) = -Q$. (25.5)

But Z itself need not change.

Measurement rotation answers a different question about the same state.

25.4 Commitment

Commitment occurs when a financial consequence passes a gate and becomes ledgered.

Represent the gate as:

$G_P(\Delta R, X, L)$. (25.6)

If the event is admitted:

$L_{(k+1)} = L_k + \mathrm{Trace}(\Delta R)$. (25.7)

Commitment may alter future states through backreaction.

Neither state evolution nor measurement rotation alone guarantees ledger commitment.

25.5 The three operations in one table

Operation	What changes?	Can it create P&L?	Can it create ledger trace?
State evolution	Z	Yes	Not automatically
Measurement rotation	readout basis	No	No
Commitment gate	ledger status	Recognizes existing consequence	Yes

The distinction is foundational for the rest of the article.

25.6 Active rotation

An active rotation changes the state:

$$Z \rightarrow \exp(i\Delta\theta)Z. \quad (25.8)$$

The measurement axis remains fixed.

The real coordinate changes:

$$R \rightarrow R\cos\Delta\theta - Q\sin\Delta\theta. \quad (25.9)$$

This can represent actual valuation movement.

25.7 Passive rotation

A passive rotation leaves Z fixed but changes the measurement orientation:

$$M_0(Z) \rightarrow M_{\varphi}(Z). \quad (25.10)$$

No economic P&L is produced merely by changing the question.

The observer has changed frame.

The asset has not necessarily changed state.

25.8 Commitment after active movement

The rigorous sequence is:

Z
→ Active Movement
→ ΔR
→ Gate

→ Ledger Trace. (25.11)

The passive measurement may identify the relevant exposure before the movement:

Z
→ M_π/2(Z) = −Q
→ Exposure Report. (25.12)

These two sequences must not be conflated.

---

26. Why a Complex State Is More Than Two Spreadsheet Columns

26.1 An ordered pair is not yet a complex structure

A spreadsheet may contain:

Column 1 = R. (26.1)

Column 2 = Q. (26.2)

This creates an ordered pair:

(R,Q). (26.3)

But two columns alone do not specify how one coordinate transforms into the other.

They do not imply:

$\mathcal{J}$R = −Q. (26.4)

They do not imply:

$\mathcal{J}$Q = R. (26.5)

They do not imply:

$\mathcal{J}^2 = -I$. (26.6)

A complex structure adds these transformation rules.

26.2 The role of $\mathcal{J}$

The operator:

$\mathcal{J} = [0\ -1;\ 1\ 0]$ (26.7)

specifies:

- orientation;
- orthogonality;
- quarter-turn transformation;

- periodic closure.

Therefore:

Complex State = Ordered Pair + Compatible Complex Structure. (26.8)

The symbol i is a compact representation of $\mathcal{J}$.

26.3 Compatibility with the norm

The Euclidean norm is:

$A^2 = R^2 + Q^2$. (26.9)

The operator preserves that norm:

$\|\mathcal{J}x\| = \|x\|$. (26.10)

Therefore measurement rotation changes orientation without changing total magnitude.

This distinguishes rotation from haircut subtraction.

The transformation:

$R \to R - Q$ (26.11)

does not preserve the same geometry.

26.4 Conjugacy

R and Q are not merely correlated variables.

They satisfy:

$dR/d\theta = -Q$. (26.12)

$dQ/d\theta = R$. (26.13)

Each coordinate governs the angular movement of the other.

This reciprocal relation is the sense in which they are conjugate within the declared valuation plane.

26.5 Closure

The structure closes:

$\mathcal{J}^4 = I$. (26.14)

Therefore every measurement orientation belongs to a coherent rotational family.

Without $\mathcal{J}$, the four entries:

$R, -Q, -R, Q$ (26.15)

could be an arbitrary list.

With $\mathcal{J}$, they are ordered consequences of one transformation law.

26.6 Why the imaginary unit matters

The value of i is not that finance requires fictional numbers.

Its value is that it packages:

- two coordinates;
- one orientation;
- one norm;
- one quarter-turn operator;
- one closed measurement cycle.

Thus:

Z = R + iQ (26.16)

is more than compressed notation when the operator structure is actively used.

If the structure is never used, ordinary vector notation may be sufficient.

---

27. The Deeper Financial Meaning of $i^2 = -1$

27.1 The elementary identity

The imaginary unit satisfies:

$i^2 = -1$. (27.1)

In the completed valuation plane:

$iZ = -Q + iR$. (27.2)

Applying i again:

$i^2Z = -R - iQ$. (27.3)

Therefore:

$i^2Z = -Z$. (27.4)

27.2 First quarter-turn

The first quarter-turn gives:

$R \to -Q$. (27.5)

Financially:

Mark Measurement → Conjugate Exposure Measurement. (27.6)

The real-axis readout changes type.

It moves from current admitted value to the signed sensitivity of that value to positive phase movement.

27.3 Second quarter-turn

The second quarter-turn gives:

$-Q \to -R$. (27.7)

Financially:

Conjugate Exposure → Opposite Signed Mark. (27.8)

In a linear position space, this may be interpreted as a change from long orientation to short orientation.

But the operation itself is first a measurement-orientation reversal.

27.4 Why the negative sign is structural

The negative sign in:

$i^2 = -1$ (27.9)

does not mean destruction.

It indicates opposite orientation on the same axis.

Examples of signed financial reversals include:

- long versus short;
- asset versus opposite claim;
- receive-fixed versus pay-fixed;
- positive versus negative notional.

The sign is relational.

Its economic meaning depends on the declared position space.

27.5 Third and fourth turns

The third turn gives:

$-R \to Q$. (27.10)

This is:

Opposite Mark → Opposite Phase Exposure. (27.11)

The fourth gives:

Q → R. (27.12)

This returns to the original mark.

Therefore:

$i^2 = -1$ (27.13)

belongs to a complete cycle:

1,i,−1,−i,1. (27.14)

27.6 Not a double-loss law

A mistaken interpretation would be:

First i = hidden risk becomes loss. (27.15)

Second i = loss occurs again and becomes −R. (27.16)

That is not the mathematical structure.

The correct interpretation is:

First i = measurement quarter-turn. (27.17)

Second i = orientation reversal. (27.18)

Loss requires actual state movement:

$\Delta R \neq 0$. (27.19)

And recognition requires a gate.

27.7 The strongest finance-first statement

The most disciplined interpretation is:

> In the CAPM complex valuation plane, i represents a quarter-turn between mark and conjugate phase exposure, while i^2 represents reversal of the signed valuation orientation.

This statement is mathematically exact within the declared geometry.

It does not depend on a physical analogy.

---

28. Quadrature Measurement without Physics

28.1 In-phase and quadrature channels

Engineering often represents a signal using two orthogonal components:

- an in-phase component;
- a quadrature component.

The two components are separated by 90°.

A generic representation is:

Z = X + iY. (28.1)

Here X and Y are both real measurable quantities.

The imaginary unit records their phase relation.

The CAPM valuation plane has an analogous form:

Z = R + iQ. (28.2)

R is the admitted-value channel.

Q is the conjugate phase-exposure channel.

28.2 Phase-sensitive measurement

A phase-sensitive detector may read:

M_φ(Z) = R cos φ − Q sin φ. (28.3)

Changing φ changes which mixture of the two channels becomes visible.

This supplies a useful measurement-equipment analogy.

A financial risk system may report:

- mark;
- rotated stress readout;
- opposite-position mark;
- opposite exposure.

The state need not physically rotate for the reporting basis to change.

28.3 Analytic-signal analogy

In signal analysis, a real signal s(u) may be completed into an analytic signal:

z(u) = s(u) + i$\mathscr{H}$s. (28.4)

Where $\mathscr{H}$ is the Hilbert transform.

Under suitable conditions:

$\mathscr{H}^2[s] = -s.$ (28.5)

This produces a quadrature relation similar to:

$\mathcal{J}^2 = -I.$ (28.6)

The analogy is mathematically informative.

Both structures contain:

- a real channel;
- a quadrature channel;
- a quarter-turn operator;
- a sign reversal after two applications.

28.4 CAPM Q is not a Hilbert transform of price

The constructions are not identical.

In analytic-signal theory:

$\mathscr{H}[s]$ (28.7)

is derived from a signal across its domain, often calendar time or frequency.

In CAPM geometry:

$Q = \sqrt{(A^2 - R^2)}$ (28.8)

is derived from a valuation ratio and a declared norm.

Therefore:

$Q \neq \mathscr{H}[R]$ (28.9)

as a general financial identity.

The shared feature is the complex-structure algebra, not the method of construction.

28.5 Shared mathematics without shared ontology

The comparison should be stated as:

CAPM Phase Geometry
↔ Quadrature Measurement Grammar. (28.10)

It should not be stated as:

Asset Price = Physical Wave Signal. (28.11)

The analogy is useful because it demonstrates that complex numbers can encode two real,

operationally meaningful measurement channels without requiring either channel to be fictional.

28.6 What the analogy contributes

The quadrature analogy clarifies four points.

First, Q need not be a second market price to be measurable.

Second, the sign change under quarter-turn rotation is normal complex algebra.

Third, changing the measurement phase need not change the underlying state.

Fourth, a complete complex representation can support measurements unavailable from the real channel alone.

28.7 What the analogy does not prove

It does not prove that:

- financial prices are waves;
- market risk obeys linear signal dynamics;
- a Hilbert transform will recover CAPM Q;
- phase rotation creates causal market movement;
- finance is quantum mechanical.

The analogy provides engineering intuition for the measurement structure established in Part I.

It does not replace financial validation.

29. Relative Valuation Frames

29.1 A value is always measured under a protocol

Part I treated A, R, Q, and θ as coordinates generated by one declared valuation protocol.

That protocol includes at least:

- a cash-flow model;
- a baseline discount rule;
- a risk-adjusted discount rule;
- a horizon;
- a metric;
- an orientation convention.

Different financial observers may adopt different protocols even when they refer to the same underlying claim.

Examples include:

- an equity analyst using CAPM;
- a credit analyst using spread-adjusted discounting;
- a dealer using a funding curve;
- an accountant using fair-value recognition rules;
- a regulator using capital and liquidity constraints;
- a strategic buyer using synergy-adjusted cash flows.

The underlying claim may be the same.

The admitted valuation state may not be.

29.2 Frame-indexed valuation states

Let observer or protocol a assign:

$Z_a = R_a + iQ_a.$ (29.1)

Let observer or protocol b assign:

$Z_b = R_b + iQ_b.$ (29.2)

In general:

$Z_a \neq Z_b.$ (29.3)

The difference may arise because:

$A_a \neq A_b,$ (29.4)

$\theta_a \neq \theta_b,$ (29.5)

or both.

Therefore relative-frame analysis must distinguish two cases.

Case 1 — Same magnitude, different orientation

$A_a = A_b.$ (29.6)

$\theta_a \neq \theta_b.$ (29.7)

Case 2 — Different magnitude and different orientation

$A_a \neq A_b.$ (29.8)

$\theta_a \neq \theta_b.$ (29.9)

The first case can be represented by a pure rotation.

The second requires both radial scaling and rotation.

29.3 Pure relative rotation

Suppose two valuation frames share the same amplitude:

$A_a = A_b = A.$ (29.10)

Let their phase difference be:

$\varphi_{ab} = \theta_b - \theta_a.$ (29.11)

Then:

$Z_b = \exp(i\varphi_{ab})Z_a.$ (29.12)

The coordinate transformation is:

$R_b = R_a \cos \varphi_{ab} - Q_a \sin \varphi_{ab}.$ (29.13)

$Q_b = R_a \sin \varphi_{ab} + Q_a \cos \varphi_{ab}.$ (29.14)

The inverse transformation is:

$Z_a = \exp(-i\varphi_{ab})Z_b.$ (29.15)

Therefore:

$\varphi_{ba} = -\varphi_{ab}.$ (29.16)

The relative phase behaves as an oriented difference between valuation frames.

29.4 General scaling and rotation

If the two frames also assign different amplitudes, define the scale ratio:

$\lambda_{ab} = A_b/A_a.$ (29.17)

Then:

$Z_b = \lambda_{ab} \exp(i\varphi_{ab})Z_a.$ (29.18)

Therefore:

$R_b = \lambda_{ab}(R_a \cos \varphi_{ab} - Q_a \sin \varphi_{ab}).$ (29.19)

$Q_b = \lambda_{ab}(R_a \sin \varphi_{ab} + Q_a \cos \varphi_{ab}).$ (29.20)

This decomposition separates:

- a change in declared economic magnitude;
- a change in valuation orientation.

Without this distinction, a disagreement in value may be incorrectly attributed entirely to risk preference or entirely to cash-flow expectation.

29.5 Benchmark and numeraire dependence

A valuation may change when the benchmark or numeraire changes.

Examples include:

- domestic currency versus foreign currency;
- nominal value versus inflation-linked value;
- risk-free discounting versus funding-adjusted discounting;
- market-value reporting versus regulatory-capital reporting.

Let N_a and N_b be two numeraires.

A value may be represented as:

V_a = V/N_a. (29.21)

V_b = V/N_b. (29.22)

Changing numeraire alters the scalar value before any phase completion is applied.

Therefore numeraire transformation should not automatically be represented as a pure rotation.

It may change both A and R.

The complex formalism is useful only after the numeraire protocol has been declared.

29.6 Funding curves as frame differences

Suppose an uncollateralized observer discounts at r_u, while a collateralized observer discounts at r_c.

Then:

R_u = CF_t/(1 + r_u)^t. (29.23)

R_c = CF_t/(1 + r_c)^t. (29.24)

If both use the same baseline amplitude A:

cos θ_u = R_u/A. (29.25)

cos θ_c = R_c/A. (29.26)

The relative phase is:

φ_uc = θ_c − θ_u. (29.27)

This phase difference compactly represents the directional gap between two discount protocols.

But the economic explanation remains:

- funding basis;
- collateral agreement;
- liquidity;
- counterparty terms.

The angle summarizes the valuation relation.

It does not replace the institutional cause.

29.7 Accounting and regulatory frames

A market valuation and an accounting valuation may read the same state differently.

For example:

R_market = quoted or model-implied fair value. (29.28)

R_accounting = recognized carrying amount. (29.29)

A regulatory frame may instead emphasize:

R_regulatory = value admitted after capital and liquidity constraints. (29.30)

The associated Q-coordinates may differ because the relevant gates differ.

One protocol may admit a market movement immediately.

Another may defer it.

A third may convert it into a capital requirement rather than an accounting loss.

Thus relative valuation frames are not merely different opinions.

They may be different institutional measurement systems.

29.8 Readout transformation versus state reassignment

A crucial distinction is:

Readout transformation

The underlying completed state is held fixed, and only the measurement orientation changes.

State reassignment

A different observer constructs a different A, θ, and Q.

In the first case:

$Z_b = \exp(i\varphi_{ab}) Z_a$. (29.31)

In the second case:

$Z_b = F_{ab}(Z_a, P_a, P_b)$. (29.32)

Where F_{ab} may include:

- rescaling;
- rotation;
- nonlinear filtering;
- gate changes;
- revised cash flows;
- revised horizons.

Not every observer disagreement can be represented by a simple unitary rotation.

29.9 Frame invariants

A useful relative-frame theory should identify quantities that remain stable across admissible transformations.

Under a pure rotation:

$|Z_b| = |Z_a|$. (29.33)

Therefore:

$A_b = A_a$. (29.34)

Under a scale-and-rotation transformation:

$|Z_b| / |Z_a| = \lambda_{ab}$. (29.35)

Possible invariants or covariants include:

- rank ordering of phase;
- sign of exposure;
- normalized ratio Q/A;
- relative phase differences;
- zero-sum long–short structure;
- residual size after frame reconciliation.

These must be tested rather than assumed.

29.10 Frame consistency

For three compatible frames a, b, and c:

$\varphi_{ac} = \varphi_{ab} + \varphi_{bc}$. (29.36)

Therefore:

$\exp(i\varphi_{ac}) = \exp(i\varphi_{bc})\exp(i\varphi_{ab})$. (29.37)

This is the consistency condition for sequential frame translation.

If empirical frame mappings violate this relation materially, possible explanations include:

- nonlinear protocol differences;
- path dependence;
- inconsistent baselines;
- residual transaction costs;
- institutional asymmetry;
- model error.

The failure of simple phase additivity can therefore reveal where pure rotational comparison is insufficient.

29.11 Relative valuation as a reconciliation problem

A practical reconciliation can be organized as:

Z_a
→ Declare protocol difference
→ Separate scale difference λ_{ab}
→ Estimate phase difference φ_{ab}
→ Transform into frame b
→ Compare transformed and observed Z_b
→ Record residual ε_{ab}. (29.38)

Define:

$\varepsilon_{ab} = Z_b - \lambda_{ab} \exp(i\varphi_{ab})Z_a$. (29.39)

A small ε_{ab} suggests that scale and rotation capture most of the frame difference.

A large ε_{ab} indicates additional structure.

This residual may represent:

- omitted factors;
- nonlinearity;
- contractual asymmetry;
- path dependence;

- incompatible gates.

---

30. Derivatives and Composite Financial States

30.1 A single complex valuation state is not yet a composite system

The one-state construction:

Z = R + iQ (30.1)

contains a mark and its conjugate phase exposure.

This is not yet derivative entanglement.

It is a two-coordinate representation of one valuation state.

A derivative relationship requires at least two coupled sectors:

- an underlying sector;
- a contractual derivative sector.

The relevant state space is therefore larger than one complex plane.

30.2 Underlying and derivative states

Let the underlying state be:

$Z_U = R_U + iQ_U$. (30.2)

Let the derivative state be:

$Z_D = R_D + iQ_D$. (30.3)

The joint state is not necessarily the simple sum:

$Z_{joint} \neq Z_U + Z_D$ (30.4)

as a complete description.

The derivative value depends functionally on the underlying and contract:

$R_D = V(R_U,\sigma,r,t,K,\ldots)$. (30.5)

Its phase exposure may depend on both the derivative valuation phase and the underlying state.

Therefore the joint system should be represented schematically as:

$\Psi_{UD} = \Psi(Z_U,Z_D,C)$. (30.6)

Where C denotes the contractual coupling.

This notation does not assert a physical wavefunction.

It indicates that the two sectors cannot always be analysed independently.

30.3 Contractual coupling

A derivative contract links outcomes across sectors.

For a European call:

$$\text{Payoff}_T = \max(S_T - K, 0). \quad (30.7)$$

For a put:

$$\text{Payoff}_T = \max(K - S_T, 0). \quad (30.8)$$

The derivative's future cash flow depends on the underlying state.

Thus:

$$CF_{D,T} = F(S_T, K). \quad (30.9)$$

The coupling is not metaphorical.

It is written into the contract.

The derivative state therefore inherits structural dependence from the underlying.

30.4 Local valuation maps

An observer may measure only the underlying:

$$M_U(\Psi_{UD}). \quad (30.10)$$

Another may measure only the derivative:

$$M_D(\Psi_{UD}). \quad (30.11)$$

A portfolio observer may measure the joint hedge:

$$M_{joint}(\Psi_{UD}). \quad (30.12)$$

These readouts can differ even when they refer to the same composite financial arrangement.

A local observer may not reconstruct the complete joint state from one marginal readout.

That limited access creates an observer-bound structure.

30.5 Conditional update

Suppose the underlying moves:

$$Z_U \rightarrow Z_U'. \quad (30.13)$$

The derivative must be revalued:

$Z_D \to Z_D' = F_D(Z_U', C, P_D)$. (30.14)

Therefore a local event in the underlying sector produces a conditional update in the derivative sector.

The update may involve:

- delta;
- gamma;
- vega;
- funding;
- collateral;
- exercise probability;
- residual model risk.

The derivative state is relational.

Its meaning cannot be specified independently of the underlying and contract.

30.6 Greek decomposition inside the composite state

For a derivative value V:

$dV = \Delta dS + \tfrac{1}{2}\Gamma(dS)^2 + \text{Vega}d\sigma + \text{Rho}dr + \Theta dt + \cdots$. (30.15)

If the underlying itself is represented by:

$dR_U = -Q_U d\theta_U + R_U g_{A,U}$, (30.16)

then:

$dV = -\Delta Q_U d\theta_U + \Delta R_U g_{A,U} + \tfrac{1}{2}\Gamma(dR_U)^2 + \cdots$. (30.17)

This does not eliminate ordinary Greeks.

It allows the underlying movement entering the Greek expansion to be decomposed into radial and angular components.

30.7 Derivative phase exposure

The derivative may possess its own completed state:

$Z_D = A_D \exp(i\theta_D)$. (30.18)

Then:

$\partial R_D / \partial \theta_D = -Q_D$. (30.19)

But θ_D need not equal θ_U.

The derivative phase may depend on:

- moneyness;
- volatility;
- time to maturity;
- skew;
- funding;
- collateral;
- exercise structure.

Therefore:

$$\theta_D = G(\theta_U,\sigma,t,K,C,\ldots). \quad (30.20)$$

By the chain rule:

$$dR_D = -Q_D d\theta_D + (R_D/A_D)dA_D. \quad (30.21)$$

And:

$$d\theta_D = (\partial\theta_D/\partial\theta_U)d\theta_U + (\partial\theta_D/\partial\sigma)d\sigma + \cdots. \quad (30.22)$$

This creates a multi-channel phase sensitivity.

30.8 Hedge coupling

Suppose a dealer holds one derivative and $-\Delta$ units of the underlying.

The hedge portfolio is:

$$\Pi = R_D - \Delta R_U. \quad (30.23)$$

In complex form:

$$Z_\Pi = Z_D - \Delta Z_U. \quad (30.24)$$

Therefore:

$$R_\Pi = R_D - \Delta R_U. \quad (30.25)$$

$$Q_\Pi = Q_D - \Delta Q_U. \quad (30.26)$$

A hedge may reduce ordinary delta while leaving residual phase exposure:

$$Q_\Pi \neq 0. \quad (30.27)$$

This residual may arise because:

- the phase coordinates are different;
- the hedge is only local;

- gamma remains;
- volatility moves;
- funding and collateral differ;
- the contract gate is nonlinear.

Thus a price hedge need not be a complete phase hedge.

30.9 Exercise as a gate

Before exercise, an option contains contingent value.

At the exercise gate:

$$G_{exercise}(S_T,K,C) \in \{Exercise,Expire\}. \quad (30.28)$$

For a call:

$$Exercise \text{ if } S_T > K. \quad (30.29)$$

$$Expire \text{ if } S_T \leq K. \quad (30.30)$$

The gate converts a contingent contractual state into a ledgered payoff or zero payoff.

The imaginary coordinate should not be identified mechanically with option time value.

But both structures involve value that is active before final contractual commitment.

30.10 Settlement and collateral gates

A derivative system may contain several gates:

- variation-margin gate;
- initial-margin gate;
- collateral threshold;
- exercise gate;
- default gate;
- settlement gate.

A price movement may therefore produce multiple traces:

$$\Delta R_{market}, \quad (30.31)$$

$$\Delta Collateral, \quad (30.32)$$

$$\Delta Capital, \quad (30.33)$$

$$\Delta FundingCost. \quad (30.34)$$

The same underlying event is projected into several institutional worlds.

Each world has its own real-axis admission rule.

30.11 Backreaction from derivatives to the underlying

The derivative is not always a passive observer.

Dealer hedging can alter underlying demand.

For example:

$$dPosition_U \approx -\Gamma dS \quad (30.35)$$

under simplified dynamic hedging.

Large option positions may therefore influence:

- underlying order flow;
- volatility;
- liquidity;
- price pressure;
- closing behaviour.

The derivative sector observes the underlying through valuation and simultaneously acts back upon it through hedging.

This is a genuine financial backreaction loop.

30.12 Why conjugate risk is not yet entanglement

The relation:

$$\partial R/\partial \theta = -Q \quad (30.36)$$

belongs to one completed state.

It does not imply two-system nonseparability.

Derivative entanglement-like structure requires additional conditions:

1. a joint state;
2. contractual coupling;
3. local readout restrictions;
4. conditional update;
5. persistent cross-sector dependence;
6. possible backreaction.

Therefore:

Conjugate Measurement ≠ Composite Entanglement. (30.37)

The first is a one-state geometry.

The second is a relational architecture.

30.13 Functional analogy and physical limit

A derivative–underlying system may display:

- relational state dependence;
- observer-dependent local valuation;
- conditional updates;
- joint hedge constraints;
- nonlocal contractual linkage in accounting space.

But it does not thereby demonstrate:

- Bell inequality violation;
- quantum nonlocality;
- physical superposition;
- no-signalling entanglement.

The analogy is structural and operational.

It is not material identity.

---

31. Calendar Time, Valuation Phase, and Ledger Time

31.1 Three different orderings

Finance uses several kinds of temporal ordering.

This article distinguishes:

t = calendar duration. (31.1)

θ = valuation phase. (31.2)

k = ledger-event index. (31.3)

These coordinates answer different questions.

Calendar time asks:

> How much external time has elapsed?

Valuation phase asks:

> How far has the state rotated under the declared valuation filter?

Ledger index asks:

> How many consequential commitments have been recorded?

31.2 Calendar duration

The DCF horizon t enters:

$A_t = CF_t/(1 + r_{base})^t$. (31.4)

$R_t = CF_t/(1 + r)^t$. (31.5)

Calendar duration affects:

- discount accumulation;
- cash-flow timing;
- option decay;
- credit exposure;
- funding requirements.

It is an external ordering variable.

31.3 Valuation phase

The phase is:

$\theta_t = \arccos(R_t/A_t)$. (31.6)

It depends on t, but it is not identical to t.

Two assets at the same calendar horizon may have different θ values.

The same asset may experience a rapid phase movement without much calendar time passing.

Therefore:

$\theta \neq t$. (31.7)

Phase is an internal valuation coordinate.

31.4 Ledger-event index

Let k count committed financial events:

$L_0,L_1,L_2,\ldots$ (31.8)

An event increments k only when it passes the relevant gate.

A large amount of market activity may occur without changing a particular ledger.

Conversely, one legal or accounting event may create a major ledger transition at one instant.

Therefore:

$k \neq t$. (31.9)

And:

$k \neq \theta$. (31.10)

31.5 Phase movement without ledger movement

Suppose:

$\Delta\theta \neq 0$. (31.11)

Then:

$\Delta R \neq 0$ (31.12)

may arise economically.

But if the recognition gate defers the event:

$\Delta k = 0$. (31.13)

The valuation state has moved.

The ledger history has not yet advanced under that protocol.

This is a common financial condition:

- unrealized gain or loss;
- pending impairment;
- contingent claim;
- unsettled trade;
- unexercised option.

31.6 Ledger movement without large phase movement

A gate event may also occur after a small final movement if pressure has accumulated near a threshold.

For example:

- a covenant ratio crosses its limit;
- collateral falls below a threshold;

- an option ends slightly in the money;
- a rating trigger activates.

Thus:

Small $\Delta\theta$ + Threshold Crossing → Large Ledger Consequence. (31.14)

The size of the final movement need not equal the size of the institutional transition.

31.7 Internal financial time

A phase-based internal time may be defined only after a protocol states how phase movement is accumulated.

Let $\tilde{\Phi}$ be an unwrapped phase.

A simple internal phase clock may be:

$d\tau_{phase} = |d\tilde{\Phi}|/\Omega$. (31.15)

Where Ω is a declared phase scale.

This clock advances when valuation orientation changes, not merely when calendar time passes.

A quiet market may have:

$dt > 0$, (31.16)

but:

$d\tau_{phase} \approx 0$. (31.17)

A rapid repricing episode may have:

small dt, (31.18)

but:

large $d\tau_{phase}$. (31.19)

31.8 Gate-weighted ledger time

Introduce a gate-admission weight G:

$0 \le G \le 1$. (31.20)

Define ledgered phase time:

$d\tau_L = G|d\tilde{\Phi}|/\Omega$. (31.21)

Define residual phase depth:

$dT_R = (1 - G)|d\tilde{\Phi}|/\Omega$. (31.22)

Then:

$d\tau_{total} = d\tau_L + dT_R.$ (31.23)

The first term measures phase movement admitted into consequential history.

The second records movement that remains unresolved or uncommitted.

This construction is interpretive and protocol-dependent.

It is not a second physical clock.

31.9 Event time and market activity

The distinction resembles event-time models in finance.

Market time may advance through:

- trades;
- quote changes;
- volume;
- volatility events;
- order-book transitions.

Calendar seconds are not always the most informative clock.

The phase framework adds another possibility:

> Financial time may be indexed by admitted valuation reorientation rather than by uniform external duration.

This is a research hypothesis.

31.10 Irreversibility requires trace

Pure phase rotation is reversible:

$U(-\varphi)U(\varphi) = I.$ (31.24)

A ledger update may not be.

Once an event is:

- settled;
- reported;
- taxed;
- margin-called;
- defaulted;
- legally adjudicated;

the previous institutional state may not be recoverable by simply reversing φ.

Thus:

Reversible Rotation + Irreversible Trace → Financial History. (31.25)

The ledger supplies the asymmetry that circular phase alone lacks.

31.11 Phase unwrapping

Circular phase satisfies:

θ and θ + 2π (31.26)

at the same orientation.

Therefore circular phase alone forgets completed revolutions.

To preserve cumulative chronology, define an unwrapped phase:

$\tilde{\Phi} = \theta + 2\pi n$. (31.27)

Where n counts completed phase turns under the declared tracking rule.

This permits:

- cumulative phase distance;
- regime-cycle counting;
- event sequencing.

But n must be defined operationally.

Otherwise phase unwrapping can become arbitrary.

31.12 Three-time summary

Coordinate	Meaning	Advances when
t	calendar time	external duration passes
$\tilde{\Phi}$	unwrapped valuation phase	valuation orientation changes
k or τ_L	ledger time	a gate admits consequential trace

These clocks may correlate.

They are not identical.

---

32. Residual after Projection

32.1 Projection does not exhaust the financial state

A scalar valuation reports:

R. (32.1)

The completed state preserves:

Z = R + iQ. (32.2)

The readout R does not imply that Q has ceased to exist as a model coordinate.

Likewise, reading −Q does not erase R.

A projection selects one component.

It does not necessarily destroy the rest of the state.

32.2 Residual has several meanings

The word residual should not be used for only one quantity.

At least four residuals must be distinguished.

Geometric residual

$Q = \sqrt{(A^2 - R^2)}$. (32.3)

Model residual

ε_{model} = Observed Value − Model Value. (32.4)

Gate residual

ε_{gate} = Economic Consequence − Ledgered Consequence. (32.5)

Frame residual

$\varepsilon_{ab} = Z_b - \lambda_{ab} \exp(i\varphi_{ab}) Z_a$. (32.6)

These quantities may interact but are not identical.

32.3 Geometric Q is not all omitted risk

Q is derived from the relation between A and R.

It does not automatically contain every omitted risk factor.

For example, a CAPM-derived Q may fail to represent:

- liquidity;
- credit;
- legal risk;
- operational risk;

- funding asymmetry;
- tax;
- model uncertainty.

Such risks may appear as additional residual channels.

Therefore a richer state may be written:

$Z = R + iQ_{CAPM} + \varepsilon$. (32.7)

Or, in multidimensional form:

$\mathbf{Z} = (R, Q_{CAPM}, Q_{credit}, Q_{liquidity}, Q_{funding}, \ldots)$. (32.8)

The simple complex plane is a minimal model, not a complete risk universe.

32.4 Residual after a failed gate

Suppose a market move produces:

$\Delta R_{economic}$. (32.9)

But the gate admits nothing:

$\Delta R_{ledger} = 0$. (32.10)

Then:

$\varepsilon_{gate} = \Delta R_{economic}$. (32.11)

The pressure remains outside the ledger.

It may influence:

- funding;
- behaviour;
- future valuation;
- collateral negotiations;
- risk limits.

The absence of a ledger entry does not imply absence of economic consequence.

32.5 Residual after a successful gate

Suppose the gate admits only part of the consequence:

$\Delta R_{ledger} = \alpha \Delta R_{economic}$, (32.12)

where:

$0 < \alpha < 1$. (32.13)

Then:

$\varepsilon_{gate} = (1 - \alpha)\Delta R_{economic}$. (32.14)

Examples include:

- partial impairment;
- partial collateralization;
- staged settlement;
- incomplete loss recognition;
- capped insurance recovery.

A successful gate can leave substantial residual.

32.6 Residual can change future projection

Let the next valuation state depend on the retained residual:

$Z_{k+1} = F(Z_k, L_{k+1}, \varepsilon_k)$. (32.15)

Then residual is not merely accounting noise.

It can influence future:

- discount rates;
- liquidity;
- expected cash flows;
- behaviour;
- gate thresholds;
- confidence.

This creates residual backreaction.

32.7 Residual governance

A disciplined system should declare:

1. what was projected;
2. what was admitted;
3. what remained outside;
4. how the residual was measured;

5. when it will be reviewed;
6. what intervention it can trigger.

Without residual governance, a scalar valuation can create false closure.

32.8 False completion

False completion occurs when:

Observed Result = R (32.16)

is treated as:

Complete State = R. (32.17)

The complex framework resists that collapse by preserving:

Z = R + iQ. (32.18)

A more complete model may preserve additional residual channels beyond Q.

The principle is:

Admitted Value ≠ Exhaustive Financial State. (32.19)

32.9 Residual honesty

A mature implementation should report:

- model-implied R;
- declared A;
- derived Q;
- observed market value;
- market-implied Q where admissible;
- residual ε;
- baseline sensitivity;
- gate status.

This makes the model easier to falsify.

It also prevents Q from becoming a decorative symbol that absorbs every unexplained phenomenon.

---

33. When a Valuation Model Becomes World-Like

33.1 A number is not yet a world

A scalar present value is a number.

A completed valuation state is more structured.

But even:

$$Z = R + iQ \tag{33.1}$$

is not yet a world.

For a valuation system to become world-like, it must support more than representation.

It must organize:

- states;
- transitions;
- measurements;
- admissible events;
- persistent traces;
- interventions;
- revisions.

33.2 Minimal ingredients

A world-like valuation model requires at least:

1. a state space;
2. transition laws;
3. measurement protocols;
4. gates;
5. a ledger;
6. residual storage;
7. backreaction;
8. revision rules.

These ingredients produce an operational environment in which events can occur and matter.

33.3 State space

The minimal state space is:

$\mathcal{S} = \{(A,R,Q,\theta)\}$. (33.2)

A richer state may include:

$\mathcal{S} = \{(A,R,\mathbf{Q},\theta,L,\varepsilon,P)\}$. (33.3)

Where:

$\mathbf{Q}$ = multiple pressure channels; (33.4)

L = ledger; (33.5)

ε = residual; (33.6)

P = protocol. (33.7)

The protocol is part of the effective world because it determines what can be observed and admitted.

33.4 Transition laws

A local transition may be:

$dR = Rg_A - Qd\theta + \varepsilon_R$. (33.8)

$dQ = Qg_A + Rd\theta + \varepsilon_Q$. (33.9)

These equations specify how the state moves before gate commitment.

A world requires lawful transition, not merely static description.

33.5 Measurement protocols

A measurement family is:

$M_\varphi(Z) = \mathrm{Re}[\exp(i\varphi)Z]$. (33.10)

Different φ values expose different readouts.

A broader observer may also select:

- horizon;
- numeraire;
- funding curve;
- accounting rule;
- stress scenario.

Thus observation is protocol-bound.

33.6 Admissible events

Not every mathematical movement becomes a recognized event.

The gate determines admissibility:

$G_P(X,L) \to \{\text{Admit},\text{Defer},\text{Reject}\}$. (33.11)

A valuation world therefore contains rules defining what counts as consequential.

33.7 Persistent trace

A ledger update is:

$L_{(k+1)} = \text{Update}(L_k,\text{Event}_k)$. (33.12)

The trace persists beyond the moment of observation.

It may affect:

- future risk limits;
- contractual obligations;
- capital;
- behaviour;
- model inputs.

Persistence distinguishes a world history from an isolated calculation.

33.8 Intervention

An observer may alter the state through:

- trading;
- hedging;
- collateral calls;
- policy decisions;
- accounting recognition;
- capital allocation.

Represent an intervention as u:

$X' = F(X,u)$. (33.13)

A world-like model must describe not only passive observation but also admissible action.

33.9 Backreaction

The intervention or ledger event changes future dynamics:

$X_{k+1} = \mathscr{B}(X_k,u_k,L_{k+1})$. (33.14)

Examples include:

- hedging changes market demand;
- reporting changes investor behaviour;
- margin changes liquidity;
- capital rules change lending;
- settlement changes available collateral.

The observer is therefore not always external to the financial world.

33.10 Protocol revision

A mature world model must also permit the protocol itself to change:

$P_k \to P_{k+1}$. (33.15)

This may occur because:

- a model fails;
- regulation changes;
- liquidity disappears;
- a new market regime emerges;
- accounting rules change;
- an observer adopts another horizon.

Protocol revision changes what the world can admit as real.

33.11 World-like does not mean physically fundamental

Calling a valuation system world-like does not mean that it is a universe in the physical sense.

It means that the system contains an internally organized operational domain with:

- state;
- law;
- observation;
- event;
- history;

- intervention.

The term is structural.

It does not imply independent physical ontology.

33.12 Minimal world-formation cycle

The full cycle may be written:

Possibility
→ Valuation State
→ Measurement
→ Exposure
→ Movement
→ Gate
→ Ledger Trace + Residual
→ Backreaction
→ Updated State. (33.16)

This cycle extends the static CAPM calculation into an observer-compatible financial runtime.

The next sections will examine which parts of this runtime resemble broader measurement structures and which parts remain specifically financial.

34. What Finance Can Reproduce

34.1 The purpose of the comparison

The financial construction developed in Part I does not prove that markets are quantum systems.

It establishes something more limited but still important.

A mature, non-quantum financial system can reproduce several structures that are often first encountered together in quantum theory:

- complex representation;
- relative phase;
- conjugate measurement;
- observer-dependent readout;
- incomplete local access;
- conditional updating;
- gates;
- trace;
- residual;

- backreaction.

The recurrence of these structures outside microscopic physics suggests that they should not all be treated as uniquely quantum.

The correct methodological question is:

> Which apparently strange features arise from general measurement and commitment architecture, and which require specifically quantum physics?

34.2 Complex coordinates

Finance can support a complex state:

$Z = R + iQ.$ (34.1)

Where:

R = admitted financial value. (34.2)

Q = conjugate valuation-phase coordinate. (34.3)

The imaginary coordinate is not fictional.

It represents a real-valued financial sensitivity that is not identical to the ordinary mark.

The existence of complex coordinates alone is not quantum-specific.

Complex numbers are already used in:

- electrical engineering;
- signal analysis;
- control theory;
- oscillation theory;
- fluid dynamics;
- communications.

The financial contribution is to derive Q from a mature valuation filter rather than introduce it as decorative notation.

34.3 Relative phase

Finance can support a valuation phase:

$\theta = \arccos(R/A).$ (34.4)

Different valuation protocols can produce different phases:

$\theta_a \neq \theta_b.$ (34.5)

Their relative phase is:

$$\varphi_{ab} = \theta_b - \theta_a. \quad (34.6)$$

This relative orientation may summarize differences in:

- discount protocols;
- funding assumptions;
- risk premia;
- horizons;
- recognition rules.

Relative phase is not uniquely quantum.

Any rotational or oscillatory representation may contain it.

What is distinctive in the financial model is that the phase is constructed from a valuation ratio.

34.4 Conjugate measurement

Finance can support two related readouts:

$$M_0(Z) = R. \quad (34.7)$$

$$M_{\pi/2}(Z) = -Q. \quad (34.8)$$

The first measures admitted value.

The second measures signed phase exposure.

These readouts are connected by the quarter-turn operator:

$$\mathcal{J}[R,Q]^{\mathsf{T}} = [-Q,R]^{\mathsf{T}}. \quad (34.9)$$

Therefore:

$$\mathcal{J}^2 = -I. \quad (34.10)$$

This shows that a measurement system can contain incompatible-looking but lawfully related readouts without requiring quantum mechanics.

34.5 Contextual readout

A financial state can produce different visible quantities under different measurement protocols:

$$M_\varphi(Z) = R\cos\varphi - Q\sin\varphi. \quad (34.11)$$

The observed result depends on the declared orientation φ.

Likewise, a bank, dealer, regulator, accountant, and shareholder may extract different financial consequences from the same underlying arrangement.

This is a form of contextual measurement in the broad operational sense:

Observed Result = Function(State,Measurement Protocol). (34.12)

It does not automatically imply quantum contextuality in the technical Kochen–Specker sense.

The shared feature is protocol-dependent readout.

The stronger quantum theorem requires additional mathematical conditions.

34.6 Local observer restriction

A derivative trader may observe:

- the derivative mark;
- local Greeks;
- collateral;
- hedge inventory.

The trader may not know:

- the counterparty's complete funding state;
- every investor's position;
- the full market order book;
- every future regulatory intervention.

Thus:

Observer Information < Total Financial State. (34.13)

A bounded financial observer works with a projection:

Observed State_P = Extracted Structure_P + Residual_P. (34.14)

This is a general property of bounded observation.

It does not require quantum uncertainty.

34.7 Conditional update

A derivative's value changes when the underlying state changes:

$Z_U \to Z_U'$. (34.15)

$Z_D \to F_D(Z_U', C, P_D)$. (34.16)

The derivative sector is updated conditionally upon the underlying sector.

Similarly, collateral, funding, or regulatory capital may update after market movements.

Conditional state revision is common in classical information systems.

Its presence alone does not establish physical collapse.

34.8 Commitment gates

Finance contains explicit gates.

Examples include:

- exercise conditions;
- default thresholds;
- margin triggers;
- accounting recognition rules;
- covenant tests;
- settlement procedures.

A generic gate may be written:

$$G_P(X,L) \to \{\text{Admit},\text{Defer},\text{Reject}\}. \quad (34.17)$$

The gate determines which possibilities become operative financial events.

Gate architecture is not uniquely quantum.

Legal systems, databases, organizations, and artificial agents also use admission rules.

34.9 Trace and irreversibility

After a gate admits an event:

$$L_k \to L_{k+1}. \quad (34.18)$$

A persistent record is created.

The record may change:

- legal obligations;
- capital;
- collateral;
- available liquidity;
- future beliefs;
- future prices.

The trace creates effective irreversibility.

The underlying valuation equations may be reversible, while the institutional record is not.

This resembles the broader measurement problem of how reversible state evolution yields irreversible records.

Finance does not solve the physical measurement problem.

It supplies a visible macroscopic example of how gates and ledgers generate effective irreversibility.

34.10 Residual

Finance can preserve structure that does not enter the admitted result.

The minimal geometric residual is:

$Q = \sqrt{(A^2 - R^2)}$. (34.19)

Additional residuals include:

ε_{model} = Observed Value − Model Value. (34.20)

ε_{gate} = Economic Consequence − Ledgered Consequence. (34.21)

ε_{frame} = Observed Frame Difference − Modelled Frame Difference. (34.22)

The persistence of residual after projection is a general organizational feature.

It does not prove the existence of a quantum superposition.

34.11 Backreaction

Financial observation can alter the object observed.

Examples include:

- analyst reports changing investor demand;
- credit ratings changing funding costs;
- margin calculations causing asset sales;
- option hedging changing the underlying price;
- regulation changing bank behaviour.

A schematic backreaction is:

$X_{k+1} = \mathscr{B}(X_k, M_k, L_{k+1}, u_k)$. (34.23)

Here measurement, trace, and intervention influence the next state.

Observer backreaction is therefore not uniquely quantum.

Financial systems are reflexive because observers are participants.

34.12 Composite relational states

An underlying and derivative form a coupled arrangement:

$\Psi_{UD} = \Psi(Z_U, Z_D, C)$. (34.24)

The derivative cannot be valued independently of the underlying and contract.

Local observations may fail to reconstruct the complete joint arrangement.

This produces relational dependence and local incompleteness.

But contractual dependence is not automatically quantum entanglement.

The similarity concerns structural roles:

- joint state;
- local readout;
- conditional update;
- persistent coupling.

34.13 Internal time

Finance can define a phase-based internal clock:

$d\tau_{phase} = |d\tilde{\Phi}|/\Omega$. (34.25)

And a gate-weighted ledger clock:

$d\tau_L = G|d\tilde{\Phi}|/\Omega$. (34.26)

This produces a variable internal event rate.

Market time may accelerate during rapid revaluation and slow during stable periods.

Internal-time construction is not uniquely quantum.

Many nonlinear and event-driven systems support endogenous clocks.

34.14 The financial reproduction set

The structures that finance can reproduce may be summarized as:

$\mathfrak{F}_{reproducible} = \{$Complex Coordinate, Relative Phase, Conjugate Measurement, Contextual Readout, Bounded Observation, Conditional Update, Gate, Trace, Residual, Backreaction, Composite Coupling, Internal Time$\}$. (34.27)

The existence of this set suggests:

> A significant portion of measurement strangeness may arise from general observer-bound organization rather than exclusively from microscopic quantum substance.

This is a research proposition, not a conclusion about fundamental physics.

---

35. What Remains Specifically Quantum

35.1 Why subtraction is necessary

Cross-domain comparison becomes unreliable when every resemblance is treated as equivalence.

The correct method is subtractive.

Begin with the full set of phenomena associated with quantum theory.

Remove those structures already reproducible by classical waves, bounded observers, conditional systems, institutional gates, and complex measurement.

The remaining set is the candidate specifically quantum residue.

Symbolically:

Quantum Phenomenology = General Measurement Grammar + Quantum-Specific Residue. (35.1)

The financial model may help clarify the first term.

It does not automatically explain the second.

35.2 The Born rule

Quantum theory relates complex amplitudes to outcome probabilities through the Born rule.

For a quantum state $|\psi\rangle$ and projector P_a:

$Pr(a) = \langle\psi|P_a|\psi\rangle$. (35.2)

Or, for an amplitude ψ_a:

$Pr(a) = |\psi_a|^2$. (35.3)

The CAPM complex valuation plane does not derive this probability rule.

Its norm relation:

$A^2 = R^2 + Q^2$ (35.4)

is a valuation geometry.

It is not a proof that outcome probability equals squared amplitude.

Therefore:

Financial Norm Preservation ≠ Born Rule. (35.5)

35.3 Physical superposition

A quantum state may contain a coherent superposition:

$|\psi\rangle = \alpha|a\rangle + \beta|b\rangle$. (35.6)

The coefficients can interfere at the amplitude level before probability is calculated.

A financial portfolio can also be written as a linear combination:

$Z_P = w_aZ_a + w_bZ_b$. (35.7)

But portfolio addition usually represents simultaneous holdings, aggregation, or scenario weighting.

It does not automatically represent coherent physical superposition.

Therefore:

Portfolio Combination ≠ Quantum Superposition. (35.8)

35.4 Quantum interference

Quantum alternatives may interfere through cross terms:

$|\alpha + \beta|^2 = |\alpha|^2 + |\beta|^2 + 2\mathrm{Re}(\alpha\beta^*)$. (35.9)

Financial complex vectors may also produce constructive or destructive aggregation.

But classical cancellation is sufficient to explain many such effects.

To establish a specifically quantum analogue, one would need a probability rule and experimental structure in which amplitude-level interference cannot be replaced by classical mixture.

The present framework does not provide that result.

35.5 Bell inequality violation

Quantum entanglement can produce correlations violating Bell inequalities under suitable experiments.

The derivative–underlying system contains contractual coupling and restricted local observation.

But no result in this article establishes:

$CHSH > 2$. (35.10)

Financial correlation, contractual dependence, or rapid cross-market transmission does not imply Bell nonlocality.

Therefore:

Derivative Coupling ≠ Bell-Nonlocal Entanglement. (35.11)

35.6 No-signalling structure

Quantum entanglement exhibits correlations that do not permit ordinary faster-than-light signalling.

Financial systems are connected through:

- communication;
- contracts;
- order flow;
- information networks;
- common causes.

Observed coordination normally has classical transmission channels.

The financial model does not derive no-signalling constraints of quantum theory.

35.7 No-cloning

Quantum theory forbids the creation of an identical copy of an arbitrary unknown quantum state.

Finance contains practical limits on copying:

- incomplete information;
- transaction costs;
- liquidity;
- execution delay;
- model uncertainty.

But these are not the quantum no-cloning theorem.

A financial state can often be duplicated as data or replicated approximately through contracts.

Therefore:

Replication Limits ≠ Quantum No-Cloning. (35.12)

35.8 Noncommuting physical observables

In quantum theory, observables may satisfy:

ÂB̂ ≠ B̂Â. (35.13)

Or:

$[\hat{A},\hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} \neq 0.$ (35.14)

Financial operations may also be order-dependent:

Apply Margin Gate then Sell Assets ≠ Sell Assets then Apply Margin Gate. (35.15)

But institutional path dependence does not by itself establish canonical quantum commutation relations.

One would need a precise operator algebra and empirical consequences not reducible to classical sequencing.

35.9 Quantum contextuality

Technical quantum contextuality concerns the impossibility of assigning measurement outcomes consistently and noncontextually under particular mathematical conditions.

Finance certainly contains contextual valuation.

But:

Financial Context Dependence ≠ Kochen–Specker Contextuality. (35.16)

The former may arise because observers use different models, information, rules, or incentives.

The latter is a theorem about quantum observable structures.

35.10 Physical collapse

A financial gate converts an unresolved economic situation into a ledgered event.

This is collapse-like in organizational role.

But it does not prove that a physical state vector undergoes fundamental collapse.

A financial gate may be implemented by:

- contract;
- software;
- law;
- accounting policy;
- human decision.

Therefore:

Ledger Commitment ≠ Physical Wavefunction Collapse. (35.17)

35.11 Planck's constant and physical scale

Quantum commutation relations involve the physical constant $\hbar$:

$[\hat{x},\hat{p}] = i\hbar$. (35.18)

The financial complex plane contains no independently derived analogue of $\hbar$.

The imaginary unit i alone is insufficient.

Complex algebra appears in many classical systems.

A specifically quantum theory requires more than:

$Z = R + iQ$. (35.19)

35.12 Quantum field statistics

Quantum systems distinguish bosonic and fermionic statistics.

They support structures such as:

- symmetrization;
- antisymmetrization;
- exclusion;
- particle creation and annihilation.

The present financial model does not derive these physical statistics.

Organizational exclusion or market capacity constraints may be functionally analogous in limited ways, but they are not physical fermionic statistics.

35.13 The specifically quantum residue

The candidate specifically quantum residue includes at least:

$\mathcal{Q}$_residue = {Born Rule,Coherent Superposition,Irreducible Amplitude Interference,Bell Violation,Quantum No-Signalling,No-Cloning,Canonical Noncommutation,Quantum Contextuality,Physical Collapse Structure,$\hbar$-Scaled Dynamics,Quantum Statistics}. (35.20)

Some items remain subject to interpretive debate within physics.

The financial construction does not resolve those debates.

35.14 A disciplined boundary

The strongest justified conclusion is:

> Finance can reconstruct a substantial measurement-and-commitment grammar that resembles part of quantum theory's operational structure, while leaving the specifically quantum probability, correlation, and physical-state architecture unexplained.

This boundary protects both disciplines.

It prevents finance from claiming unearned physical status.

It also prevents general observer-bound structures from being treated as mysterious merely because they are familiar from quantum theory.

---

36. The Revised Quantum-Subtraction Programme

36.1 The original temptation

A cross-domain framework may begin with the impression:

Finance looks quantum. (36.1)

That statement is too broad.

A better research programme asks:

> After reconstructing phase, conjugate measurement, bounded observation, gates, residual, trace, and backreaction inside a mature classical system, what quantum strangeness remains?

This is a subtraction programme.

36.2 The three-layer decomposition

Observed quantum strangeness may be provisionally decomposed as:

Observed Quantum Strangeness = General Complex Grammar + Observer–Gate–Trace Grammar + Irreducibly Quantum Residue. (36.2)

Layer 1 — General complex grammar

Includes:

- two-component representation;
- phase;
- rotation;
- quadrature;
- interference-like vector aggregation;
- periodic closure.

Layer 2 — Observer–gate–trace grammar

Includes:

- bounded access;
- contextual readout;

- selective commitment;
- residual;
- irreversible record;
- backreaction;
- internal event time.

Layer 3 — Irreducibly quantum residue

Includes candidate structures listed in Section 35.

This decomposition is conceptual.

Its purpose is to organize research questions, not to declare that physical quantum mechanics has been reduced to finance.

36.3 What finance contributes

Finance contributes a visible macroscopic environment in which:

- valuation protocols are explicit;
- observers have identifiable roles;
- gates are institutionally implemented;
- traces are recorded;
- residuals can be audited;
- backreaction can be observed;
- models can be falsified.

This makes finance useful as an experimental grammar laboratory.

The relevant claim is:

> Finance may help isolate which structures follow from bounded valuation and institutional commitment before one invokes specifically quantum ontology.

36.4 Why CAPM is a useful starting point

CAPM is not chosen because it is the final theory of asset pricing.

It is chosen because it provides a simple mature filter:

$\beta ERP \rightarrow r_CAPM \rightarrow R.$ (36.3)

The scalar valuation can be completed transparently:

$R/A \rightarrow \theta \rightarrow Q.$ (36.4)

The resulting geometry is simple enough to derive fully:

$\partial R/\partial\theta = -Q.$ (36.5)

$\mathcal{J}^2 = -I.$ (36.6)

This clarity makes CAPM a pedagogical and analytical starting point.

A more complete financial theory may later replace CAPM while preserving the operator logic.

36.5 Extension beyond CAPM

Let a mature finance filter produce:

$R = F_P(A,\xi).$ (36.7)

Where:

P = protocol; (36.8)

ξ = filter inputs. (36.9)

If:

$0 \le R/A \le 1,$ (36.10)

one may define:

$\theta_P = \arccos(R/A).$ (36.11)

And:

$Q_P = \sqrt{(A^2 - R^2)}.$ (36.12)

Then:

$R = A\cos\theta_P.$ (36.13)

$Q_P = A\sin\theta_P.$ (36.14)

At fixed A:

$\partial R/\partial\theta_P = -Q_P.$ (36.15)

Thus the conjugate-risk theorem is not limited mathematically to CAPM.

CAPM supplies one mature construction of the filter ratio.

36.6 Candidate mature filters

The same completion may be tested with:

- certainty-equivalent valuation;
- stochastic discount factors;

- credit-spread discounting;
- liquidity haircuts;
- risk-neutral derivative valuation;
- collateral valuation;
- regulatory capital filters;
- accounting recognition rules;
- real-option exercise filters.

Each filter generates its own:

A_P, (36.16)

R_P, (36.17)

θ_P, (36.18)

Q_P. (36.19)

The coordinates should not be mixed without protocol reconciliation.

36.7 Multi-dimensional residual geometry

A single Q may be insufficient.

Suppose mature filters identify several distinct pressure channels:

$\mathbf{Q}$ = (Q_market,Q_credit,Q_liquidity,Q_funding,Q_option,Q_model). (36.20)

A generalized norm may be:

$A^2 = R^2 + \mathbf{Q}^T G \mathbf{Q}$. (36.21)

Where G is a declared positive metric.

The simple plane becomes a higher-dimensional valuation space.

The original Q is then a one-channel reduction.

This extension may require matrix calculus but not necessarily tensor physics.

36.8 Path dependence

The present CAPM phase is constructed from the current valuation ratio.

But financial systems may remember the path.

A path-dependent state may be written:

Z_t = Z_t[History_(0:t),P_t]. (36.22)

Then the final phase may depend on the sequence of:

- prior gates;
- collateral calls;
- liquidity events;
- model revisions;
- behavioural responses.

Two systems with the same current R may possess different residual histories.

This is where ledger trace becomes essential.

36.9 Noncommuting financial operations

Although ordinary CAPM phase rotations commute:

$$U(\varphi_1)U(\varphi_2) = U(\varphi_1 + \varphi_2), \quad (36.23)$$

more complex financial operations may not.

For example:

$$\text{Funding Revaluation} \circ \text{Margin Liquidation} \neq \text{Margin Liquidation} \circ \text{Funding Revaluation}. \quad (36.24)$$

A future research programme may define financial operators:

$$\mathcal{O}_{\text{funding}}, \mathcal{O}_{\text{liquidity}}, \mathcal{O}_{\text{accounting}}, \mathcal{O}_{\text{settlement}}. \quad (36.25)$$

Then examine:

$$[\mathcal{O}_a, \mathcal{O}_b] = \mathcal{O}_a\mathcal{O}_b - \mathcal{O}_b\mathcal{O}_a. \quad (36.26)$$

Nonzero commutators would indicate path-dependent institutional structure.

They would not automatically imply quantum physics.

36.10 Curved financial geometry

If the metric depends on the state:

$$G = G(X), \quad (36.27)$$

then the meaning of orthogonality changes across the valuation space.

A movement that is a quarter-turn in one region may not remain a quarter-turn elsewhere.

This produces an effective curved financial geometry.

Possible causes include:

- changing liquidity;

- nonlinear leverage;
- state-dependent correlations;
- funding thresholds;
- regime-dependent risk premia.

The present Euclidean CAPM plane is the local flat starting point.

36.11 Gauge-like frame transport

Suppose an asset is revalued across several frames:

$a \to b \to c \to a.$ (36.28)

If the state returns unchanged:

$\text{Transport_loop}(Z_a) = Z_a,$ (36.29)

the frame system is path-independent.

If instead:

$\text{Transport_loop}(Z_a) \neq Z_a,$ (36.30)

there is a loop residual.

This may represent:

- transaction costs;
- funding basis;
- tax;
- settlement mismatch;
- model inconsistency;
- institutional arbitrage.

Such loop dependence is gauge-like in mathematical role.

It is not a claim that finance contains a physical gauge field.

36.12 Observer-bound world classification

Financial worlds may be classified by:

- state dimension;
- admissible measurements;
- gate architecture;

- ledger persistence;
- residual capacity;
- backreaction strength;
- frame-transport consistency;
- intervention set.

Define a world specification:

$\mathcal{W} = (\mathcal{S}, \square, \mathcal{G}, \mathcal{L}, \mathcal{E}, \mathcal{U}, \mathcal{B})$. (36.31)

Where:

$\mathcal{S}$ = state space; (36.32)

$\square$ = measurement family; (36.33)

$\mathcal{G}$ = gate family; (36.34)

$\mathcal{L}$ = ledger architecture; (36.35)

$\mathcal{E}$ = residual architecture; (36.36)

$\mathcal{U}$ = admissible interventions; (36.37)

$\mathcal{B}$ = backreaction law. (36.38)

This is a possible general interface for comparing financial, organizational, and artificial-observer systems.

36.13 Falsification requirement

Every extension should specify:

- what is measured;
- how it differs from existing finance;
- what observation could reject it;
- what decision it improves;
- what residual remains;
- when the analogy fails.

Without these conditions, terms such as phase, curvature, gauge, and world become decorative.

The entire programme must preserve the rule:

No Operational Gain → No Justified Additional Structure. (36.39)

---

37. Final Synthesis

37.1 The original problem

CAPM-based discounted-cash-flow valuation produces:

$R = CF_t/(1 + r_{CAPM})^t$. (37.1)

This scalar value is useful but structurally incomplete relative to a declared baseline:

$A = CF_t/(1 + r_{base})^t$. (37.2)

The admitted ratio is:

R/A. (37.3)

The complex completion defines:

$\theta = \arccos(R/A)$. (37.4)

$Q = \sqrt{(A^2 - R^2)}$. (37.5)

$Z = R + iQ = A \exp(i\theta)$. (37.6)

The central unresolved question was:

> What is Q financially?

37.2 The central answer

At fixed A:

$R = A \cos \theta$. (37.7)

Therefore:

$\partial R/\partial \theta = -A \sin \theta$. (37.8)

Since:

$Q = A \sin \theta$, (37.9)

we obtain:

$\partial R/\partial \theta = -Q$. (37.10)

Thus:

> Q is the magnitude of the first-order dollar exposure of admitted value to movement in the declared valuation phase.

The signed exposure is:

$\Delta_\theta = -Q$. (37.11)

This is the precise finance-first identity of Q.

37.3 Q is not the loss

Q is not:

$$Q = A - R. \quad (37.12)$$

The scalar haircut is:

$$H = A - R. \quad (37.13)$$

Its marginal relation is:

$$dH/d\theta = Q. \quad (37.14)$$

And:

$$H = \int_0^{\theta} Q(\varphi)d\varphi. \quad (37.15)$$

Therefore:

> Q is the marginal phase-pressure underlying the accumulated scalar haircut.

Only when A is independently established as the best foregone alternative does this become a marginal opportunity-cost interpretation.

37.4 Ordinary finance is preserved

The required-return sensitivity is:

$$dR = -[tR/(1 + r)]dr. \quad (37.16)$$

The phase sensitivity is:

$$dR = -Qd\theta. \quad (37.17)$$

Therefore:

$$Qd\theta = [tR/(1 + r)]dr. \quad (37.18)$$

The complex phase representation does not alter ordinary first-order CAPM P&L.

It changes the risk coordinate.

37.5 The operator answer

Define:

$$\mathcal{J} = [0\ -1;\ 1\ 0]. \quad (37.19)$$

Then:

$$\mathcal{J}[R,Q]^T = [-Q,R]^T. \quad (37.20)$$

And:

$\mathcal{J}^2 = -I$. (37.21)

$\mathcal{J}^4 = I$. (37.22)

The measurement family is:

$M_{\varphi}(Z) = \mathrm{Re}[\exp(i\varphi)Z]$. (37.23)

The principal readouts are:

$M_0(Z) = R$. (37.24)

$M_{\pi/2}(Z) = -Q$. (37.25)

$M_{\pi}(Z) = -R$. (37.26)

$M_{3\pi/2}(Z) = Q$. (37.27)

$M_{2\pi}(Z) = R$. (37.28)

Thus:

$R \to -Q \to -R \to Q \to R$. (37.29)

37.6 The financial meaning of the cycle

In a signed linear position space, the cycle means:

Long Mark
→ Long Phase Exposure
→ Short Mark
→ Short Phase Exposure
→ Long Mark. (37.30)

The two mark readouts are:

$R \leftrightarrow -R$. (37.31)

The two exposure readouts are:

$-Q \leftrightarrow Q$. (37.32)

Therefore $-Q$ is not the counterparty's mark.

It is the long position's conjugate phase exposure.

37.7 The meaning of $i^2 = -1$

The first quarter-turn changes measurement type:

Mark → Conjugate Exposure. (37.33)

The second quarter-turn reverses signed orientation:

Conjugate Exposure → Opposite Mark. (37.34)

Therefore:

> In this financial geometry, $i^2 = -1$ means measurement-orientation reversal.

It does not mean:

- risk happened twice;
- the asset suffered two losses;
- Q automatically became realized P&L.

37.8 Exposure, movement, and history

The correct causal sequence is:

Measurement identifies −Q. (37.35)

Actual phase movement $\Delta\theta$ generates ΔR. (37.36)

A gate determines recognition. (37.37)

A ledger records trace. (37.38)

Residual remains. (37.39)

In compact form:

Measurement → Exposure → Movement → P&L → Gate → Ledger + Residual. (37.40)

This is the complete financial runtime.

37.9 Multi-period extension

For multiple cash flows:

$Z = \Sigma_t A_t \exp(i\theta_t)$. (37.41)

$R = \Sigma_t A_t \cos \theta_t$. (37.42)

$Q = \Sigma_t A_t \sin \theta_t$. (37.43)

For a common phase rotation φ:

$d\mathrm{Re}[\exp(i\varphi)Z]/d\varphi|_{(\varphi=0)} = -Q$. (37.44)

For term-specific movements:

$dR = -\Sigma_t Q_t d\theta_t$. (37.45)

Thus the scalar conjugate exposure generalizes to a phase-exposure term structure.

37.10 The empirical limit

In a one-period static model:

$Q = \sqrt{(A^2 - R^2)}$. (37.46)

Therefore Q is not algebraically independent of A and R.

Its practical value must come from:

- dynamic attribution;
- multi-horizon structure;
- protocol comparison;
- market-implied residual;
- gate-event diagnosis;
- better risk communication;
- improved intervention.

If none of these gains appears, Q remains an optional reparameterization.

37.11 The broader structural result

Once mark and conjugate exposure are treated as one state, finance can display:

- complex coordinates;
- phase;
- quadrature;
- contextual measurement;
- local observer limits;
- gates;
- trace;
- residual;
- backreaction;
- internal event time;
- composite contractual states.

These structures make finance a useful macroscopic laboratory for observer-bound valuation.

But they do not derive the specifically quantum residue:

- Born probabilities;
- Bell violation;

- no-cloning;
- quantum no-signalling;
- irreducible coherent superposition;
- physical wavefunction collapse.

37.12 Final conclusion

The final result can be stated in three sentences.

> **Q is the conjugate valuation-phase coordinate and the magnitude of the first-order dollar exposure generated by the same declared geometry that produces the CAPM value R.**
>
> **Multiplication by i rotates the financial measurement from mark to conjugate exposure; applying it twice reverses the signed valuation orientation and produces −R.**
>
> **Exposure becomes economic profit or loss only through actual state movement, and that consequence becomes financial history only after passing a gate into a ledger while leaving an auditable residual.**

The complete sequence is therefore:

CAPM Filter
→ Admitted Value R
→ Declared Phase θ
→ Conjugate Exposure Q
→ Measurement Rotation
→ State Movement
→ P&L
→ Gate
→ Ledger Trace + Residual
→ Updated Financial World. (37.47)

The complex valuation plane does not replace mature finance.

It reveals a measurement structure already latent within mature financial filtering.

Main Text Conclusion

The main argument is now complete.

The appendices should next provide:

- full symbol definitions;
- detailed proofs;
- sign conventions;
- finite-difference implementation;

- comparison with duration, convexity, and option Greeks;
- multi-period derivations;
- empirical test templates;
- the precise limits of the quantum comparison.

Appendix A — Symbol Dictionary, Units, and Protocol Declarations

A.1 Purpose of the symbol dictionary

The complex valuation framework uses familiar financial quantities together with several derived coordinates. Because some symbols can acquire different meanings under different protocols, every implementation should declare:

- the valued claim;
- the future cash-flow model;
- the horizon;
- the baseline rate;
- the risk-adjusted valuation filter;
- the metric;
- the phase orientation;
- the measurement convention;
- the gate and ledger rules.

The same formula can carry a different economic meaning when any of these declarations changes.

A.2 Core valuation symbols

Symbol	Meaning	Typical unit
CF_t	cash flow at horizon t	currency
t	discount horizon	period count or time
r_base	declared baseline discount rate	dimensionless rate
β	CAPM market-risk loading	dimensionless
ERP	equity risk premium	dimensionless rate
r	CAPM required return	dimensionless rate
A_t	baseline-discounted cash-flow amplitude	currency
R_t	CAPM-admitted present value	currency
Q_t	conjugate valuation-phase coordinate	currency
θ_t	CAPM valuation phase	radians
Z_t	completed complex valuation state	currency-valued complex coordinate

The CAPM required return is:

$r = r_{base} + \beta ERP$. (A.1)

The baseline amplitude is:

$A_t = CF_t/(1 + r_{base})^t$. (A.2)

The admitted value is:

$R_t = CF_t/(1 + r)^t$. (A.3)

The valuation ratio is:

$c_t = R_t/A_t$. (A.4)

The phase is:

$\theta_t = \arccos(c_t)$. (A.5)

The conjugate coordinate is:

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (A.6)

The completed state is:

$Z_t = R_t + iQ_t$. (A.7)

Equivalently:

$Z_t = A_t \exp(i\theta_t)$. (A.8)

A.3 Sensitivity symbols

Symbol	Meaning	Unit
Δ_θ	signed CAPM Phase Delta	currency per radian
D_r	positive dollar exposure to required-return movement	currency per unit rate
$\mathcal{J}$	complex-structure or quarter-turn operator	dimensionless linear operator
$U(\varphi)$	finite phase-rotation operator	dimensionless linear operator
M_φ	real readout under measurement orientation φ	currency
g_A	proportional radial amplitude change	dimensionless local change
ε_R	unexplained real-coordinate residual	currency
ε_Q	unexplained conjugate-coordinate residual	currency

The signed CAPM Phase Delta is:

$\Delta_\theta \equiv \partial R/\partial\theta = -Q$. (A.9)

The positive required-return exposure coefficient is:

$D_r \equiv -\partial R/\partial r = tR/(1 + r)$. (A.10)

The bridge between the two risk coordinates is:

$Qd\theta = D_r dr$. (A.11)

A.4 Position and portfolio symbols

Symbol	Meaning
Z_{long}	completed state of a long position
Z_{short}	completed state of the opposite signed position
w_i	signed portfolio weight
Z_P	aggregate portfolio complex state
R_P	portfolio admitted value
Q_P	portfolio conjugate phase exposure
Q_t	phase exposure assigned to cash-flow horizon t
$\mathbf{Q}$	vector of term-specific phase exposures
$d\boldsymbol{\theta}$	vector of term-specific phase movements

For a linear signed position space:

$$Z_{short} = -Z_{long}. \quad (A.12)$$

For a portfolio:

$$Z_P = \Sigma_i w_i Z_i. \quad (A.13)$$

Therefore:

$$R_P = \Sigma_i w_i R_i. \quad (A.14)$$

$$Q_P = \Sigma_i w_i Q_i. \quad (A.15)$$

A.5 Gate, ledger, and residual symbols

Symbol	Meaning
P	declared protocol
G_P	gate operating under protocol P
L_k	ledger state after k committed events
ε_{gate}	unledgered portion of economic consequence
ε_{model}	observed value minus model value
ε_{ab}	residual after translating between frames a and b
$\mathscr{B}$	backreaction map
u	admissible intervention

A generic gate is:

$$G_P(\Delta R, X, L) \in \{\text{Admit}, \text{Defer}, \text{Reject}\}. \quad (A.16)$$

A ledger update is:

$$L_{(k+1)} = \text{Update}(L_k, \text{Event}_k). \quad (A.17)$$

The gate residual is:

$$\varepsilon_{gate} = \Delta R_{economic} - \Delta R_{ledger}. \quad (A.18)$$

A.6 Time symbols

Symbol	Meaning
t	external calendar horizon
θ	circular valuation phase
$\tilde{\Phi}$	unwrapped cumulative phase
k	discrete ledger-event count
τ_phase	phase-based internal time
τ_L	gate-admitted ledger time
T_R	uncommitted residual phase depth

A phase clock may be defined as:

$d\tau_{phase} = |d\tilde{\Phi}|/\Omega.$ (A.19)

A gate-weighted ledger clock may be defined as:

$d\tau_L = G|d\tilde{\Phi}|/\Omega.$ (A.20)

Residual phase depth may be written:

$dT_R = (1 - G)|d\tilde{\Phi}|/\Omega.$ (A.21)

These are protocol-defined internal coordinates, not additional physical clocks.

A.7 Minimal protocol declaration

A minimal protocol should be written:

$P = (B, CF, h, r_{base}, F, G, L, Metric, Orientation).$ (A.22)

Where:

B = system boundary. (A.23)

CF = declared cash-flow model. (A.24)

h = horizon structure. (A.25)

r_base = baseline discount rule. (A.26)

F = mature finance filter producing R. (A.27)

G = gate rule. (A.28)

L = ledger architecture. (A.29)

Metric = rule defining magnitude and orthogonality. (A.30)

Orientation = sign convention for the quarter-turn. (A.31)

No valid Q should be reported without the corresponding protocol.

Appendix B — Complete CAPM Projection and Q Derivations

B.1 CAPM required return

Begin with:

r = r_base + βERP. (B.1)

Assume:

1 + r_base > 0. (B.2)

1 + r > 0. (B.3)

For the ordinary first-quadrant construction, also assume:

r ≥ r_base. (B.4)

B.2 Baseline and admitted valuations

For a positive future cash flow CF_t:

A_t = CF_t/(1 + r_base)^t. (B.5)

R_t = CF_t/(1 + r)^t. (B.6)

The admitted-value ratio is:

R_t/A_t = [CF_t/(1 + r)^t]/[CF_t/(1 + r_base)^t]. (B.7)

Cancel CF_t:

R_t/A_t = [(1 + r_base)/(1 + r)]^t. (B.8)

Define:

c_t = [(1 + r_base)/(1 + r)]^t. (B.9)

Then:

R_t = A_tc_t. (B.10)

B.3 Definition of phase

Define the principal phase:

θ_t = arccos(c_t). (B.11)

Therefore:

cos θ_t = c_t. (B.12)

Hence:

$R_t = A_t \cos \theta_t$. (B.13)

This proves the CAPM projection identity.

B.4 Definition of Q

Using the Euclidean norm:

$A_t^2 = R_t^2 + Q_t^2$. (B.14)

Solve for Q_t:

$Q_t^2 = A_t^2 - R_t^2$. (B.15)

In the first quadrant:

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (B.16)

Because:

$R_t = A_t \cos \theta_t$, (B.17)

we obtain:

$Q_t = \sqrt{[A_t^2 - A_t^2 \cos^2 \theta_t]}$. (B.18)

Therefore:

$Q_t = A_t \sqrt{(1 - \cos^2 \theta_t)}$. (B.19)

In the first quadrant:

$Q_t = A_t \sin \theta_t$. (B.20)

B.5 Fully expanded CAPM formula

Substitute:

$A_t = CF_t/(1 + r_{base})^t$. (B.21)

And:

$\cos \theta_t = [(1 + r_{base})/(1 + r)]^t$. (B.22)

Then:

$Q_t = [CF_t/(1 + r_{base})^t] \sqrt{\{1 - [(1 + r_{base})/(1 + r)]^{(2t)}\}}$. (B.23)

Using:

$r = r_{base} + \beta ERP$, (B.24)

we obtain:

$Q_t = [CF_t/(1 + r_{base})^t] \sqrt{\{1 - [(1 + r_{base})/(1 + r_{base} + \beta ERP)]^{(2t)}\}}$. (B.25)

B.6 Ratio relations

Because:

$R = A \cos \theta$, (B.26)

and:

$Q = A \sin \theta$, (B.27)

we have:

$Q/R = \tan \theta$. (B.28)

Also:

$R/A = \cos \theta$. (B.29)

$Q/A = \sin \theta$. (B.30)

Therefore:

$\theta = \arctan(Q/R)$. (B.31)

Within the principal domain:

$\theta = \arccos(R/A) = \arcsin(Q/A) = \arctan(Q/R)$. (B.32)

B.7 Recovering A

Given R and Q:

$A = \sqrt{(R^2 + Q^2)}$. (B.33)

Thus the representations:

(A,θ), (B.34)

and:

(R,Q) (B.35)

contain the same static information under the declared sign and domain conventions.

B.8 Small-risk approximation

Define:

$x = \beta ERP/(1 + r_base)$. (B.36)

Then:

$1 + r = (1 + r_base)(1 + x)$. (B.37)

Therefore:

cos θ = (1 + x)^(−t). (B.38)

For small x:

(1 + x)^(−t) ≈ 1 − tx + t(t + 1)x²/2 + O(x³). (B.39)

For small θ:

cos θ ≈ 1 − θ²/2 + θ⁴/24 + O(θ⁶). (B.40)

At leading order:

1 − θ²/2 ≈ 1 − tx. (B.41)

Therefore:

θ² ≈ 2tx. (B.42)

Hence:

θ ≈ √[2tβERP/(1 + r_base)]. (B.43)

The leading relation is nonlinear:

θ ∝ √(βERP). (B.44)

B.9 Small-angle approximations for Q and H

For small θ:

Q = A sin θ ≈ Aθ. (B.45)

The scalar haircut is:

H = A − R = A(1 − cos θ). (B.46)

Therefore:

H ≈ Aθ²/2. (B.47)

Eliminate θ:

H ≈ Q²/(2A). (B.48)

The exact identity is:

H = Q²/(A + R). (B.49)

Proof:

H = A − R. (B.50)

Multiply numerator and denominator by A + R:

H = (A − R)(A + R)/(A + R). (B.51)

Therefore:

$$H = (A^2 - R^2)/(A + R). \quad (B.52)$$

Since:

$$A^2 - R^2 = Q^2, \quad (B.53)$$

we obtain:

$$H = Q^2/(A + R). \quad (B.54)$$

Appendix C — Complete Conjugate Risk Derivation

C.1 Fixed-amplitude orbit

Let:

$$Z(\theta) = A \exp(i\theta). \quad (C.1)$$

Assume A is fixed.

Then:

$$R(\theta) = A \cos \theta. \quad (C.2)$$

$$Q(\theta) = A \sin \theta. \quad (C.3)$$

Differentiate:

$$dR/d\theta = -A \sin \theta. \quad (C.4)$$

Therefore:

$$dR/d\theta = -Q. \quad (C.5)$$

Similarly:

$$dQ/d\theta = A \cos \theta. \quad (C.6)$$

Therefore:

$$dQ/d\theta = R. \quad (C.7)$$

C.2 Tangent vector

The state vector is:

$$x(\theta) = [R(\theta),Q(\theta)]^T. \quad (C.8)$$

Its derivative is:

$$dx/d\theta = [-Q,R]^T. \quad (C.9)$$

The radial vector is:

$$x = [R,Q]^T. \quad (C.10)$$

Their Euclidean inner product is:

$x^T(dx/d\theta) = R(-Q) + Q(R)$. (C.11)

Therefore:

$x^T(dx/d\theta) = 0$. (C.12)

The phase derivative is orthogonal to the radial state vector.

C.3 Preservation of magnitude

Differentiate:

$A^2 = R^2 + Q^2$. (C.13)

At fixed A:

$d(R^2 + Q^2)/d\theta = 0$. (C.14)

Therefore:

$2R(dR/d\theta) + 2Q(dQ/d\theta) = 0$. (C.15)

Substitute:

$dR/d\theta = -Q$, (C.16)

and:

$dQ/d\theta = R$. (C.17)

Then:

$2R(-Q) + 2Q(R) = 0$. (C.18)

The derivative law preserves the norm.

C.4 Required-return derivative

For:

R = CF_t/(1 + r)^t, (C.19)

differentiate:

∂R/∂r = −tCF_t/(1 + r)^(t+1). (C.20)

Since:

R = CF_t/(1 + r)^t, (C.21)

we obtain:

∂R/∂r = −tR/(1 + r). (C.22)

C.5 Phase derivative with respect to r

Start from:

$\cos \theta = R/A.$ (C.23)

Hold A fixed.

Differentiate with respect to r:

$-\sin \theta \cdot \partial\theta/\partial r = (1/A)\partial R/\partial r.$ (C.24)

Since:

$\sin \theta = Q/A,$ (C.25)

we obtain:

$-(Q/A)\partial\theta/\partial r = (1/A)\partial R/\partial r.$ (C.26)

Therefore:

$-Q\partial\theta/\partial r = \partial R/\partial r.$ (C.27)

Substitute:

$\partial R/\partial r = -tR/(1 + r).$ (C.28)

Hence:

$Q\partial\theta/\partial r = tR/(1 + r).$ (C.29)

Thus:

$\partial\theta/\partial r = tR/[(1 + r)Q].$ (C.30)

C.6 Chain-rule recovery of Q

By the chain rule:

$\partial R/\partial\theta = (\partial R/\partial r)/(\partial\theta/\partial r).$ (C.31)

Substitute equations (C.22) and (C.30):

$\partial R/\partial\theta = [-tR/(1 + r)]/[tR/((1 + r)Q)].$ (C.32)

Therefore:

$\partial R/\partial\theta = -Q.$ (C.33)

C.7 Differential P&L equivalence

The required-return representation is:

$dR = -[tR/(1 + r)]dr.$ (C.34)

The phase representation is:

$dR = -Qd\theta$. (C.35)

Therefore:

$Qd\theta = [tR/(1 + r)]dr$. (C.36)

This equation proves that the two local risk representations generate the same first-order P&L.

C.8 Beta derivative

Holding ERP fixed:

$\partial r/\partial \beta = ERP$. (C.37)

Therefore:

$\partial R/\partial \beta = -tERP \cdot R/(1 + r)$. (C.38)

And:

$\partial \theta/\partial \beta = tERP \cdot R/[(1 + r)Q]$. (C.39)

Thus:

$(\partial R/\partial \beta)/(\partial \theta/\partial \beta) = -Q$. (C.40)

C.9 ERP derivative

Holding β fixed:

$\partial r/\partial ERP = \beta$. (C.41)

Therefore:

$\partial R/\partial ERP = -t\beta R/(1 + r)$. (C.42)

And:

$\partial \theta/\partial ERP = t\beta R/[(1 + r)Q]$. (C.43)

Thus:

$(\partial R/\partial ERP)/(\partial \theta/\partial ERP) = -Q$. (C.44)

C.10 Total differential when A changes

When A is not fixed:

$R = A \cos \theta$. (C.45)

Therefore:

$$dR = \cos\theta\, dA - A \sin\theta\, d\theta. \quad (C.46)$$

Using:

$$\cos\theta = R/A, \quad (C.47)$$

and:

$$A \sin\theta = Q, \quad (C.48)$$

we obtain:

$$dR = (R/A)dA - Qd\theta. \quad (C.49)$$

Similarly:

$$Q = A \sin\theta. \quad (C.50)$$

Therefore:

$$dQ = \sin\theta\, dA + A \cos\theta\, d\theta. \quad (C.51)$$

Hence:

$$dQ = (Q/A)dA + Rd\theta. \quad (C.52)$$

Define:

$$g_A = dA/A. \quad (C.53)$$

Then:

$$dR = Rg_A - Qd\theta. \quad (C.54)$$

$$dQ = Qg_A + Rd\theta. \quad (C.55)$$

These equations separate radial and angular movement.

C.11 Complex differential form

Because:

$$Z = A \exp(i\theta), \quad (C.56)$$

take the logarithmic differential:

$$dZ/Z = dA/A + i\, d\theta. \quad (C.57)$$

Therefore:

$$dZ = Z(g_A + i\, d\theta). \quad (C.58)$$

Expanding:

$$dZ = (R + iQ)(g_A + i\, d\theta). \quad (C.59)$$

Hence:

$$dZ = (Rg_A - Qd\theta) + i(Qg_A + Rd\theta). \quad (C.60)$$

The real and imaginary components reproduce equations (C.54) and (C.55).

Appendix D — Sign, Orientation, and Measurement Conventions

D.1 Why conventions matter

Complex rotation can be written using either:

$$\exp(i\varphi), \quad (D.1)$$

or:

$$\exp(-i\varphi). \quad (D.2)$$

Both are mathematically valid.

They correspond to opposite orientation conventions.

The sign of the quarter-turn readout depends on the selected convention.

A financial implementation must declare the convention before interpreting Q.

D.2 Adopted state convention

This article defines:

$$Z = R + iQ. \quad (D.3)$$

With:

$$Q \geq 0 \quad (D.4)$$

in the ordinary first-quadrant CAPM construction.

The positive orientation is counterclockwise.

The quarter-turn operator is:

$$\mathcal{J}_+ = [0\ -1;\ 1\ 0]. \quad (D.5)$$

Thus:

$$\mathcal{J}_+[R,Q]^T = [-Q,R]^T. \quad (D.6)$$

D.3 Adopted measurement convention

The measurement family is:

$$M_\varphi(Z) = \mathrm{Re}[\exp(i\varphi)Z]. \quad (D.7)$$

Therefore:

$M_{\varphi}(Z) = R\cos\varphi - Q\sin\varphi.$ (D.8)

The special readouts are:

$M_0(Z) = R.$ (D.9)

$M_{\pi/2}(Z) = -Q.$ (D.10)

$M_{\pi}(Z) = -R.$ (D.11)

$M_{3\pi/2}(Z) = Q.$ (D.12)

$M_{2\pi}(Z) = R.$ (D.13)

D.4 Opposite convention

If instead one defines:

$\tilde{M}_{\varphi}(Z) = \mathrm{Re}[\exp(-i\varphi)Z],$ (D.14)

then:

$\tilde{M}_{\varphi}(Z) = R\cos\varphi + Q\sin\varphi.$ (D.15)

Therefore:

$\tilde{M}_{\pi/2}(Z) = Q.$ (D.16)

The cycle becomes:

$R \to Q \to -R \to -Q \to R.$ (D.17)

This is the same geometry with opposite orientation.

D.5 Why the article adopts $R \to -Q$

The adopted convention aligns positive θ movement with stronger valuation filtering.

Because:

$R = A\cos\theta,$ (D.18)

and:

$Q = A\sin\theta,$ (D.19)

we have:

$dR/d\theta = -Q.$ (D.20)

Therefore the signed exposure to positive θ movement is naturally:

$-Q.$ (D.21)

The cycle:

$$R \to -Q \to -R \to Q \to R \quad (D.22)$$

preserves this derivative sign.

D.6 Positive and negative Q

The principal CAPM construction uses:

$$0 \le \theta < \pi/2. \quad (D.23)$$

Therefore:

$$Q \ge 0. \quad (D.24)$$

A wider financial state space may allow:

$$Q < 0. \quad (D.25)$$

A negative Q may represent the opposite phase orientation, such as:

- protective exposure;
- hedge-dominant orientation;
- negative sensitivity under the declared phase direction.

But this interpretation requires a protocol extending beyond the principal CAPM ratio.

It should not be inferred solely from the positive-cash-flow CAPM formula.

D.7 Positive and negative R

The ordinary positive-cash-flow CAPM formula gives:

$$R > 0. \quad (D.26)$$

Negative R enters through a wider signed state space, such as:

- short positions;
- liabilities;
- negative cash-flow claims;
- opposite contractual orientation;
- measurement half-turn.

Therefore:

$$R < 0 \quad (D.27)$$

is not generally produced by merely increasing the positive discount rate.

It requires a signed claim or a broader measurement orientation.

D.8 Active and passive sign conventions

An active rotation changes the state:

$Z' = \exp(i\varphi)Z.$ (D.28)

A passive rotation may instead be written as a basis change:

$\text{Coordinates_new} = \exp(-i\varphi)\text{Coordinates_old}.$ (D.29)

The apparent sign reversal arises because rotating the state and rotating the basis are inverse operations.

A financial article should state explicitly whether:

- the valuation state moves;
- the measurement frame moves;
- both move relative to a third frame.

D.9 Long–short convention

For a linear signed position:

$Z_{short} = -Z_{long}.$ (D.30)

If:

$Z_{long} = R + iQ,$ (D.31)

then:

$Z_{short} = -R - iQ.$ (D.32)

The marks are:

$V_{long} = R.$ (D.33)

$V_{short} = -R.$ (D.34)

The phase exposures are:

$\Delta_{\theta,long} = -Q.$ (D.35)

$\Delta_{\theta,short} = Q.$ (D.36)

This sign structure should not be transferred automatically to non-linear institutional relationships.

Appendix E — Matrix Algebra of the Complex Operator

E.1 Real representation of multiplication by i

Let:

$x = [R,Q]^T$. (E.1)

Multiplication of:

$Z = R + iQ$ (E.2)

by i gives:

$iZ = -Q + iR$. (E.3)

The corresponding real matrix is:

$\mathcal{J} = [0\ -1;\ 1\ 0]$. (E.4)

Thus:

$\mathcal{J}x = [-Q,R]^T$. (E.5)

E.2 Square of the operator

Calculate:

$\mathcal{J}^2 = [0\ -1;\ 1\ 0][0\ -1;\ 1\ 0]$. (E.6)

Therefore:

$\mathcal{J}^2 = [-1\ 0;\ 0\ -1]$. (E.7)

Hence:

$\mathcal{J}^2 = -I$. (E.8)

E.3 Higher powers

From:

$\mathcal{J}^2 = -I$, (E.9)

we obtain:

$\mathcal{J}^3 = -\mathcal{J}$. (E.10)

And:

$\mathcal{J}^4 = I$. (E.11)

More generally:

$\mathcal{J}^(4n) = I$. (E.12)

$\mathcal{J}^(4n+1) = \mathcal{J}$. (E.13)

$\mathcal{J}^{(4n+2)} = -I$. (E.14)

$\mathcal{J}^{(4n+3)} = -\mathcal{J}$. (E.15)

E.4 Eigenvalues

Solve:

$\det(\mathcal{J} - \lambda I) = 0$. (E.16)

Therefore:

$\det[-\lambda\ -1;\ 1\ -\lambda] = \lambda^2 + 1$. (E.17)

Hence:

$\lambda^2 = -1$. (E.18)

The eigenvalues are:

$\lambda_+ = i$. (E.19)

$\lambda_- = -i$. (E.20)

The operator has no nonzero real eigenvector.

This is expected: a real vector cannot remain on its own line after a 90° rotation.

E.5 Orthogonality

The transpose is:

$\mathcal{J}^T = [0\ 1;\ -1\ 0]$. (E.21)

Therefore:

$\mathcal{J}^T = -\mathcal{J}$. (E.22)

And:

$\mathcal{J}^T\mathcal{J} = I$. (E.23)

Thus $\mathcal{J}$ is orthogonal and skew-symmetric.

It preserves the Euclidean norm.

E.6 Finite rotation

Define:

$U(\varphi) = \exp(\varphi\mathcal{J})$. (E.24)

Because:

$\mathcal{J}^2 = -I$, (E.25)

the exponential becomes:

$U(\varphi) = I \cos \varphi + \mathcal{J} \sin \varphi$. (E.26)

Therefore:

$U(\varphi) = [\cos \varphi \; -\sin \varphi; \sin \varphi \; \cos \varphi]$. (E.27)

This satisfies:

$U(\varphi_1)U(\varphi_2) = U(\varphi_1 + \varphi_2)$. (E.28)

And:

$U(\varphi)^{-1} = U(-\varphi)$. (E.29)

E.7 Rotation of the state

Apply $U(\varphi)$:

$x' = U(\varphi)x$. (E.30)

Therefore:

$R' = R \cos \varphi - Q \sin \varphi$. (E.31)

$Q' = R \sin \varphi + Q \cos \varphi$. (E.32)

The norm is invariant:

$R'^2 + Q'^2 = R^2 + Q^2$. (E.33)

E.8 Generator equation

Let:

$x(\theta) = U(\theta)x(0)$. (E.34)

Differentiate:

$dx/d\theta = \mathcal{J}U(\theta)x(0)$. (E.35)

Therefore:

$dx/d\theta = \mathcal{J}x(\theta)$. (E.36)

This reproduces:

$dR/d\theta = -Q$. (E.37)

$dQ/d\theta = R$. (E.38)

E.9 Second-order equation

Differentiate again:

$d^2x/d\theta^2 = \mathcal{J}^2x$. (E.39)

Since:

$\mathcal{J}^2 = -I$, (E.40)

we obtain:

$d^2x/d\theta^2 = -x$. (E.41)

In components:

$d^2R/d\theta^2 = -R$. (E.42)

$d^2Q/d\theta^2 = -Q$. (E.43)

The valuation coordinates satisfy harmonic-oscillator equations in phase space.

This is a mathematical consequence of circular geometry, not a claim that the asset physically oscillates.

E.10 Complex differential equation

The complex state satisfies:

$dZ/d\theta = iZ$. (E.44)

The solution is:

$Z(\theta) = Z(0)\exp(i\theta)$. (E.45)

Differentiating twice:

$d^2Z/d\theta^2 = -Z$. (E.46)

This is the compact complex form of the matrix relations.

Appendix F — Finite-Rotation P&L and Taylor Expansion

F.1 Exact rotation

Begin with:

$Z = R + iQ$. (F.1)

Apply an active phase movement $\Delta\theta$:

$Z_{new} = \exp(i\Delta\theta)Z$. (F.2)

The new real value is:

$R_{new} = R \cos \Delta\theta - Q \sin \Delta\theta$. (F.3)

Therefore:

$\Delta R = R_{new} - R$. (F.4)

Hence:

$\Delta R = R(\cos \Delta\theta - 1) - Q \sin \Delta\theta$. (F.5)

F.2 Exact conjugate-coordinate change

The new conjugate coordinate is:

$Q_{new} = R \sin \Delta\theta + Q \cos \Delta\theta$. (F.6)

Therefore:

$\Delta Q = Q_{new} - Q$. (F.7)

Hence:

$\Delta Q = R \sin \Delta\theta + Q(\cos \Delta\theta - 1)$. (F.8)

F.3 Taylor series

Use:

$\sin \Delta\theta = \Delta\theta - (\Delta\theta)^3/6 + (\Delta\theta)^5/120 - O((\Delta\theta)^7)$. (F.9)

And:

$\cos \Delta\theta = 1 - (\Delta\theta)^2/2 + (\Delta\theta)^4/24 - (\Delta\theta)^6/720 + O((\Delta\theta)^8)$. (F.10)

Substitute into ΔR:

$\Delta R = -Q\Delta\theta - (R/2)(\Delta\theta)^2 + (Q/6)(\Delta\theta)^3 + (R/24)(\Delta\theta)^4 - (Q/120)(\Delta\theta)^5 - (R/720)(\Delta\theta)^6 + O((\Delta\theta)^7)$. (F.11)

Similarly:

$\Delta Q = R\Delta\theta - (Q/2)(\Delta\theta)^2 - (R/6)(\Delta\theta)^3 + (Q/24)(\Delta\theta)^4 + (R/120)(\Delta\theta)^5 - (Q/720)(\Delta\theta)^6 + O((\Delta\theta)^7)$. (F.12)

F.4 First-order approximation

For sufficiently small $\Delta\theta$:

$\Delta R \approx -Q\Delta\theta$. (F.13)

$\Delta Q \approx R\Delta\theta$. (F.14)

The approximation error begins at second order.

F.5 Second-order approximation

Retaining terms through second order:

$\Delta R \approx -Q\Delta\theta - (R/2)(\Delta\theta)^2$. (F.15)

$\Delta Q \approx R\Delta\theta - (Q/2)(\Delta\theta)^2$. (F.16)

This is the phase delta–curvature approximation.

F.6 Radial and angular finite movement

If the amplitude also changes from A to A + ΔA:

$Z_{new} = (A + \Delta A)\exp[i(\theta + \Delta\theta)]$. (F.17)

Relative to:

$Z = A \exp(i\theta)$, (F.18)

we have:

$Z_{new} = (1 + \Delta A/A)\exp(i\Delta\theta)Z$. (F.19)

The exact real value is:

$R_{new} = (1 + \Delta A/A)[R \cos \Delta\theta - Q \sin \Delta\theta]$. (F.20)

Therefore:

$\Delta R = (1 + \Delta A/A)[R \cos \Delta\theta - Q \sin \Delta\theta] - R$. (F.21)

To first order:

$\Delta R \approx (R/A)\Delta A - Q\Delta\theta$. (F.22)

F.7 P&L attribution residual

Given an observed change ΔR_{obs}, define the first-order explained change:

$\Delta R_{explained} = (R/A)\Delta A - Q\Delta\theta$. (F.23)

Define the residual:

$\varepsilon_R = \Delta R_{obs} - \Delta R_{explained}$. (F.24)

At second order:

$\Delta R_{explained,2} = (R/A)\Delta A - Q\Delta\theta - (R/2)(\Delta\theta)^2 - (Q/A)\Delta A\Delta\theta$. (F.25)

The residual should decline only if the geometry genuinely improves attribution.

Appendix G — Practical Bump-and-Revalue Implementation

G.1 General procedure

To estimate Q numerically:

1. declare the base model;
2. calculate A, R, θ, and Q;
3. select a mature financial input x;
4. apply symmetric bumps x + ε and x − ε;
5. recalculate R_+, R_−, θ_+, and θ_−;
6. estimate ∂R/∂θ.

The central estimator is:

∂R/∂θ ≈ (R_+ − R_−)/(θ_+ − θ_−). (G.1)

The model predicts:

(R_+ − R_−)/(θ_+ − θ_−) ≈ −Q. (G.2)

G.2 Required-return bump

Let:

r_+ = r + ε. (G.3)

r_− = r − ε. (G.4)

Then:

R_+ = CF_t/(1 + r_+)^t. (G.5)

R_− = CF_t/(1 + r_−)^t. (G.6)

Using fixed A:

θ_+ = arccos(R_+/A). (G.7)

θ_− = arccos(R_−/A). (G.8)

Estimate:

Q_est = −(R_+ − R_−)/(θ_+ − θ_−). (G.9)

G.3 Beta bump

Let:

$\beta_+ = \beta + \varepsilon$. (G.10)

$\beta_- = \beta - \varepsilon$. (G.11)

Then:

$r_+ = r_{base} + \beta_+ ERP$. (G.12)

$r_- = r_{base} + \beta_- ERP$. (G.13)

Proceed as in equations (G.5)–(G.9).

G.4 ERP bump

Let:

$ERP_+ = ERP + \varepsilon$. (G.14)

$ERP_- = ERP - \varepsilon$. (G.15)

Then:

$r_+ = r_{base} + \beta ERP_+$. (G.16)

$r_- = r_{base} + \beta ERP_-$. (G.17)

Again calculate $R_\pm$ and $\theta_\pm$.

G.5 Choice of bump size

The bump ε must balance:

- truncation error;
- floating-point error;
- model discontinuity;
- numerical instability near $Q = 0$.

A useful diagnostic is to repeat the estimate across several bump sizes:

$\varepsilon_1 > \varepsilon_2 > \varepsilon_3$. (G.18)

A stable implementation should show convergence:

$Q_{est}(\varepsilon_n) \to Q$ (G.19)

as ε_n decreases within the numerically reliable region.

G.6 Instability near $\theta = 0$

When:

$\theta \to 0$, (G.20)

we have:

$Q \to 0$. (G.21)

But:

$\partial\theta/\partial r = tR/[(1 + r)Q]$ (G.22)

can become large.

This means that the angular coordinate becomes sensitive near the baseline axis.

The identity:

$Qd\theta = D_rdr$ (G.23)

remains finite because the small Q and large $d\theta$ offset one another.

Numerical implementations should avoid dividing by an extremely small Q without stabilization.

G.7 Multi-period implementation

For term-specific cash flows:

$Q_t = -\partial R/\partial\theta_t$. (G.24)

A common rate bump generates:

$\Delta\theta_t = \theta_t(r + \varepsilon) - \theta_t(r - \varepsilon)$. (G.25)

The total finite-difference P&L is:

$\Delta R \approx -\Sigma_t\, Q_t\Delta\theta_t$. (G.26)

For a common artificial phase shock φ:

$R(\varphi) = \mathrm{Re}[\exp(i\varphi)Z]$. (G.27)

Then:

$Q_total \approx -[R(\varepsilon) - R(-\varepsilon)]/(2\varepsilon)$. (G.28)

G.8 Minimum audit output

Every implementation should report:

- CF_t;
- t;
- r_base;
- β;

- ERP;
- r;
- A;
- R;
- Q;
- θ;
- bump variable;
- bump size;
- finite-difference estimate;
- analytic estimate;
- approximation error;
- domain warnings.

This prevents Q from becoming an unauditable derived score.

Appendix H — Comparison with Duration, DV01, and Convexity

H.1 Why comparison is necessary

The CAPM phase construction produces:

$\Delta_\theta = \partial R/\partial\theta = -Q.$ (H.1)

Because this resembles ordinary financial sensitivity analysis, Q must be compared carefully with established measures.

The relevant question is not:

> Is Q another name for duration?

The correct question is:

> How does the same value response appear under the required-return coordinate r and the valuation-phase coordinate θ?

H.2 Required-return sensitivity

For a single future cash flow:

$R = CF_t/(1 + r)^t.$ (H.2)

The first derivative is:

$$\partial R/\partial r = -tR/(1 + r). \quad (H.3)$$

Define the positive dollar sensitivity:

$$D_r = -\partial R/\partial r. \quad (H.4)$$

Therefore:

$$D_r = tR/(1 + r). \quad (H.5)$$

This is the dollar-value response to a unit movement in the required return.

H.3 Modified-duration form

Define modified duration relative to r:

$$Dur_{mod} = -(1/R)(\partial R/\partial r). \quad (H.6)$$

For the single cash flow:

$$Dur_{mod} = t/(1 + r). \quad (H.7)$$

Therefore:

$$D_r = R \cdot Dur_{mod}. \quad (H.8)$$

The ordinary local P&L is:

$$dR = -R \cdot Dur_{mod} \cdot dr. \quad (H.9)$$

H.4 Phase sensitivity

The phase representation gives:

$$dR = -Qd\theta. \quad (H.10)$$

Therefore:

$$Q = D_r(dr/d\theta). \quad (H.11)$$

Equivalently:

$$Q = R \cdot Dur_{mod} \cdot (dr/d\theta). \quad (H.12)$$

Using:

$$d\theta/dr = tR/[(1 + r)Q], \quad (H.13)$$

we recover:

$$dr/d\theta = (1 + r)Q/(tR). \quad (H.14)$$

Thus equation (H.12) reduces identically to Q.

H.5 Q is not ordinary duration

Duration has units of time or inverse rate, depending on convention.

Q has units of currency.

Therefore:

$Q \neq Dur_mod.$ (H.15)

Nor is Q generally equal to dollar duration:

$Q \neq D_r.$ (H.16)

Their relation is coordinate-dependent:

$Qd\theta = D_rdr.$ (H.17)

Q is the exposure coefficient under θ.

D_r is the exposure coefficient under r.

H.6 DV01 comparison

DV01 measures the approximate change in value for a one-basis-point movement in yield or required return.

Let:

$1\ bp = 0.0001.$ (H.18)

Then:

$DV01 \approx D_r \times 0.0001.$ (H.19)

The corresponding phase movement is:

$d\theta_1bp = (d\theta/dr)0.0001.$ (H.20)

Therefore:

$DV01 = Qd\theta_1bp.$ (H.21)

This is an exact local coordinate translation.

Q is not DV01.

Q becomes a one-basis-point P&L only after multiplication by the phase movement induced by one basis point.

H.7 Convexity in the required-return coordinate

Differentiate equation (H.3):

$\partial^2 R/\partial r^2 = t(t + 1)R/(1 + r)^2.$ (H.22)

This quantity is positive when:

$R > 0$. (H.23)

The ordinary second-order approximation is:

$\Delta R \approx -D_r \Delta r + \frac{1}{2} C_r (\Delta r)^2$. (H.24)

Where:

$C_r = \partial^2 R / \partial r^2$. (H.25)

Thus:

$C_r = t(t + 1)R/(1 + r)^2$. (H.26)

H.8 Curvature in the phase coordinate

Under the phase coordinate:

$R = A \cos \theta$. (H.27)

Therefore:

$\partial^2 R / \partial \theta^2 = -R$. (H.28)

The phase expansion is:

$\Delta R \approx -Q\Delta\theta - \frac{1}{2} R(\Delta\theta)^2$. (H.29)

The second derivative is negative in the first quadrant.

This does not contradict positive rate convexity.

The two derivatives are taken with respect to different coordinates.

H.9 Chain rule for second derivatives

For a nonlinear coordinate transformation $\theta = \theta(r)$:

$\partial^2 R / \partial r^2 = (\partial^2 R / \partial \theta^2)(\partial \theta / \partial r)^2 + (\partial R / \partial \theta)(\partial^2 \theta / \partial r^2)$. (H.30)

Substitute:

$\partial^2 R / \partial \theta^2 = -R$, (H.31)

and:

$\partial R / \partial \theta = -Q$. (H.32)

Then:

$\partial^2 R / \partial r^2 = -R(\partial \theta / \partial r)^2 - Q(\partial^2 \theta / \partial r^2)$. (H.33)

The positive rate convexity arises from the complete nonlinear transformation, including the second derivative of θ with respect to r.

Therefore:

Phase Curvature ≠ Rate Convexity. (H.34)

H.10 Why phase curvature may still matter

The phase coordinate compresses the valuation relation into a circular orbit:

$R^2 + Q^2 = A^2$. (H.35)

Within that coordinate:

$\partial^2 R/\partial\theta^2 = -R$ (H.36)

has an unusually simple form.

This simplicity may support:

- common-phase scenario construction;
- closed-form higher-order expansions;
- normalized comparison across filters;
- operator-based risk decomposition.

The empirical question is whether that simplicity improves financial analysis.

H.11 Duration table

Quantity	Definition	Unit	Interpretation
Dur_mod	$-(1/R)\partial R/\partial r$	inverse rate or time-like	proportional sensitivity to r
D_r	$-\partial R/\partial r$	currency per unit rate	dollar required-return exposure
DV01	D_r × 0.0001	currency	P&L for one-basis-point move
Q	$-\partial R/\partial\theta$	currency per radian	dollar phase exposure
−R	$\partial^2 R/\partial\theta^2$	currency per radian²	phase curvature

The measures are related but not interchangeable.

H.12 Multi-period duration comparison

For multiple cash flows:

R = Σ_t CF_t/(1 + r)^t. (H.37)

The required-return sensitivity is:

D_r = Σ_t tR_t/(1 + r). (H.38)

The term-specific phase exposures are:

Q_t = $-\partial R/\partial\theta$_t. (H.39)

A common rate movement produces:

$$dR = -\Sigma_t Q_t d\theta_t. \quad (H.40)$$

Where:

$$d\theta_t = tR_t dr/[(1 + r)Q_t]. \quad (H.41)$$

Substituting gives:

$$dR = -\Sigma_t [tR_t/(1 + r)]dr. \quad (H.42)$$

Thus the phase term structure reproduces ordinary multi-period duration exactly under the corresponding maturity-specific phase movements.

H.13 Conditions for practical advantage

Phase exposure may add practical value if it:

1. creates more stable normalized comparisons across claims;
2. improves decomposition across multiple filters;
3. supports a useful common-phase stress scenario;
4. exposes term-structure concentration hidden by one duration number;
5. produces interpretable residuals after conventional sensitivity has been applied.

If none of these advantages appears:

Q should remain a derived reporting coordinate rather than a replacement for duration.

Appendix I — Comparison with Option Delta, Gamma, Vega, and Rho

I.1 Why option Greeks are the closest methodological comparison

The identity:

$$\Delta_\theta = \partial R/\partial\theta = -Q \quad (I.1)$$

places Q inside the general family of local sensitivity measures.

Option Greeks are mature examples of this method.

For a derivative value:

$$V = V(S,\sigma,r,T,K,\ldots), \quad (I.2)$$

the principal Greeks include:

$$\text{Delta} = \partial V/\partial S. \quad (I.3)$$

Gamma = $\partial^2 V/\partial S^2$. (I.4)

Vega = $\partial V/\partial \sigma$. (I.5)

Rho = $\partial V/\partial r$. (I.6)

Theta_option = $\partial V/\partial T$. (I.7)

The word theta in option pricing refers to calendar-time decay and must not be confused with valuation phase θ.

I.2 Phase Delta is a coordinate sensitivity

For a completed derivative valuation state:

$Z_D = R_D + iQ_D = A_D \exp(i\theta_D)$, (I.8)

define:

Phase Delta_D = $\partial R_D/\partial \theta_D$. (I.9)

Therefore:

Phase Delta_D = $-Q_D$. (I.10)

This is a sensitivity to the derived valuation coordinate θ_D.

It is not a replacement for Delta, Gamma, Vega, Rho, or calendar Theta.

I.3 Chain-rule relation to established Greeks

Suppose:

$\theta_D = \theta_D(S,\sigma,r,T,K,\ldots)$. (I.11)

Then:

$d\theta_D = (\partial\theta_D/\partial S)dS + (\partial\theta_D/\partial\sigma)d\sigma + (\partial\theta_D/\partial r)dr + (\partial\theta_D/\partial T)dT + \cdots$. (I.12)

At fixed amplitude:

$dR_D = -Q_D d\theta_D$. (I.13)

Therefore:

$dR_D = -Q_D[(\partial\theta_D/\partial S)dS + (\partial\theta_D/\partial\sigma)d\sigma + (\partial\theta_D/\partial r)dr + \cdots]$. (I.14)

Comparing coefficients gives:

Delta = $-Q_D(\partial\theta_D/\partial S)$ + radial terms. (I.15)

Vega = $-Q_D(\partial\theta_D/\partial\sigma)$ + radial terms. (I.16)

Rho = $-Q_D(\partial\theta_D/\partial r)$ + radial terms. (I.17)

The phrase radial terms matters because A_D may also vary with the underlying risk factors.

I.4 Full derivative decomposition

Because:

$R_D = A_D \cos \theta_D$, (I.18)

the derivative with respect to any risk factor x is:

$\partial R_D/\partial x = (R_D/A_D)(\partial A_D/\partial x) - Q_D(\partial \theta_D/\partial x)$. (I.19)

This gives:

Traditional Greek_x = Radial Contribution_x + Angular Contribution_x. (I.20)

Where:

Radial Contribution_x = $(R_D/A_D)(\partial A_D/\partial x)$. (I.21)

Angular Contribution_x = $-Q_D(\partial \theta_D/\partial x)$. (I.22)

This decomposition may be more useful than treating Q as an additional independent Greek.

I.5 Option Delta

For the underlying price S:

Delta = $\partial R_D/\partial S$. (I.23)

Therefore:

Delta = $(R_D/A_D)(\partial A_D/\partial S) - Q_D(\partial \theta_D/\partial S)$. (I.24)

The first term represents movement in derivative amplitude.

The second represents movement in valuation orientation.

A conventional Delta hedge may neutralize the net result without separately neutralizing both components.

I.6 Option Gamma

Differentiate equation (I.24) again.

The full Gamma contains:

- second derivative of A_D;
- second derivative of θ_D;
- cross terms;
- phase curvature $-R_D$.

Schematically:

Gamma = Radial Curvature + Angular Curvature + Radial–Angular Cross Terms. (I.25)

The angular curvature contribution contains:

$-R_D(\partial\theta_D/\partial S)^2 - Q_D(\partial^2\theta_D/\partial S^2)$. (I.26)

Therefore the simple identity:

$\partial^2 R_D/\partial\theta_D^2 = -R_D$ (I.27)

is only one component of ordinary option Gamma.

I.7 Vega

For volatility σ:

Vega = $\partial R_D/\partial\sigma$. (I.28)

The phase decomposition is:

Vega = $(R_D/A_D)(\partial A_D/\partial\sigma) - Q_D(\partial\theta_D/\partial\sigma)$. (I.29)

An option may gain value from volatility because:

- the amplitude A_D rises;
- the phase rotates toward a more admitted orientation;
- both occur.

The sign and interpretation depend on the declared baseline construction.

I.8 Rho

For the interest-rate coordinate r:

Rho = $\partial R_D/\partial r$. (I.30)

Therefore:

Rho = $(R_D/A_D)(\partial A_D/\partial r) - Q_D(\partial\theta_D/\partial r)$. (I.31)

If A_D is held fixed under the selected rate bump:

Rho = $-Q_D(\partial\theta_D/\partial r)$. (I.32)

This resembles the CAPM bridge equation.

I.9 Calendar Theta versus valuation phase θ

Option Theta is conventionally:

$\text{Theta}_{\text{option}} = \partial V/\partial T$. (I.33)

The valuation phase is:

θ_D = arccos(R_D/A_D). (I.34)

The symbols must not be confused.

A change in calendar maturity may produce:

dθ_D = (∂θ_D/∂T)dT. (I.35)

But:

θ_D ≠ T. (I.36)

A clear article should use:

Theta_option

for the Greek and:

θ_D

for valuation phase.

I.10 Hedge interpretation

Consider a hedge portfolio:

Π = R_D − ΔR_U. (I.37)

Its phase exposure is:

Q_Π = Q_D − ΔQ_U. (I.38)

A Delta-neutral portfolio satisfies:

∂Π/∂S = 0. (I.39)

It does not necessarily satisfy:

Q_Π = 0. (I.40)

Nor does:

Q_Π = 0 (I.41)

necessarily imply Delta neutrality.

The two conditions neutralize different coordinates.

This difference may reveal residual structure in apparently hedged portfolios.

I.11 Phase hedge

A pure phase hedge would select hedge weights h_j such that:

$Q_D + \Sigma_j h_j Q_j = 0$. (I.42)

Under a common phase movement $d\varphi$:

$dR_{portfolio} = -[Q_D + \Sigma_j h_j Q_j]d\varphi$. (I.43)

Therefore:

$dR_{portfolio} = 0$ (I.44)

to first order under that declared common phase shock.

This hedge is meaningful only if the instruments share a compatible phase protocol.

I.12 Empirical test against Greeks

A derivative implementation should compare:

1. conventional Greek attribution;
2. radial–angular attribution;
3. residual after each attribution;
4. stability under large moves;
5. hedge performance.

The complex decomposition is justified only if it improves at least one of these tasks.

Appendix J — Multi-Period Phase-Exposure Term Structure

J.1 Cash-flow decomposition

For horizons $t = 1,\ldots,T$:

$Z_t = A_t \exp(i\theta_t)$. (J.1)

The aggregate state is:

$Z = \Sigma_t Z_t$. (J.2)

Therefore:

$R = \Sigma_t R_t$. (J.3)

$Q = \Sigma_t Q_t$. (J.4)

Where:

$R_t = A_t \cos \theta_t$. (J.5)

$Q_t = A_t \sin \theta_t$. (J.6)

J.2 Phase-exposure vector

Define:

$\mathbf{Q} = (Q_1, Q_2, \ldots, Q_T)^T$. (J.7)

Define:

$d\boldsymbol{\theta} = (d\theta_1, d\theta_2, \ldots, d\theta_T)^T$. (J.8)

At fixed amplitudes:

$dR = -\mathbf{Q}^T d\boldsymbol{\theta}$. (J.9)

This is the term-structure phase P&L equation.

J.3 Common phase shock

If:

$d\theta_t = d\varphi$ (J.10)

for every t, then:

$dR = -[\Sigma_t Q_t]d\varphi$. (J.11)

Therefore:

$Q_{common} = \Sigma_t Q_t$. (J.12)

The aggregate Q is the correct exposure to a common phase rotation.

J.4 Parallel rate shock

For a common required-return movement dr:

$d\theta_t = tR_t dr/[(1 + r)Q_t]$. (J.13)

Therefore:

$dR = -\Sigma_t [tR_t/(1 + r)]dr$. (J.14)

This reproduces ordinary dollar duration.

The phase movements are maturity-dependent even when the rate shock is parallel.

J.5 Key-rate phase exposures

Suppose the yield or required-return curve is divided into key maturities.

Let x_j denote the jth key-rate coordinate.

The value sensitivity is:

$$\partial R/\partial x_j = -\Sigma_t Q_t(\partial \theta_t/\partial x_j) + \Sigma_t (R_t/A_t)(\partial A_t/\partial x_j). \quad (J.15)$$

If amplitudes are held fixed:

$$\partial R/\partial x_j = -\Sigma_t Q_t(\partial \theta_t/\partial x_j). \quad (J.16)$$

This is the phase equivalent of key-rate duration.

J.6 Q-weighted maturity

Define:

$$T_Q = [\Sigma_t tQ_t]/[\Sigma_t Q_t]. \quad (J.17)$$

This quantity measures the average horizon of oriented phase exposure.

It is valid when:

$$\Sigma_t Q_t \neq 0. \quad (J.18)$$

It should not be interpreted as ordinary duration.

It answers:

> At what average horizon is the aggregate phase exposure concentrated?

J.7 Absolute Q-weighted maturity

Signed Q values may cancel.

To measure gross concentration, define:

$$T_{|Q|} = [\Sigma_t t|Q_t|]/[\Sigma_t |Q_t|]. \quad (J.19)$$

This distinguishes:

- net phase exposure;
- gross phase exposure.

A portfolio may have:

$$\Sigma_t Q_t \approx 0, \quad (J.20)$$

while:

$$\Sigma_t |Q_t| \quad (J.21)$$

is large.

Such a portfolio is phase-neutral only under a common shock, not exposure-free.

J.8 Phase dispersion

Define normalized phase weights:

$w_t^Q = |Q_t|/\Sigma_s |Q_s|$. (J.22)

Then define phase-horizon dispersion:

$Var_Q(t) = \Sigma_t w_t^Q(t - T_{|Q|})^2$. (J.23)

A high value indicates phase exposure distributed across distant maturities.

A low value indicates concentrated exposure.

J.9 Phase coherence across cash flows

Define the amplitude-weighted complex coherence:

$C = [\Sigma_t A_t \exp(i\theta_t)]/[\Sigma_t A_t]$. (J.24)

Write:

$C = \rho \exp(i\Phi)$. (J.25)

Where:

$\rho = |C|$. (J.26)

$\Phi = \arg(C)$. (J.27)

The coherence satisfies:

$0 \le \rho \le 1$. (J.28)

If all phases align:

$\rho = 1$. (J.29)

If phases disperse strongly:

$\rho < 1$. (J.30)

This is a classical vector-coherence measure.

J.10 Effective aggregate phase

If:

$Z \neq 0$, (J.31)

define:

$\Phi = \arg(Z)$. (J.32)

Then:

$R = |Z| \cos \Phi$. (J.33)

$Q = |Z| \sin \Phi$. (J.34)

The aggregate state therefore has an effective phase even when the individual θ_t differ.

But:

$|Z| \neq \Sigma_t A_t$ (J.35)

in general.

The effective aggregate amplitude includes cancellation among differently oriented cash-flow vectors.

J.11 Term-structure residual

A simplified one-phase approximation may write:

$Z_{approx} = A_{total} \exp(i\theta_{eff})$. (J.36)

Where:

$A_{total} = \Sigma_t A_t$. (J.37)

Define the residual:

$\varepsilon_{term} = Z - Z_{approx}$. (J.38)

A large ε_{term} indicates that one global phase fails to represent the cash-flow term structure adequately.

J.12 Scenario matrix

For multiple phase shocks:

$d\boldsymbol{\theta} = Bdf$. (J.39)

Where:

df = vector of underlying risk-factor movements; (J.40)

B = phase-loading matrix. (J.41)

Then:

$dR = -\mathbf{Q}^{T}Bdf$. (J.42)

Define the factor exposure vector:

$\mathbf{E}_f = B^{T}\mathbf{Q}$. (J.43)

Therefore:

$dR = -\mathbf{E}_f{}^{T}df$. (J.44)

This connects the phase term structure to conventional multi-factor sensitivity analysis.

J.13 Portfolio extension

For asset i and horizon t:

$Q_{i,t} = A_{i,t} \sin \theta_{i,t}$. (J.45)

For portfolio weights w_i:

$Q_{P,t} = \Sigma_i w_i Q_{i,t}$. (J.46)

The portfolio phase-exposure surface is:

$\{Q_{P,t}\}$. (J.47)

It may reveal maturity concentrations hidden by one aggregate portfolio Q.

J.14 Practical reporting table

A term-structure implementation should report:

Horizon	A_t	R_t	Q_t	θ_t	Rate sensitivity	Phase sensitivity
1	…	…	…	…	…	…
2	…	…	…	…	…	…
…	…	…	…	…	…	…

The report should also show:

- total R;
- total Q;
- gross |Q|;
- T_Q;
- $T_{|Q|}$;
- phase coherence ρ;
- effective phase Φ;
- term-structure residual.

Appendix K — Long–Short, Counterparty, and Stakeholder Conditions

K.1 The clean signed-position identity

For a claim with value R under a linear position map:

$V(n) = nR.$ (K.1)

The opposite position satisfies:

$V(-n) = -V(n).$ (K.2)

In complex form:

$Z(-n) = -Z(n).$ (K.3)

Therefore:

$Z_{short} = -Z_{long}.$ (K.4)

This is the cleanest realization of the half-turn:

$Z \to -Z.$ (K.5)

K.2 Conditions for exact long–short symmetry

Exact symmetry requires:

1. the same underlying claim;
2. the same notional magnitude;
3. the same valuation timestamp;
4. the same currency;
5. the same discount and funding protocol;
6. no bid–ask spread;
7. no transaction cost;
8. no counterparty asymmetry;
9. no tax asymmetry;
10. no nonlinear position constraint.

Under these conditions:

$R_{long} + R_{short} = 0.$ (K.6)

And:

$Q_{long} + Q_{short} = 0.$ (K.7)

K.3 Bilateral derivative symmetry

For a clean derivative contract between A and B:

$V_A = -V_B.$ (K.8)

If both parties use the same completed geometry:

$Z_A = -Z_B$. (K.9)

Therefore:

$R_A = -R_B$. (K.10)

$Q_A = -Q_B$. (K.11)

The phase deltas are:

$\Delta_{\theta,A} = -Q_A$. (K.12)

$\Delta_{\theta,B} = -Q_B = Q_A$. (K.13)

The bilateral system clears both mark and phase exposure.

K.4 Funding asymmetry breaks exact symmetry

Suppose party A discounts using r_A and party B uses r_B.

Then:

$R_A = CF_A/(1 + r_A)^t$. (K.14)

$R_B = CF_B/(1 + r_B)^t$. (K.15)

Even if:

$CF_B = -CF_A$, (K.16)

it may follow that:

$R_A + R_B \neq 0$. (K.17)

The difference may represent:

- funding valuation adjustment;
- credit valuation adjustment;
- debit valuation adjustment;
- collateral terms;
- liquidity basis.

The clean half-turn becomes an approximation rather than an exact bilateral identity.

K.5 Counterparty credit asymmetry

Let:

V_{clean} (K.18)

be the clean derivative value.

Then party A may report:

$V_A = V_{clean} - CVA_A + DVA_A + FVA_A + \cdots$. (K.19)

Party B may report a different adjusted value.

Therefore:

$V_A \neq -V_B$ (K.20)

in practice.

The frame residual is:

$\varepsilon_{AB} = Z_A + Z_B$. (K.21)

This residual may be economically meaningful rather than a mathematical error.

K.6 Clearinghouse and collateral frames

A centrally cleared derivative may have:

- clean mark;
- variation margin;
- initial margin;
- default-fund contribution;
- capital charge.

These are different financial readouts.

The opposite party's clean mark may be $-R$, while the collateral or capital quantities are not simply $\pm Q$.

Each requires its own protocol.

K.7 Shareholder and company

A shareholder's position is an asset.

The issuing company does not generally record the negative of the shareholder's market value as a matching liability.

Therefore:

$Z_{company} \neq -Z_{shareholder}$ (K.22)

as a general identity.

The relationship passes through:

- share capital;
- retained earnings;
- market capitalization;
- enterprise value;
- cost of equity;
- dividend obligations;
- governance rights.

A separate transformation is required.

K.8 Borrower and lender

For a simple loan with symmetric valuation:

R_lender = −R_borrower. (K.23)

But after incorporating:

- default risk;
- funding;
- collateral;
- regulatory capital;
- tax;
- liquidity;

the two reported states may differ.

The legal cash-flow symmetry remains.

The institutional valuation symmetry may not.

K.9 Insurer and insured

An insurance premium is a cost to the insured and income to the insurer.

But the two full states are not simple negatives because:

- the insurer pools many risks;
- reserves are nonlinear;
- capital requirements apply;
- claims probabilities differ across information sets;

- administrative costs exist.

Therefore a premium payment may create opposite real cash flows without creating exactly opposite complete complex states.

K.10 Regulator and regulated institution

A regulator's capital charge is not the negative of the bank's asset value.

The regulator and bank occupy different measurement protocols.

The regulator may convert risk into:

- capital requirement;
- liquidity requirement;
- leverage limit;
- reporting obligation.

These are gate outputs, not opposite-position marks.

K.11 Employee and employer

A salary expense to the employer is income to the employee.

But:

$Z_{employer} \neq -Z_{employee}$ (K.24)

for the complete employment relation.

The two parties have different:

- opportunity costs;
- taxes;
- risk exposures;
- contractual rights;
- future benefits.

Cash-flow opposition does not imply full-state negation.

K.12 Stakeholder interpretation rule

A stakeholder can be interpreted as the $-R$ side only when:

1. the stakeholder holds the exact opposite signed claim;
2. the same valuation protocol applies;

3. the state space is linear;

4. no asymmetric adjustment intervenes.

Otherwise:

−R is a mathematical opposite orientation, not an identified stakeholder value.

K.13 Why −Q is even more restricted

Because −Q is a sensitivity readout:

$$-Q = \partial R/\partial \theta. \quad (K.25)$$

It should not be assigned to another stakeholder merely because that stakeholder may suffer when the asset performs well or badly.

A stakeholder's exposure must be derived from that stakeholder's own value function:

$$\Delta_{\theta,j} = \partial R_j/\partial \theta. \quad (K.26)$$

Only if:

$$R_j = -R \quad (K.27)$$

under the same phase coordinate does:

$$\Delta_{\theta,j} = Q. \quad (K.28)$$

follow.

K.14 Stakeholder mapping table

Relationship	Is −R naturally the other side?	Is +Q naturally the other side's phase exposure?
Clean long–short security	Yes	Yes
Clean bilateral derivative	Yes	Yes
Funded bilateral derivative	Not exactly	Not exactly
Shareholder–company	No	No
Lender–borrower	Sometimes approximately	Protocol-dependent
Insurer–insured	No for full state	No for full state
Bank–regulator	No	No
Buyer–seller after settlement	Cash-flow opposition only	Not automatically

---

K.15 Final stakeholder conclusion

The closed operator cycle remains exact mathematically:

R → −Q → −R → Q → R. (K.29)

Its strongest direct financial realization is:

Long Mark
→ Long Phase Exposure
→ Short Mark
→ Short Phase Exposure
→ Long Mark. (K.30)

Other stakeholder interpretations require separate institutional mappings and should not be inferred from the operator alone.

Appendix L — Conditions for Opportunity-Cost Interpretation

L.1 Why the distinction matters

The baseline valuation A and the admitted valuation R generate the scalar difference:

$H = A - R.$ (L.1)

The complex completion generates:

$Q = \sqrt{(A^2 - R^2)}.$ (L.2)

Because both H and Q arise from the same pair (A,R), it is tempting to describe Q as hidden opportunity cost.

That interpretation is valid only under additional conditions.

The geometry alone establishes:

$Q = dH/d\theta.$ (L.3)

It does not establish:

H = Economic Opportunity Cost. (L.4)

L.2 Standard opportunity cost

Let the available alternatives be:

$\mathcal{A} = \{a_1, a_2, \ldots, a_n\}.$ (L.5)

Let:

$V(a_j)$ (L.6)

be the value assigned to alternative a_j under a common decision protocol.

Suppose alternative a_c is chosen.

Define the best foregone alternative:

$a_* = \arg\max_{a_j \neq a_c} V(a_j)$. (L.7)

Then the opportunity cost is:

$OC = V(a_*) - V(a_c)$. (L.8)

This definition requires:

- a feasible choice set;
- mutually exclusive or resource-constrained alternatives;
- a common valuation protocol;
- an identified chosen alternative;
- an identified best rejected alternative.

Without these elements, the phrase opportunity cost is incomplete.

L.3 CAPM filter haircut

In the CAPM construction:

$A = CF_t/(1 + r_{base})^t$. (L.9)

$R = CF_t/(1 + r_{CAPM})^t$. (L.10)

The same cash flow is valued under two discount protocols.

Therefore:

$H_{CAPM} = A - R$ (L.11)

is initially:

> the reduction in admitted present value created by replacing the baseline discount rule with the CAPM required-return rule.

This is a filter haircut.

It is not necessarily the difference between two available investments.

L.4 Sufficient conditions for opportunity-cost equivalence

The equality:

$OC = A - R$ (L.12)

is justified when all of the following hold.

Condition 1 — A represents a feasible alternative

The baseline valuation must correspond to an alternative actually available to the decision

maker.

Condition 2 — A is the best foregone alternative

The alternative represented by A must dominate all other unchosen alternatives under the declared decision protocol.

Condition 3 — R represents the chosen alternative

The CAPM-admitted state must correspond to the investment or action actually selected.

Condition 4 — Both values use compatible units and timing

The two alternatives must be compared at the same date, currency, and scale.

Condition 5 — External effects are included consistently

Taxes, liquidity, transaction costs, option value, and institutional constraints must not be included in one value but excluded from the other without adjustment.

When these conditions hold:

$$OC = A - R = H. \quad (L.13)$$

L.5 Marginal opportunity cost

Under the opportunity-cost conditions:

$$OC(\theta) = A - R(\theta). \quad (L.14)$$

At fixed A:

$$dOC/d\theta = -dR/d\theta. \quad (L.15)$$

Since:

$$dR/d\theta = -Q, \quad (L.16)$$

we obtain:

$$dOC/d\theta = Q. \quad (L.17)$$

Thus:

> Q is the marginal increase in opportunity cost generated by one additional unit of valuation phase.

This is an exact local interpretation under the stated conditions.

L.6 Accumulated opportunity cost

Since:

$$OC(0) = 0, \quad (L.18)$$

we have:

$OC(\theta) = \int_0^{\theta} Q(\varphi)d\varphi$. (L.19)

Therefore Q is not the accumulated opportunity cost.

It is the phase derivative of that accumulated amount.

The distinction is:

$Q(\theta)$ = Marginal Opportunity-Cost Pressure. (L.20)

$OC(\theta)$ = Accumulated Opportunity Cost. (L.21)

L.7 Opportunity cost under changing A

If the value of the best foregone alternative changes:

$A = A(\tau)$, (L.22)

then:

$OC = A - R$. (L.23)

The total differential is:

$dOC = dA - dR$. (L.24)

Using:

$dR = (R/A)dA - Qd\theta$, (L.25)

we obtain:

$dOC = [1 - R/A]dA + Qd\theta$. (L.26)

Since:

$1 - R/A = H/A$, (L.27)

we have:

$dOC = (H/A)dA + Qd\theta$. (L.28)

Thus opportunity cost changes through both:

- movement in the alternative value A;
- movement in valuation phase θ.

Q captures only the angular component.

L.8 Multiple alternatives

Suppose several alternatives are available:

$A_j = V(a_j)$. (L.29)

The best alternative may change through time:

$a_*(\tau) = \arg\max_j A_j(\tau)$. (L.30)

Then the opportunity-cost baseline may switch discontinuously.

The opportunity cost becomes:

$OC(\tau) = \max_j A_j(\tau) - R(\tau)$. (L.31)

At a switching boundary:

$A_j = A_k$, (L.32)

the derivative of OC may be non-smooth.

A single circular phase model may not capture this switching geometry without a gate selecting the active alternative.

L.9 Opportunity-set gate

Define the alternative-selection gate:

$G_{alt}(\{A_j\}) = a_*$. (L.33)

The active amplitude is:

$A_{active} = V[G_{alt}(\{A_j\})]$. (L.34)

Then:

$OC = A_{active} - R$. (L.35)

The gate may change when:

$A_j > A_{active}$ (L.36)

for another feasible alternative.

Thus opportunity cost is partly a gate-dependent construct.

L.10 Shadow price interpretation

In constrained optimization, a shadow price measures the marginal value of relaxing a constraint.

Suppose the decision problem is:

$\max_x V(x)$ (L.37)

subject to:

$g(x) \le c$. (L.38)

Let λ be the Lagrange multiplier associated with the constraint.

Then:

$\lambda = \partial V^*/\partial c$. (L.39)

Q is not automatically λ.

Q is:

$Q = \partial H/\partial \theta$. (L.40)

A shadow-price interpretation requires θ to parameterize the relevant constraint and H to represent the associated value loss.

If:

$\theta = \theta(c)$, (L.41)

then:

$\partial H/\partial c = Q(\partial \theta/\partial c)$. (L.42)

Only this transformed quantity is the shadow value per unit constraint.

L.11 Capital-allocation interpretation

Suppose A represents the value available under unconstrained capital allocation, while R represents value admitted after a capital charge.

Then:

$H = A - R$ (L.43)

may be interpreted as the scalar value cost of the capital filter.

And:

$Q = dH/d\theta$ (L.44)

is the marginal pressure of that capital filter with respect to its phase coordinate.

This may resemble opportunity cost, but the correct term depends on whether the unconstrained allocation was genuinely feasible.

L.12 Liquidity interpretation

Suppose:

A = value under frictionless liquidation. (L.45)

R = value under current market liquidity. (L.46)

Then:

H = liquidity haircut. (L.47)

And:

Q = marginal liquidity-haircut pressure per unit liquidity phase. (L.48)

Calling Q opportunity cost would be inappropriate unless the frictionless liquidation alternative were actually available.

L.13 Regulatory interpretation

Suppose:

A = economic value before regulatory admissibility. (L.49)

R = value admitted for regulatory capital purposes. (L.50)

Then:

H = regulatory admission haircut. (L.51)

And:

Q = marginal regulatory-filter pressure. (L.52)

This is not necessarily opportunity cost.

It may instead represent institutional exclusion.

L.14 Opportunity-cost classification table

Construction of A	Meaning of A – R	Meaning of Q
Risk-free discount value of same cash flow	CAPM filter haircut	marginal CAPM haircut pressure
Best feasible foregone investment	opportunity cost	marginal opportunity-cost pressure
Frictionless liquidation value	liquidity haircut	marginal liquidity pressure
Unconstrained capital value	capital-filter haircut	marginal capital-filter pressure
Regulatory pre-filter value	regulatory haircut	marginal regulatory pressure
Accounting pre-recognition value	recognition gap	marginal recognition pressure

The formula remains the same.

The economic meaning changes with the protocol.

L.15 Final opportunity-cost rule

The correct rule is:

Q = Marginal Opportunity Cost (L.53)

only when:

A = Best Feasible Foregone Alternative Value. (L.54)

Otherwise:

Q = Marginal Declared-Filter Haircut Pressure. (L.55)

This more general description should be treated as the default.

Appendix M — Measurement, Gate, Trace, and Residual

M.1 Four-layer architecture

A complete valuation runtime contains four logically separate layers:

1. state and measurement;
2. economic movement;
3. gate and commitment;
4. trace and residual.

The architecture is:

State Z
→ Measurement M
→ Exposure
→ Movement
→ Economic Consequence
→ Gate G
→ Ledger Trace L + Residual ε. (M.1)

Each arrow represents a different operation.

M.2 State layer

The minimal completed state is:

Z = R + iQ. (M.2)

More generally:

X = (A,R,Q,θ,P,E). (M.3)

Where:

P = valuation protocol. (M.4)

E = relevant environment. (M.5)

The state exists before a particular readout is selected.

M.3 Measurement layer

A measurement protocol m produces:

$y_m = M_m(X)$. (M.6)

For the complex plane:

$M_\varphi(Z) = \mathrm{Re}[\exp(i\varphi)Z]$. (M.7)

The mark measurement is:

$y_0 = R$. (M.8)

The conjugate-exposure measurement is:

$y_{\pi/2} = -Q$. (M.9)

Measurement changes what becomes visible to the observer.

It does not necessarily change the state.

M.4 Movement layer

An economic or valuation shock u changes the state:

$X' = F(X,u)$. (M.10)

For radial and angular movement:

$dR = Rg_A - Qd\theta$. (M.11)

$dQ = Qg_A + Rd\theta$. (M.12)

The economic consequence is:

$\Delta R_{econ} = R' - R$. (M.13)

Movement is distinct from measurement.

M.5 Gate layer

A gate determines whether the consequence becomes operative under a declared institutional protocol.

Let:

$g = G_P(X,\Delta R_{econ},L,\mathrm{Context})$. (M.14)

The gate output may be:

$g \in \{\text{Admit},\text{Partially Admit},\text{Defer},\text{Reject}\}$. (M.15)

A quantitative gate may return an admission fraction:

$0 \le \alpha \le 1$. (M.16)

Then:

$\Delta R_{admitted} = \alpha \Delta R_{econ}$. (M.17)

M.6 Ledger layer

The admitted consequence updates the ledger:

$L_{k+1} = L_k + \mathrm{Trace}(\Delta R_{admitted})$. (M.18)

The ledger may represent:

- accounting entries;
- legal obligations;
- settled cash flows;
- collateral transfers;
- capital consumption;
- margin records;
- regulatory status.

The ledger creates persistent historical consequence.

M.7 Residual layer

Define:

$\varepsilon_{gate} = \Delta R_{econ} - \Delta R_{admitted}$. (M.19)

The residual may remain:

- economically active;
- legally unresolved;
- unrecognized;
- unhedged;
- model-dependent.

The updated state may depend on both trace and residual:

$X_{k+1} = \mathscr{B}(X_k, L_{k+1}, \varepsilon_{gate}, u_k)$. (M.20)

M.8 Measurement does not collapse the full state by

necessity

In the passive readout model:

$M_0(Z) = R$ (M.21)

does not destroy Q.

Likewise:

$M_{\pi/2}(Z) = -Q$ (M.22)

does not erase R.

The complete state remains available to another compatible measurement.

This is unlike a literal destructive physical measurement.

The financial model therefore supports selective projection without requiring ontological state destruction.

M.9 Commitment can reduce the feasible state space

A gate event may nevertheless alter what states remain possible.

Before exercise, an option may have two future branches:

Exercise. (M.23)

Expire. (M.24)

After the exercise gate is crossed, the unchosen branch is no longer contractually available.

Thus commitment can produce an effective reduction of the feasible state space.

This is a legal and institutional closure, not necessarily a physical collapse.

M.10 Trace formation

A trace should satisfy at least three properties.

Persistence

The trace remains after the triggering event.

Accessibility

Authorized observers can read or verify it.

Consequence

The trace influences future rights, obligations, or decisions.

A temporary calculation that leaves no persistent consequence is not yet a full ledger trace.

M.11 Weak and strong gates

Weak gate

A weak gate changes reporting without materially changing future possibilities.

Example:

- an internal risk classification.

Strong gate

A strong gate changes rights, obligations, or feasible actions.

Examples:

- contract settlement;
- default declaration;
- collateral seizure;
- legal judgment;
- option exercise.

Strong gates create more irreversible financial worlds.

M.12 Soft and hard admission

A soft admission may use:

$0 < \alpha < 1$. (M.25)

Examples include:

- partial impairment;
- probabilistic reserve;
- staged capital charge.

A hard gate uses:

$\alpha \in \{0,1\}$. (M.26)

Examples include:

- exercise versus expiry;
- covenant passed versus breached;
- trade accepted versus rejected.

Financial systems commonly combine both types.

M.13 Gate hysteresis

A gate may use different thresholds for entry and exit.

Let:

$\theta_{enter} > \theta_{exit}$. (M.27)

Then the system exhibits hysteresis.

For example:

- a downgrade may occur at one threshold;
- an upgrade may require a stronger recovery threshold.

Hysteresis creates path dependence.

The current state cannot be inferred from the present θ alone without ledger history.

M.14 Gate-memory state

Introduce a gate-memory variable m_k:

$m_{(k+1)} = H(m_k, X_k, Event_k)$. (M.28)

Then the gate becomes:

$G_P(X_k, L_k, m_k)$. (M.29)

This allows the model to represent:

- prior breaches;
- probation periods;
- cumulative losses;
- repeated warnings;
- historical defaults.

A world with gate memory is more than a static threshold system.

M.15 Residual accumulation

Residual may accumulate:

$E_{(k+1)} = E_k + \varepsilon_k - Dissipation_k - Resolution_k$. (M.30)

Where E is a residual ledger.

This residual ledger may influence future gates:

$G_{(k+1)} = G(X_{(k+1)}, L_{(k+1)}, E_{(k+1)})$. (M.31)

Unresolved pressure can therefore build until a later threshold is crossed.

M.16 Residual dissipation

Not all residual becomes future crisis.

Residual may dissipate through:

- time;
- hedging;
- liquidity recovery;
- legal clarification;
- operational repair;
- cash-flow realization.

Represent dissipation as:

$D_k \geq 0$. (M.32)

Then:

$E_{(k+1)} = E_k + \varepsilon_k - D_k$. (M.33)

A residual model must account for both accumulation and release.

M.17 False ledger closure

False closure occurs when:

$\Delta R_{admitted} = \Delta R_{econ}$ (M.34)

is assumed merely because a ledger entry exists.

In practice:

$\varepsilon_{gate} \neq 0$ (M.35)

may remain after recognition.

Examples include:

- partial settlement;
- remaining litigation;
- future tax impact;

- ongoing liquidity pressure;
- reputational damage.

A ledger entry is evidence of commitment, not proof of total resolution.

M.18 Gate-dependent reality

Different institutions may apply different gates to the same economic movement.

For one state change ΔR:

G_accounting(ΔR) = Defer. (M.36)

G_margin(ΔR) = Admit. (M.37)

G_regulatory(ΔR) = Partially Admit. (M.38)

Thus one economic event can enter several institutional worlds differently.

The disagreement is not necessarily observational error.

It may reflect different constitutive rules.

M.19 Cross-ledger consistency

Let:

L_accounting, (M.39)

L_collateral, (M.40)

L_regulatory, (M.41)

and:

L_tax (M.42)

be distinct ledgers.

A reconciliation map should satisfy:

C(L_accounting,L_collateral,L_regulatory,L_tax) = Residual_Reconciliation. (M.43)

Large reconciliation residuals may indicate:

- inconsistent timing;
- hidden leverage;
- regulatory arbitrage;
- data mismatch;
- unresolved obligations.

M.20 Backreaction loop

A complete financial loop is:

Z_k
$\rightarrow M_k$
$\rightarrow \Delta R_k$
$\rightarrow G_k$
$\rightarrow L_{(k+1)} + \varepsilon_k$
$\rightarrow$ Behaviour_k
$\rightarrow$ Market Impact_k
$\rightarrow Z_{(k+1)}$. (M.44)

This loop explains why observation and reporting may become economically causal.

The causal force comes through behaviour and institutional action, not from complex notation alone.

M.21 Gate–trace proposition

Proposition M.1 — Gate–Trace Separation

A valuation exposure becomes ledgered financial history only when:

1. an actual state movement produces economic consequence;
2. a declared gate admits some or all of that consequence;
3. a persistent ledger records the admitted result. (M.45)

Therefore:

Measurement Rotation ≠ Economic Event. (M.46)

Economic Event ≠ Ledgered History. (M.47)

Ledgered History ≠ Exhaustive Resolution. (M.48)

M.22 Minimal audit record

Every gate event should record:

- pre-event state;
- measurement protocol;
- identified exposure;
- realized movement;
- economic consequence;
- gate rule;

- admitted amount;
- deferred amount;
- residual;
- ledger update;
- backreaction action;
- protocol version.

This is the operational bridge from complex valuation to auditable institutional use.

Appendix N — Quadrature, Analytic Signals, and the Hilbert-Transform Boundary

N.1 Purpose of the comparison

The CAPM complex state:

Z = R + iQ (N.1)

resembles an analytic-signal representation:

z(u) = x(u) + iy(u). (N.2)

In both cases, two real channels form one complex state.

The mathematical resemblance is useful.

But the method generating the second channel is different.

N.2 Analytic signal

For a real signal x(u), the analytic signal is often defined as:

z_a(u) = x(u) + i$\mathscr{H}$x. (N.3)

Where $\mathscr{H}$ denotes the Hilbert transform.

The Hilbert transform is:

$\mathscr{H}$x = (1/π)PV∫_−∞^∞ x(v)/(u − v) dv. (N.4)

Here PV denotes the Cauchy principal value.

The quadrature component depends on the entire signal domain.

N.3 Hilbert-transform quarter-turn

Under suitable function classes:

$\mathscr{H}^2[x] = -x$. (N.5)

Therefore the Hilbert transform behaves like a 90° phase-shift operator.

The sequence is:

$x \to \mathscr{H}[x] \to -x \to -\mathscr{H}[x] \to x$. (N.6)

This resembles the financial cycle:

$R \to -Q \to -R \to Q \to R$. (N.7)

The sign difference reflects convention.

N.4 CAPM construction of Q

The CAPM Q is defined by:

$Q = \sqrt{(A^2 - R^2)}$. (N.8)

Or:

$Q = A \sin \theta$. (N.9)

Where:

$\theta = \arccos(R/A)$. (N.10)

This construction is pointwise in the declared valuation state.

It does not require the full historical path of R.

Therefore:

$Q_{CAPM} \neq \mathscr{H}[R]$ (N.11)

as a general identity.

N.5 Two different meanings of quadrature

Signal quadrature

The second channel is derived from a real signal through a transform over its domain.

Valuation quadrature

The second channel is derived from a declared amplitude and projection relation.

The common structure is:

- orthogonal pairing;
- phase orientation;
- quarter-turn algebra;

- norm-preserving representation.

The generating protocols differ.

N.6 Instantaneous amplitude and phase

For an analytic signal:

$z_a(u) = a(u)\exp[i\varphi(u)]$. (N.12)

The instantaneous amplitude is:

$a(u) = |z_a(u)|$. (N.13)

The instantaneous phase is:

$\varphi(u) = \arg[z_a(u)]$. (N.14)

For the CAPM state:

$Z = A \exp(i\theta)$. (N.15)

A is the declared baseline valuation amplitude.

θ is the CAPM valuation phase.

The forms are identical.

Their semantics are not.

N.7 Signal frequency and financial phase velocity

In signal analysis:

$\omega(u) = d\varphi/du$. (N.16)

may be interpreted as instantaneous angular frequency.

In financial phase analysis:

$\omega_\theta(\tau) = d\theta/d\tau$ (N.17)

may be interpreted as valuation-phase velocity.

Then:

$dR/d\tau = -Q\omega_\theta + Rg_A$. (N.18)

This can be useful dynamically.

But ω_θ is not a physical oscillation frequency unless a specific periodic financial process has been established.

N.8 Calendar-time Hilbert approach

A separate research programme could construct:

$Z_H(t) = R(t) + i\mathcal{H}R$. (N.19)

This would create a signal-derived financial quadrature channel.

It would not generally equal the CAPM state:

$Z_{CAPM}(t) = R(t) + iQ_{CAPM}(t)$. (N.20)

The two imaginary coordinates answer different questions.

Q_CAPM

How much conjugate phase exposure is implied by the declared valuation filter?

$\mathcal{H}$[R]

What quadrature component is implied by the historical signal shape of R(t)?

N.9 Comparison residual

Define:

$\varepsilon_{HQ}(t) = \mathcal{H}R - Q_{CAPM}(t)$. (N.21)

This residual may reveal disagreement between:

- model-implied valuation orientation;
- history-implied signal orientation.

A large residual could indicate:

- trend;
- cycle;
- model misspecification;
- regime change;
- nonstationarity;
- inadequate baseline.

This is an empirical proposal, not an established identity.

N.10 Causality limitation

The standard Hilbert transform is nonlocal and may use future as well as past values in an offline calculation.

Therefore it is not automatically causal for real-time risk management.

A real-time implementation would require:

- causal filters;
- finite-window approximation;
- phase delay correction;
- robustness testing.

The CAPM Q does not have this specific noncausality problem because it is calculated from current declared valuation inputs.

N.11 Boundary effects

Finite-window Hilbert transforms can suffer from:

- edge distortion;
- leakage;
- trend sensitivity;
- window dependence.

Therefore a comparison between Q_CAPM and $\mathscr{H}[R]$ must report:

- sample window;
- detrending method;
- padding rule;
- filter order;
- endpoint handling.

Otherwise apparent phase relations may be artifacts.

N.12 Nonstationarity

Financial time series are often nonstationary.

The analytic phase of a signal containing:

- multiple trends;
- jumps;
- changing variance;
- regime shifts;

may be unstable or difficult to interpret.

The CAPM phase is also model-dependent but for different reasons.

Neither phase should be treated as self-validating.

N.13 Narrowband limitation

Instantaneous phase is most interpretable for reasonably narrowband oscillatory signals.

A broad financial series may not satisfy this condition.

The CAPM phase avoids the narrowband assumption because it is not derived as an oscillatory signal phase.

However, it also lacks the direct temporal-wave interpretation of analytic phase.

N.14 Classical phasor comparison

In AC engineering:

$Z_{AC} = R_{electrical} + iX.$ (N.22)

Where:

$R_{electrical}$ = resistance. (N.23)

X = reactance. (N.24)

The phase angle satisfies:

$\tan \varphi = X/R_{electrical}.$ (N.25)

In finance:

$Z_{fin} = R_{fin} + iQ.$ (N.26)

And:

$\tan \theta = Q/R_{fin}.$ (N.27)

The analogy is:

Real admitted channel ↔ in-phase channel. (N.28)

Conjugate pressure channel ↔ quadrature channel. (N.29)

But financial Q is not electrical reactance.

The equations share form, not material content.

N.15 Complex impedance and financial filtering

Electrical impedance relates voltage and current:

$V = Z_{AC} I.$ (N.30)

A financial filter may schematically relate baseline amplitude and admitted value:

R = Filter_P(A). (N.31)

The financial complex state does not yet establish an input–output law equivalent to impedance.

To claim a financial impedance, one would need:

- a declared input;
- a declared output;
- a transfer relation;
- frequency or phase dependence;
- empirical calibration.

The present Z is primarily a state representation, not a transfer function.

N.16 Transform theorem boundary

The Hilbert transform satisfies mathematical results tied to:

- linearity;
- Fourier representation;
- function-space conditions;
- integral kernels.

The CAPM quarter-turn operator satisfies:

$\mathcal{J}^2 = -I$ (N.32)

because the valuation state is declared on a two-dimensional Euclidean orbit.

The equality of operator squares does not make the operators identical.

N.17 Safe statement

The safe comparison is:

> CAPM Q and Hilbert quadrature instantiate the same abstract complex-structure grammar while being generated by different financial and mathematical protocols.

The unsafe statement is:

> CAPM risk is the Hilbert transform of price.

No such general result has been established.

N.18 Research comparison table

Feature	CAPM Q	Hilbert quadrature
Input	A, R, valuation protocol	full real signal x(u)
Formula	$\sqrt{(A^2 - R^2)}$	$\mathscr{H}[x]$
Pointwise?	Yes, after A and R are given	No, generally nonlocal
Depends on history?	Not necessarily	Yes
Requires oscillatory interpretation?	No	Often helpful
Quarter-turn algebra	Yes	Yes
Physical wave implication	No	No by itself
Primary use	valuation exposure	signal phase analysis

N.19 Hybrid research programme

A hybrid model may compare:

$\theta_CAPM(t) = \arccos[R(t)/A(t)]$. (N.33)

$\varphi_signal(t) = \arg[R(t) + i\mathscr{H}R]$. (N.34)

Define phase disagreement:

$\Delta\varphi(t) = \varphi_signal(t) - \theta_CAPM(t)$. (N.35)

Possible interpretations include:

- market trend running ahead of CAPM valuation;
- valuation filter tighter than price dynamics;
- regime transition;
- baseline failure.

These interpretations require empirical validation.

N.20 Final quadrature conclusion

The quadrature comparison strengthens the interpretation of i as a measurement-rotation operator.

It does not change the financial identity of Q:

$Q = -\partial R/\partial\theta$. (N.36)

Nor does it convert the CAPM model into a signal-processing theorem.

The correct hierarchy is:

Financial Construction First
→ Complex Structure Second
→ Cross-Domain Analogy Third. (N.37)

Appendix O — Empirical Test Matrix and Falsification

Harness

O.1 Purpose

The mathematical construction is exact under its declarations.

The empirical claims are not.

The purpose of this appendix is to separate:

- identities that cannot fail once defined;
- hypotheses that can and should fail under data.

O.2 Non-falsifiable internal identities

The following are construction identities:

$A^2 = R^2 + Q^2$. (O.1)

$R = A \cos \theta$. (O.2)

$Q = A \sin \theta$. (O.3)

$\partial R/\partial \theta = -Q$. (O.4)

$\mathcal{J}^2 = -I$. (O.5)

These identities test implementation correctness.

They do not establish economic usefulness.

O.3 Falsifiable claim classes

The framework produces at least six falsifiable claim classes.

Class 1 — Diagnostic improvement

Q improves interpretation of valuation differences.

Class 2 — Attribution improvement

Radial–angular decomposition reduces unexplained P&L residual.

Class 3 — Forecast improvement

Q or phase dynamics predict future events after conventional controls.

Class 4 — Hedge improvement

Phase hedging improves performance under declared shocks.

Class 5 — Frame-reconciliation improvement

Scale-and-phase translation reduces disagreement between valuation protocols.

Class 6 — Gate-event improvement

Q and residual measures improve prediction of commitment thresholds.

O.4 Test 1 — Static diagnostic value

Hypothesis

Assets with equal or similar R but different Q exhibit meaningfully different future sensitivity or institutional pressure.

Sample design

Select pairs satisfying:

$|R_i - R_j| < \delta_R$. (O.6)

Require:

$|Q_i - Q_j| > \delta_Q$. (O.7)

Outcomes

Compare:

- future drawdown;
- required-return sensitivity;
- liquidity stress;
- capital consumption;
- forecast error;
- gate-event incidence.

Rejection condition

If Q differences do not correspond to any reproducible outcome after controlling for A and standard risk variables, the diagnostic claim is weakened.

O.5 Test 2 — Dynamic P&L attribution

Benchmark model

$\Delta R_t = -D_{(r,t)}\Delta r_t + \varepsilon_{(r,t)}$. (O.8)

Phase model

$\Delta R_t = R_{tg}(A,t) - Q_t\Delta\theta_t + \varepsilon_{(\theta,t)}$. (O.9)

Evaluation

Compare:

$RMSE_r = \sqrt{E[\varepsilon_r^2]}$. (O.10)

$RMSE_\theta = \sqrt{E[\varepsilon_\theta^2]}$. (O.11)

Success condition

The phase model must reduce out-of-sample error or produce materially more interpretable residuals.

Rejection condition

If:

$RMSE_\theta \geq RMSE_r$ (O.12)

and residual interpretation does not improve, the phase attribution has no demonstrated advantage.

O.6 Test 3 — Multi-period term structure

Hypothesis

The Q-term structure contains useful information beyond aggregate duration.

Predictors

- Q_{total};
- T_Q;
- $T_{|Q|}$;
- phase dispersion;
- coherence ρ;
- term-structure residual.

Benchmarks

- Macaulay duration;
- modified duration;
- convexity;

- key-rate duration.

Outcomes

- yield-curve stress P&L;
- spread shock P&L;
- hedge error;
- liquidity need.

Rejection condition

If Q-term measures do not improve stress attribution or hedge design beyond established term-structure measures, the extension should be reduced.

O.7 Test 4 — Market-implied residual

Define:

$Q_{market} = \sqrt{(A^2 - P^2)}$. (O.13)

$Q_{model} = \sqrt{(A^2 - R_{model}^2)}$. (O.14)

Residual:

$\Delta Q = Q_{market} - Q_{model}$. (O.15)

Hypothesis

ΔQ predicts model correction, repricing, or gate events.

Controls

- valuation spread P − R_model;
- beta;
- volatility;
- liquidity;
- momentum;
- credit spread;
- analyst disagreement.

Rejection condition

If ΔQ contains no incremental information beyond P − R_model and controls, the complex residual adds no empirical value.

O.8 Test 5 — Phase hedge

Construct a portfolio satisfying:

$$\Sigma_i w_{iQ_i} = 0. \quad (O.16)$$

Hypothesis

The portfolio is locally protected against a declared common phase shock.

Comparison

Compare with:

- delta-neutral hedge;
- duration-neutral hedge;
- beta-neutral hedge;
- minimum-variance hedge.

Outcome

Measure realized P&L under scenarios approximating the declared common phase movement.

Rejection condition

If phase neutrality does not correspond to reduced P&L under the target scenario, the common-phase assumption is invalid or operationally weak.

O.9 Test 6 — Gate-event prediction

Let $Y_{(t+1)}$ be a binary gate event.

Examples:

- margin call;
- covenant breach;
- downgrade;
- impairment;
- exercise;
- default.

Estimate:

$$\Pr(Y_{(t+1)}=1) = F(Q_t,\Delta Q_t,\theta_t,\varepsilon_t,\text{controls}). \quad (O.17)$$

Benchmark

Use conventional variables only.

Success condition

Q-related variables improve:

- out-of-sample AUC;
- calibration;
- precision at operational thresholds;
- lead time.

Rejection condition

No stable out-of-sample improvement.

O.10 Test 7 — Frame transport

Estimate:

$$Z_b = \lambda_{ab} \exp(i\varphi_{ab}) Z_a + \varepsilon_{ab}. \quad (O.18)$$

Hypothesis

Scale and phase explain a meaningful portion of cross-frame valuation difference.

Metrics

- residual norm;
- stability of λ_{ab};
- stability of φ_{ab};
- loop consistency.

For three frames:

$$\varepsilon_{loop} = Z_{a'} - Z_a. \quad (O.19)$$

Rejection condition

If transformation parameters are unstable or residuals remain large, pure scale-and-rotation reconciliation is inadequate.

O.11 Test 8 — Protocol robustness

Repeat all tests using several baselines:

$$A^{(1)}, A^{(2)}, \ldots, A^{(m)}. \quad (O.20)$$

Measure rank stability:

$$\rho_{rank}[Q^{(j)},Q^{(k)}]. \quad (O.21)$$

Measure sign stability:

$$\Pr[\operatorname{sign}(\Delta Q^{(j)}) = \operatorname{sign}(\Delta Q^{(k)})]. \quad (O.22)$$

Rejection condition

If results disappear under small reasonable baseline changes, the claim is protocol-fragile.

O.12 Test 9 — Residual governance

Hypothesis

Explicit reporting of Q and residual improves decisions.

Experimental design

Compare two decision groups:

Group A receives R only.

Group B receives R, A, Q, θ, and residual diagnostics.

Outcomes

- forecast calibration;
- risk-limit quality;
- intervention timing;
- post-event surprise;
- explanation accuracy.

Rejection condition

If the richer report worsens or does not improve decisions, complexity may be harmful.

O.13 Test 10 — Hybrid signal-phase comparison

Compare:

$$\theta_{CAPM}(t) \quad (O.23)$$

with:

$$\varphi_{signal}(t). \quad (O.24)$$

Define:

$\Delta\varphi(t) = \varphi_{signal}(t) - \theta_{CAPM}(t)$. (O.25)

Hypothesis

Large phase disagreement precedes regime shifts or model failure.

Rejection condition

No stable predictive relation after correcting for trend, volatility, and look-ahead bias.

O.14 Minimum benchmark set

Every empirical paper should compare against at least:

- raw admitted value R;
- baseline difference A − R;
- beta;
- required return;
- dollar duration;
- convexity;
- volatility;
- model spread;
- simple nonlinear transformations of R/A.

This prevents Q from appearing useful merely because weak benchmarks were selected.

O.15 Avoiding mechanical multicollinearity

Since:

$Q = \sqrt{(A^2 - R^2)}$, (O.26)

including A, R, H, θ, and Q simultaneously can create severe deterministic dependence.

Researchers should use:

- orthogonalized variables;
- nested model comparison;
- regularization;
- principal components;
- carefully selected representations.

Statistical significance alone is unreliable when variables are mechanically related.

O.16 Out-of-sample protocol

A proper test should separate:

Training Window. (O.27)

Validation Window. (O.28)

Test Window. (O.29)

The following must be frozen before the test window:

- baseline rule;
- cash-flow estimator;
- CAPM inputs;
- transformation;
- gate definition;
- benchmark models;
- evaluation metric.

O.17 Failure taxonomy

A failed result should be classified.

Mathematical implementation failure

Equations do not close numerically.

Domain failure

R/A lies outside the declared real-phase domain.

Protocol failure

Results depend excessively on baseline choice.

Attribution failure

Phase decomposition does not reduce residual.

Forecast failure

Q has no incremental predictive value.

Decision failure

Additional information does not improve action.

Analogy failure

Cross-domain language adds confusion rather than insight.

O.18 Audit footer

Every empirical result should report:

- dataset;
- sampling frequency;
- valuation timestamp;
- cash-flow source;
- baseline definition;
- CAPM specification;
- metric;
- orientation;
- phase domain;
- gate definition;
- residual definition;
- benchmark set;
- training and test dates;
- code version;
- failure conditions.

O.19 Falsifiability principle

The central empirical rule is:

> The identities establish internal coherence; only independent tests can establish financial usefulness.

Or:

Mathematical Closure ≠ Empirical Validation. (O.30)

Appendix P — Limits of the Quantum Comparison

P.1 Purpose

The complex valuation framework shares mathematical and operational structures with parts of quantum theory.

Those similarities are valuable only when their limits are explicit.

This appendix provides the final boundary conditions.

P.2 Complex numbers are not uniquely quantum

The existence of:

$Z = R + iQ$ (P.1)

does not imply quantum mechanics.

Complex numbers occur in many classical systems.

Therefore:

Complex Representation ≠ Quantum Ontology. (P.2)

P.3 Phase is not automatically physical phase

The CAPM phase is:

$\theta = \arccos(R/A)$. (P.3)

It is derived from a valuation ratio.

It is not directly:

- electromagnetic phase;
- matter-wave phase;
- quantum action divided by $\hbar$;
- phase of a physical wavefunction.

Therefore:

Valuation Phase ≠ Quantum Phase. (P.4)

P.4 The norm is not a probability rule

The valuation norm is:

$A^2 = R^2 + Q^2$. (P.5)

A quantum probability rule may use:

$Pr(a) = |\psi_a|^2$. (P.6)

No derivation in this article connects A^2 to probability normalization.

Therefore:

Valuation Norm ≠ Born Probability. (P.7)

P.5 Measurement rotation is not quantum collapse

The financial measurement family is:

$M_{\varphi}(Z) = \mathrm{Re}[\exp(i\varphi)Z]$. (P.8)

This is a rotated readout.

A financial gate may then commit an event.

Neither operation proves fundamental wavefunction collapse.

Therefore:

Rotated Readout ≠ Physical Collapse. (P.9)

Ledger Gate ≠ Physical Collapse. (P.10)

P.6 Long–short duality is not particle–antiparticle duality

The financial relation:

$Z_{short} = -Z_{long}$ (P.11)

represents an opposite signed position.

It does not establish:

- antiparticles;
- charge conjugation;
- annihilation;
- relativistic field symmetry.

Therefore:

Short Position ≠ Antiparticle. (P.12)

P.7 Derivative coupling is not Bell entanglement

An underlying and derivative are contractually coupled.

Their values may update conditionally.

But no Bell test has been derived.

Therefore:

Contractual Nonseparability ≠ Bell Entanglement. (P.13)

P.8 Financial contextuality is not technical quantum contextuality

Financial values depend on:

- model;
- observer;
- regulation;
- funding;
- horizon;
- information.

This is broad contextual dependence.

Technical quantum contextuality is a stronger mathematical property.

Therefore:

Protocol Dependence ≠ Kochen–Specker Contextuality. (P.14)

P.9 Financial noncommutativity is not canonical quantization

Institutional operations may be path-dependent:

$\mathcal{O}_a\mathcal{O}_b \neq \mathcal{O}_b\mathcal{O}_a$. (P.15)

This does not imply:

$[\hat{x},\hat{p}] = i\hbar$. (P.16)

No physical constant analogous to $\hbar$ has been derived.

Therefore:

Operational Order Dependence ≠ Canonical Quantum Noncommutation. (P.17)

P.10 Irreversible ledger trace is not microscopic irreversibility

A financial ledger may create irreversible institutional history.

Quantum dynamics may be unitary while measurements create records.

The analogy concerns the emergence of trace.

It does not prove that the same physical mechanism operates.

Therefore:

Institutional Irreversibility ≠ Fundamental Physical Irreversibility. (P.18)

P.11 Internal time is not a new physical time dimension

The phase clock:

$d\tau_phase = |d\tilde{\Phi}|/\Omega$ (P.19)

indexes financial reorientation.

It does not imply an additional spacetime coordinate.

Therefore:

Financial Internal Time ≠ Physical Proper Time. (P.20)

P.12 Observer-bound does not mean conscious observer

A financial observer may be:

- a trader;
- an algorithm;
- a regulator;
- an accounting system;
- a legal protocol;
- a risk engine.

The framework does not require human consciousness.

Observer means:

> a bounded system possessing a measurement and admission protocol.

P.13 Residual is not hidden-variable proof

Q and other residuals preserve structure outside the admitted scalar.

This does not prove that quantum outcomes are generated by classical hidden variables.

Bell-type constraints remain relevant to physical hidden-variable theories.

Therefore:

Financial Residual ≠ Quantum Hidden Variable. (P.21)

P.14 Classical reproduction does not eliminate quantum mystery

If finance reproduces:

- phase;
- conjugate measurement;
- gates;
- traces;
- backreaction;

then those features alone are not sufficient markers of quantum physics.

But specifically quantum structures may still remain.

The correct inference is:

Generic Features Identified → Quantum Residue Narrowed. (P.22)

Not:

Generic Features Identified → Quantum Theory Explained Away. (P.23)

P.15 Functional homology

The strongest permitted relation is often functional homology.

Two systems are functionally homologous when corresponding structures play similar operational roles.

For example:

Financial Gate ↔ Outcome-Commitment Function. (P.24)

Financial Ledger ↔ Persistent Trace Function. (P.25)

Conjugate Exposure ↔ Orthogonal Readout Function. (P.26)

Functional homology does not imply common material composition.

P.16 Analogy-strength ladder

Quantum comparisons should be classified by strength.

Level 1 — Metaphorical resemblance

A loose explanatory analogy.

Level 2 — Formal similarity

The same equation form appears.

Level 3 — Structural isomorphism

Relations and transformations correspond under an explicit mapping.

Level 4 — Operational equivalence

The systems produce equivalent observable consequences under a declared protocol.

Level 5 — Physical identity

The systems instantiate the same physical mechanism.

The present article establishes mostly Levels 2 and 3 for selected structures.

It does not establish Level 5.

P.17 Quantum comparison checklist

Before making a quantum claim, ask:

1. What is the financial state space?
2. What is the physical state space?
3. What is the mapping between them?
4. What operator correspondence is claimed?
5. What observable consequence follows?
6. Could a classical model reproduce the same result?
7. What specifically quantum theorem is satisfied?
8. What would falsify the correspondence?

Without answers, the claim should remain metaphorical.

P.18 Residue table

Structure	Finance reproduces?	Specifically quantum?
Complex coordinates	Yes	No
Phase rotation	Yes	No
Conjugate readout	Yes	No
Protocol-dependent measurement	Yes	Not by itself
Gates and trace	Yes	No

Structure	Finance reproduces?	Specifically quantum?
Backreaction	Yes	No
Contractual composite states	Yes	No
Born rule	No	Yes
Bell violation	No	Yes
No-cloning theorem	No	Yes
$\hbar$-scaled commutation	No	Yes
Quantum statistics	No	Yes
Coherent amplitude interference	Not established	Yes in quantum form

---

P.19 Final boundary statement

The financial framework supports the following claim:

> Complex valuation shows that several apparently strange measurement structures can arise naturally from mature filtering, bounded observation, conjugate readout, selective commitment, and persistent trace.

It does not support the stronger claim:

> Quantum mechanics is merely finance in disguise.

Appendix Q — Compact Theorem and Proposition Register

Q.1 Theorem register

Theorem 1 — CAPM Projection Theorem

$R_t = A_t \cos \theta_t$. (Q.1)

Theorem 2 — Conjugate Risk Theorem

$\partial R/\partial\theta = -Q$. (Q.2)

$\partial Q/\partial\theta = R$. (Q.3)

Theorem 3 — Operator Closure Theorem

$\mathcal{J}^2 = -I$. (Q.4)

$\mathcal{J}^4 = I$. (Q.5)

Theorem 4 — Measurement-Cycle Theorem

$R \to -Q \to -R \to Q \to R$. (Q.6)

Theorem 5 — Common-Phase Aggregation Theorem

$dRe[\exp(i\varphi)Z]/d\varphi|_{(\varphi=0)} = -Q$. (Q.7)

Q.2 Proposition register

Proposition 1 — Static Non-Independence

Q contains no algebraically independent one-period information beyond declared A and R. (Q.8)

Proposition 2 — Quarter-Turn Uniqueness

The Euclidean length-preserving linear operator satisfying $\mathcal{J}^2 = -I$ is unique up to orientation. (Q.9)

Proposition 3 — Long–Short Duality

Long = (R,−Q). (Q.10)

Short = (−R,Q). (Q.11)

Proposition 4 — Marginal Haircut Relation

d(A − R)/dθ = Q. (Q.12)

Proposition 5 — Gate–Trace Separation

Measurement, economic movement, gate admission, and ledger trace are distinct operations. (Q.13)

Appendix R — One-Page Formula Summary

R.1 CAPM filter

r = r_base + βERP. (R.1)

A_t = CF_t/(1 + r_base)^t. (R.2)

R_t = CF_t/(1 + r)^t. (R.3)

R.2 Phase completion

cos θ_t = R_t/A_t. (R.4)

θ_t = arccos(R_t/A_t). (R.5)

Q_t = √(A_t² − R_t²). (R.6)

Z_t = R_t + iQ_t = A_t exp(iθ_t). (R.7)

R.3 Conjugate exposure

$\partial R/\partial\theta = -Q$. (R.8)

$\partial Q/\partial\theta = R$. (R.9)

$dR = -Q d\theta$. (R.10)

R.4 CAPM sensitivity bridge

$\partial R/\partial r = -tR/(1 + r)$. (R.11)

$\partial\theta/\partial r = tR/[(1 + r)Q]$. (R.12)

$Qd\theta = [tR/(1 + r)]dr$. (R.13)

R.5 Operator

$\mathcal{J} = [0\ -1;\ 1\ 0]$. (R.14)

$\mathcal{J}[R,Q]^T = [-Q,R]^T$. (R.15)

$\mathcal{J}^2 = -I$. (R.16)

$\mathcal{J}^4 = I$. (R.17)

R.6 Measurement family

$M_\varphi(Z) = \mathrm{Re}[\exp(i\varphi)Z]$. (R.18)

$M_\varphi(Z) = R\cos\varphi - Q\sin\varphi$. (R.19)

$R \to -Q \to -R \to Q \to R$. (R.20)

R.7 Exact finite P&L

$R_{new} = R\cos\Delta\theta - Q\sin\Delta\theta$. (R.21)

$\Delta R = R(\cos\Delta\theta - 1) - Q\sin\Delta\theta$. (R.22)

$\Delta R \approx -Q\Delta\theta - (R/2)(\Delta\theta)^2$. (R.23)

R.8 Radial–angular attribution

$dR = (R/A)dA - Qd\theta$. (R.24)

$dQ = (Q/A)dA + Rd\theta$. (R.25)

R.9 Haircut

$H = A - R$. (R.26)

$dH/d\theta = Q$. (R.27)

$H = \int_0^{\theta} Q(\varphi)d\varphi$. (R.28)

$H = Q^2/(A + R)$. (R.29)

R.10 Multi-period state

$Z = \Sigma_t A_t \exp(i\theta_t)$. (R.30)

$R = \Sigma_t A_t \cos \theta_t$. (R.31)

$Q = \Sigma_t A_t \sin \theta_t$. (R.32)

$dR = -\Sigma_t Q_t d\theta_t$. (R.33)

R.11 Runtime

Measurement
→ Exposure
→ Movement
→ P&L
→ Gate
→ Ledger + Residual
→ Backreaction. (R.34)

Closing Note

The article, including its main text and formal appendices, is now structurally complete.

Its central result remains:

$\partial R/\partial\theta = -Q$. (R.35)

Everything deeper follows from interpreting this identity carefully rather than treating Q as an unexplained hidden loss:

- Q is a conjugate exposure;
- A − R is the accumulated scalar haircut;
- i generates measurement rotation;
- i^2 reverses signed orientation;
- actual $\Delta\theta$ produces P&L;
- a gate determines commitment;
- a ledger creates trace;

- residual preserves what commitment does not exhaust.

Appendix S — Common Objections and Formal Replies

S.1 Objection: Q is only a relabelled function of A and R

The objection is:

$$Q = \sqrt{(A^2 - R^2)}. \quad (S.1)$$

Therefore Q contains no information not already present in A and R.

This objection is correct in the one-period static construction.

The article does not claim:

$$\text{Information}(Q) > \text{Information}(A,R). \quad (S.2)$$

The claim is instead that the completed representation introduces a useful transformation law:

$$\partial R/\partial \theta = -Q. \quad (S.3)$$

And:

$$\partial Q/\partial \theta = R. \quad (S.4)$$

Thus Q may provide:

- a sensitivity interpretation;
- a conjugate measurement channel;
- a closed rotation structure;
- a common phase-exposure aggregation rule;
- a term-structure representation.

The distinction is:

$$\text{New Static Information} \neq \text{New Operational Structure}. \quad (S.5)$$

A coordinate transformation can be operationally useful without creating new facts.

Duration, yield, log return, and principal components also reorganize information rather than manufacture it.

The empirical burden remains:

> Does the reorganization improve analysis, prediction, hedging, reconciliation, or governance?

S.2 Objection: θ is chosen precisely so that the result becomes circular

The phase is defined by:

$\theta = \arccos(R/A)$. (S.6)

Therefore:

$R = A \cos \theta$. (S.7)

Differentiating then gives:

$\partial R/\partial \theta = -A \sin \theta$. (S.8)

And defining:

$Q = A \sin \theta$ (S.9)

produces:

$\partial R/\partial \theta = -Q$. (S.10)

The objection is that this result is guaranteed by construction.

That is also correct.

The theorem establishes internal mathematical structure, not empirical superiority.

Its value lies in identifying a coordinate in which:

1. the admitted value and conjugate exposure form an orthogonal pair;
2. the sensitivity hierarchy closes after four derivatives;
3. the corresponding finite transformations form rotations;
4. multi-period exposure aggregates under common phase shocks.

The theorem should therefore be classified as:

Mathematical Reconstruction, not Empirical Discovery. (S.11)

Its financial importance must be established separately.

S.3 Objection: Any monotonic transformation can define a new Greek

Let:

$u = f(r)$. (S.12)

Then:

$\partial R/\partial u = (\partial R/\partial r)(dr/du)$. (S.13)

Therefore infinitely many exposure measures can be manufactured by choosing different coordinates u.

This is true.

The phase coordinate is privileged only relative to the declared geometry:

$A^2 = R^2 + Q^2$. (S.14)

$R = A \cos \theta$. (S.15)

$Q = A \sin \theta$. (S.16)

The resulting coordinate has several special properties:

- bounded angular domain for positive claims;
- norm preservation;
- orthogonal completion;
- a unique quarter-turn operator up to orientation;
- four-step closure;
- simple multi-period common-rotation aggregation.

The appropriate claim is not:

θ is the only legitimate risk coordinate. (S.17)

It is:

θ is the canonical angular coordinate of the declared Euclidean valuation completion. (S.18)

S.4 Objection: A is arbitrary

The baseline amplitude is:

$A = CF_t/(1 + r_{base})^t$. (S.19)

A different r_{base} produces a different A, θ, and Q.

Therefore Q is baseline-dependent.

This is a genuine model dependence.

The framework must not hide it.

The response has three parts.

First, many mature financial measures are protocol-dependent.

Examples include:

- fair value;
- cost of capital;
- credit spread;
- regulatory capital;

- impairment;
- funding valuation adjustment.

Second, the protocol dependence can be audited.

Researchers can report:

$Q = Q(r_{base})$. (S.20)

And test stability across admissible baselines.

Third, baseline sensitivity may itself be informative.

If small baseline changes produce large Q changes, the valuation geometry is fragile.

Thus:

Baseline Dependence = Limitation + Diagnostic Opportunity. (S.21)

S.5 Objection: Why preserve A as a Euclidean magnitude?

The model assumes:

$A^2 = R^2 + Q^2$. (S.22)

A critic may ask why A should be the hypotenuse rather than another function of R and Q.

There is no theorem proving that all finance must use this norm.

The Euclidean choice is adopted because it provides:

- the simplest norm-preserving completion;
- standard complex-number structure;
- unique quarter-turn rotation;
- transparent analytic derivations;
- direct comparison with quadrature systems.

A more general metric may be:

$A^2 = [R,Q]G[R,Q]^T$. (S.23)

Where G is positive definite.

The Euclidean model uses:

$G = I$. (S.24)

Therefore the current framework is best understood as:

> the minimal flat complex completion of a mature valuation filter.

It is a starting geometry, not the final possible geometry.

S.6 Objection: Q can be larger than the value haircut, so it exaggerates risk

The scalar haircut is:

$H = A - R.$ (S.25)

The conjugate coordinate is:

$Q = \sqrt{(A^2 - R^2)}.$ (S.26)

For small θ:

$H \approx A\theta^2/2.$ (S.27)

$Q \approx A\theta.$ (S.28)

Therefore:

$Q \gg H$ (S.29)

may occur when θ is small.

This does not mean Q exaggerates the scalar loss.

Q and H measure different orders.

H is an accumulated same-axis displacement.

Q is a first-order phase exposure.

The exact relationship is:

$H = Q^2/(A + R).$ (S.30)

Comparing their magnitudes without accounting for their roles is analogous to comparing:

- an option's price;
- its delta exposure.

Neither should be expected to equal the other.

S.7 Objection: Q has currency units but no independent cash-flow meaning

Q is measured in currency because:

$Q = -\partial R/\partial\theta.$ (S.31)

And θ is dimensionless.

Currency-valued sensitivity does not require Q to be a separate cash flow.

For example:

Dollar Duration = $-\partial R/\partial r$ (S.32)

also has monetary sensitivity units.

Likewise:

Option Delta × Unit Underlying Movement (S.33)

produces monetary P&L without Delta itself being a cash balance.

The proper classification is:

Q = Currency-Valued Exposure Coefficient. (S.34)

Not:

Q = Independently Payable Amount. (S.35)

S.8 Objection: The operator $\mathcal{J}$ is imported from complex analysis rather than finance

The operator is:

$\mathcal{J} = [0\ -1;\ 1\ 0]$. (S.36)

It is indeed the standard complex structure on a real two-dimensional Euclidean plane.

The article does not claim finance independently discovered this matrix.

The financial contribution is to show that the completed valuation coordinates satisfy:

$d/d\theta\ [R,Q]^T = \mathcal{J}[R,Q]^T$. (S.37)

Thus the standard mathematical operator acquires a finance-specific interpretation:

Mark → Signed Phase Exposure. (S.38)

Its mathematical origin is general.

Its financial semantics come from the CAPM construction.

S.9 Objection: The R → −Q → −R cycle confuses value and sensitivity

This objection targets the fact that R and Q are readouts of different types.

The article accepts that they are different financial quantities.

The cycle is not a chronological equality of economic objects.

It is a cycle of measurements generated by the same complex structure:

$M_0(Z) = R$. (S.39)

$M_{\pi/2}(Z) = -Q$. (S.40)

$M_{\pi}(Z) = -R$. (S.41)

$M_{3\pi/2}(Z) = Q$. (S.42)

The operator changes the readout type and orientation.

Therefore:

Measurement Closure ≠ Economic-Type Identity. (S.43)

The cycle is meaningful because the operator law connects the readouts, not because mark and exposure are interchangeable.

S.10 Objection: −R cannot always represent the short side

Correct.

The relation:

$Z_{short} = -Z_{long}$ (S.44)

is exact only in a signed linear position space under one common valuation protocol.

It may fail under:

- funding asymmetry;
- credit adjustments;
- tax differences;
- collateral asymmetry;
- nonlinear constraints;
- different institutional roles.

Therefore:

−R = Opposite Signed Mark (S.45)

is the general mathematical interpretation.

It becomes a specific short-position mark only when the required financial conditions hold.

S.11 Objection: The framework hides all unexplained phenomena inside “residual”

A theory can become unfalsifiable if every failure is called residual.

To prevent this, residuals must be:

- explicitly defined;
- quantitatively measured;
- separated by type;
- subjected to independent testing;
- prevented from silently re-entering the model.

The principal residuals are:

ε_{model} = Observed Value − Predicted Value. (S.46)

ε_{gate} = Economic Consequence − Ledgered Consequence. (S.47)

ε_{frame} = Observed Frame State − Transformed State. (S.48)

A valid model should state:

- the expected residual range;
- the conditions producing residual growth;
- the threshold at which the model is rejected.

Residual is not a license to avoid failure.

It is an auditable record of incompleteness.

S.12 Objection: The quantum comparison is unnecessary

For readers concerned only with finance, Part I is sufficient.

The identities:

$\partial R/\partial\theta = -Q$, (S.49)

$\mathcal{J}^2 = -I$, (S.50)

and:

$dR = -Q d\theta$ (S.51)

stand without quantum interpretation.

Part II serves a different purpose.

It asks which structures often associated with quantum observation also arise in mature classical valuation systems.

The quantum comparison is therefore optional for applying the financial model.

It is not required to justify Q.

S.13 Objection: “Conjugate risk” may be confused with canonical conjugate variables

In Hamiltonian mechanics, canonical conjugate variables such as position and momentum possess a specific symplectic structure.

The term conjugate in this article means:

> R and Q are paired by the declared complex-structure operator $\mathcal{J}$.

Specifically:

$\mathcal{J}R = -Q.$ (S.52)

$\mathcal{J}Q = R.$ (S.53)

The article does not claim that R and Q are canonical conjugates in Hamiltonian mechanics.

A terminology note should therefore state:

Complex Conjugate Pairing ≠ Canonical Symplectic Conjugacy. (S.54)

The phrase conjugate phase exposure is safer than canonical risk conjugate.

S.14 Objection: Phase Delta conflicts with option Theta terminology

The phase variable is denoted θ.

Option Theta is a conventional Greek measuring calendar-time sensitivity.

To avoid ambiguity, the article uses:

CAPM Phase Delta = $\partial R/\partial\theta$. (S.55)

Option Theta = $\partial V/\partial T$. (S.56)

The two should never be abbreviated into the same symbol without qualification.

S.15 Objection: The model works only when R ≤ A

The real first-quadrant construction requires:

$0 \le R/A \le 1.$ (S.57)

If:

$R > A,$ (S.58)

then:

$Q = \sqrt{(A^2 - R^2)}$ (S.59)

is not real.

This is not merely a numerical inconvenience.

It indicates that at least one of the following has occurred:

- the baseline A is too low;
- the market includes optionality absent from A;
- the Euclidean positive-claim completion is unsuitable;
- a wider signed or hyperbolic geometry is needed;
- the observation lies outside the model domain.

The correct response is not to force Q into the real plane.

The domain failure should be reported.

S.16 Objection: Negative cash flows break the construction

For:

$CF_t < 0$, (S.60)

both A and R may be negative under ordinary discounting.

The principal phase construction based on:

$0 \le R/A \le 1$ (S.61)

can still produce a positive ratio when A and R share sign, but the signed orientation requires careful treatment.

A liability may be represented by:

$Z_{liability} = -|A|\exp(i\theta)$. (S.62)

Or by assigning:

$A_{abs} = |CF_t|/(1 + r_{base})^t$ (S.63)

and storing the position sign separately.

The article's main derivation is intentionally restricted to positive claims before extending to signed positions.

S.17 Objection: A common phase shock has no conventional market meaning

A common phase shock:

$d\theta_t = d\varphi$ (S.64)

across horizons is not generally the same as a parallel yield shock.

It is a constructed stress scenario.

Its practical value depends on whether it corresponds to a recognizable cross-asset or cross-horizon reorientation.

Possible uses include:

- common risk-aversion shock;
- common filter-tightening shock;
- common admission-pressure scenario.

If no observable event approximates such a movement, the common-phase scenario may remain a mathematical stress test rather than a market factor.

S.18 Objection: The framework is over-engineered for routine valuation

For routine valuation, the scalar CAPM result may be sufficient.

The complex completion is most relevant when analysts need:

- sensitivity decomposition;
- multiple valuation frames;
- multi-period phase exposure;
- gate and residual analysis;
- derivative composite-state modelling;
- cross-domain measurement research.

The framework should not be imposed where a scalar present value already answers the operational question.

The principle is:

Use the Smallest Structure That Preserves the Required Distinctions. (S.65)

Appendix T — Implementation Blueprint for a CAPM Complex Valuation Lab

T.1 Objective

A practical implementation should allow users to change:

- cash flow CF_t;
- horizon t;

- beta β;
- baseline rate r_base;
- equity risk premium ERP.

It should display immediately:

- r_CAPM;
- A;
- R;
- Q;
- θ;
- scalar haircut H;
- Phase Delta −Q;
- required-return exposure D_r.

The implementation should also visualize:

- the complex valuation state;
- the measurement cycle;
- phase sensitivity;
- finite phase P&L;
- parameter surfaces;
- multi-period term structure.

T.2 Core calculation sequence

The application should calculate:

r = r_base + βERP. (T.1)

A = CF_t/(1 + r_base)^t. (T.2)

R = CF_t/(1 + r)^t. (T.3)

c = R/A. (T.4)

θ = arccos(c). (T.5)

Q = √(A² − R²). (T.6)

H = A − R. (T.7)

D_r = tR/(1 + r). (T.8)

$\Delta_\theta = -Q$. (T.9)

T.3 Domain validation

Before calculating θ and Q, validate:

$CF_t > 0$. (T.10)

$t \geq 0$. (T.11)

$1 + r_{base} > 0$. (T.12)

$1 + r > 0$. (T.13)

$0 \leq R/A \leq 1$. (T.14)

If:

$R/A > 1$, (T.15)

display:

> The declared baseline is below the admitted value. The real first-quadrant complex completion is not available under this protocol.

If:

$R/A < 0$, (T.16)

display:

> The claim or position requires a signed-state extension.

T.4 Numerical stabilization

Because floating-point error may produce:

$A^2 - R^2 \approx -\varepsilon$ (T.17)

near $\theta = 0$, calculate:

$Q = \sqrt{\max(A^2 - R^2, 0)}$. (T.18)

But only clamp small numerical errors.

A materially negative value must trigger a domain warning.

Define a tolerance:

$\varepsilon_{num} = \kappa \cdot \max(A^2, R^2, 1)$. (T.19)

If:

$-\varepsilon_{num} \leq A^2 - R^2 < 0$, (T.20)

set $Q = 0$.

If:

$A^2 - R^2 < -\varepsilon_{num}$, (T.21)

report domain failure.

T.5 Value boxes

The principal dashboard may contain eight value boxes.

Box 1 — CAPM Required Return

$r = r_{base} + \beta ERP$. (T.22)

Box 2 — Baseline Amplitude

$A = CF_t/(1 + r_{base})^t$. (T.23)

Box 3 — Admitted Value

$R = CF_t/(1 + r)^t$. (T.24)

Box 4 — Conjugate Coordinate

$Q = \sqrt{(A^2 - R^2)}$. (T.25)

Box 5 — Valuation Phase

$\theta = \arccos(R/A)$. (T.26)

Box 6 — CAPM Haircut

$H = A - R$. (T.27)

Box 7 — Phase Delta

$\Delta_\theta = -Q$. (T.28)

Box 8 — Required-Return Exposure

$D_r = tR/(1 + r)$. (T.29)

T.6 Complex-plane chart

Plot the state:

$Z = (R,Q)$. (T.30)

The chart should show:

- horizontal R-axis;

- vertical Q-axis;
- radius A;
- phase angle θ;
- vector from origin to Z;
- tangent vector [−Q,R];
- projection from Z to R-axis.

The tangent vector may be normalized:

v_tangent = [−Q,R]/A. (T.31)

The radial vector is:

v_radial = [R,Q]/A. (T.32)

Their inner product should display:

v_radial · v_tangent = 0. (T.33)

T.7 Measurement-cycle chart

Display four nodes:

Node 0 = R. (T.34)

Node 1 = −Q. (T.35)

Node 2 = −R. (T.36)

Node 3 = Q. (T.37)

Connect them in the sequence:

R → −Q → −R → Q → R. (T.38)

Each node should include its financial label:

- long mark;
- long phase exposure;
- opposite mark;
- opposite phase exposure.

A warning should state:

> These are rotated readouts, not a historical price sequence.

T.8 Parameter-response charts

The lab should plot the following as functions of beta:

$R(\beta)$. (T.39)

$Q(\beta)$. (T.40)

$\theta(\beta)$. (T.41)

$H(\beta)$. (T.42)

And as functions of horizon:

$R(t)$. (T.43)

$Q(t)$. (T.44)

$\theta(t)$. (T.45)

The user should be able to compare:

- scalar haircut growth;
- conjugate exposure growth;
- phase accumulation.

T.9 P&L scenario chart

For a selected phase shock $\Delta\theta$, calculate:

$R_{new} = R\cos\Delta\theta - Q\sin\Delta\theta$. (T.46)

Exact P&L:

$\Delta R_{exact} = R(\cos\Delta\theta - 1) - Q\sin\Delta\theta$. (T.47)

First-order P&L:

$\Delta R_1 = -Q\Delta\theta$. (T.48)

Second-order P&L:

$\Delta R_2 = -Q\Delta\theta - (R/2)(\Delta\theta)^2$. (T.49)

Plot the three values across a range of $\Delta\theta$.

This makes approximation error visible.

T.10 Required-return bump verification

For a selected Δr:

$R_{bumped} = CF_t/(1 + r + \Delta r)^t$. (T.50)

Exact rate P&L:

$\Delta R_{rate} = R_{bumped} - R$. (T.51)

Calculate the corresponding phase:

$\theta_{bumped} = \arccos(R_{bumped}/A)$. (T.52)

Therefore:

$\Delta\theta = \theta_{bumped} - \theta$. (T.53)

The phase P&L is:

$\Delta R_{phase} = R\cos\Delta\theta - Q\sin\Delta\theta - R$. (T.54)

The tool should verify:

$\Delta R_{rate} = \Delta R_{phase}$ (T.55)

within numerical tolerance.

This is an important consistency test.

T.11 Derivative-cycle display

Display:

$\partial^0 R/\partial\theta^0 = R$. (T.56)

$\partial R/\partial\theta = -Q$. (T.57)

$\partial^2 R/\partial\theta^2 = -R$. (T.58)

$\partial^3 R/\partial\theta^3 = Q$. (T.59)

$\partial^4 R/\partial\theta^4 = R$. (T.60)

A rotating indicator may show the quarter-turn movement between derivative orders.

T.12 Multi-period mode

Allow the user to enter cash flows:

$CF_1, CF_2, \ldots, CF_T$. (T.61)

For each horizon calculate:

A_t, R_t, Q_t, θ_t. (T.62)

Display a table containing:

t CF_t A_t R_t Q_t θ_t

Calculate:

$Z_{total} = \Sigma_t Z_t$. (T.63)

$R_{total} = \Sigma_t R_t$. (T.64)

$Q_{total} = \Sigma_t Q_t$. (T.65)

$A_{effective} = |Z_{total}|$. (T.66)

$\Phi_{effective} = \arg(Z_{total})$. (T.67)

Also report:

$\Sigma_t A_t - A_{effective}$ (T.68)

as the vector-alignment gap.

T.13 Term-structure charts

Plot:

Q_t against t. (T.69)

θ_t against t. (T.70)

R_t against t. (T.71)

The lab should report:

$T_Q = [\Sigma_t tQ_t]/[\Sigma_t Q_t]$. (T.72)

$T_{|Q|} = [\Sigma_t t|Q_t|]/[\Sigma_t |Q_t|]$. (T.73)

Phase coherence:

$\rho = |\Sigma_t A_t \exp(i\theta_t)|/\Sigma_t A_t$. (T.74)

T.14 Long–short mode

Display:

$Z_{long} = R + iQ$. (T.75)

$Z_{short} = -R - iQ$. (T.76)

Show:

Long Mark = R. (T.77)

Long Phase Delta = −Q. (T.78)

Short Mark = −R. (T.79)

Short Phase Delta = Q. (T.80)

The interface should state that exact opposition assumes one shared valuation protocol and a linear signed position space.

T.15 Gate-and-ledger simulator

A simplified gate may use a phase threshold θ_G.

Define:

Gate = Admit if $\theta \geq \theta_G$. (T.81)

Otherwise:

Gate = Defer. (T.82)

For a movement from θ_0 to θ_1, calculate:

$\Delta R_{economic} = A \cos \theta_1 - A \cos \theta_0$. (T.83)

If admitted:

$\Delta R_{ledger} = \Delta R_{economic}$. (T.84)

If deferred:

$\Delta R_{ledger} = 0$. (T.85)

Residual:

$\varepsilon_{gate} = \Delta R_{economic} - \Delta R_{ledger}$. (T.86)

This simulator should be described as an institutional example, not a standard CAPM rule.

T.16 Audit panel

The audit panel should display:

- formula version;
- protocol ID;
- input timestamp;
- units;
- baseline definition;
- phase domain;
- orientation;
- numerical tolerance;
- active warnings;
- residuals.

The user should be able to export the complete protocol together with results.

T.17 Recommended warning messages

Warning 1 — Static Dependence

> Q is algebraically derived from A and R in this static model. It does not constitute an independent market observation.

Warning 2 — Exposure Is Not Loss

> Q is a phase-exposure magnitude. Economic P&L requires an actual phase movement.

Warning 3 — Baseline Dependence

> Changing r_base changes A, θ, and Q.

Warning 4 — Domain Failure

> R exceeds A under the current protocol. The real Euclidean completion is not valid.

Warning 5 — Quantum Boundary

> Complex coordinates and phase rotation do not imply literal quantum behaviour.

T.18 Minimal pseudocode

The computational kernel is:

Input CF,t,β,r_base,ERP. (T.87)

r ← r_base + βERP. (T.88)

A ← CF/(1 + r_base)^t. (T.89)

R ← CF/(1 + r)^t. (T.90)

c ← R/A. (T.91)

Validate 0 ≤ c ≤ 1. (T.92)

θ ← arccos(c). (T.93)

Q ← √max(A² − R²,0). (T.94)

H ← A − R. (T.95)

PhaseDelta ← −Q. (T.96)

RateExposure ← tR/(1 + r). (T.97)

Return all values and warnings. (T.98)

Appendix U — Research Programme and Development

Stages

U.1 Stage 0 — Mathematical verification

The first stage verifies only internal correctness.

Required tests include:

$A^2 = R^2 + Q^2$. (U.1)

$R = A \cos \theta$. (U.2)

$Q = A \sin \theta$. (U.3)

$\partial R/\partial \theta = -Q$. (U.4)

$\partial Q/\partial \theta = R$. (U.5)

$\mathcal{J}^2 = -I$. (U.6)

$U(\varphi_1)U(\varphi_2) = U(\varphi_1 + \varphi_2)$. (U.7)

This stage establishes implementation integrity.

It does not establish financial usefulness.

U.2 Stage 1 — Finance-language validation

The framework should be presented to traditional finance scholars without quantum language.

Questions include:

- Is the definition of A financially acceptable?
- Is Q correctly interpreted as an exposure?
- Is Phase Delta useful or redundant?
- Are the sign conventions intuitive?
- Does the haircut relation improve explanation?
- Does the long–short table prevent category errors?

The objective is terminological and conceptual stability.

U.3 Stage 2 — Comparison with mature sensitivities

The next stage compares Q with:

- dollar duration;
- DV01;

- convexity;
- beta exposure;
- option Greeks;
- key-rate duration;
- factor sensitivities.

The framework must identify where it is:

- equivalent;
- a reparameterization;
- a useful decomposition;
- genuinely different.

No novelty should be claimed where exact equivalence already exists.

U.4 Stage 3 — Multi-period valuation tests

Research questions include:

1. Does the Q-term structure improve scenario attribution?
2. Does phase coherence identify unusual cash-flow structures?
3. Does T_Q reveal concentration missed by duration?
4. Do common phase shocks correspond to meaningful market episodes?
5. Does the term residual identify model failures?

The benchmark must include conventional term-structure risk measures.

U.5 Stage 4 — Market-implied completion

Given a declared A and market price P:

$\theta_{market} = \arccos(P/A)$. (U.8)

$Q_{market} = \sqrt{(A^2 - P^2)}$. (U.9)

Compare with:

θ_{model}, (U.10)

and:

Q_{model}. (U.11)

Define:

$\Delta\theta_{residual} = \theta_{market} - \theta_{model}$. (U.12)

$\Delta Q_{residual} = Q_{market} - Q_{model}$. (U.13)

Research should determine whether these residuals predict:

- valuation correction;
- liquidity stress;
- regime change;
- gate events;
- model breakdown.

U.6 Stage 5 — Dynamic radial–angular attribution

The central model is:

$\Delta R_t = R_{tg_{(A,t)}} - Q_t\Delta\theta_t + \varepsilon_t$. (U.14)

Questions include:

- Can g_A be estimated independently?
- Can $\Delta\theta$ be specified ex ante?
- Does the residual decline?
- Are radial and angular shocks economically interpretable?
- Does the decomposition remain stable across regimes?

This stage is essential because static identities alone cannot establish usefulness.

U.7 Stage 6 — Portfolio and hedge tests

Construct portfolios neutral under different criteria:

Beta Neutral. (U.15)

Duration Neutral. (U.16)

Delta Neutral. (U.17)

Phase Neutral. (U.18)

Compare their performance under:

- rate shocks;
- volatility shocks;
- funding shocks;

- common phase shocks;
- gate events.

The objective is not to show that phase neutrality dominates every hedge.

It is to identify which scenario class it actually addresses.

U.8 Stage 7 — Relative-frame reconciliation

Select several real valuation frames:

- analyst DCF;
- market price;
- accounting value;
- regulatory value;
- funding-adjusted value.

Estimate:

$Z_b \approx \lambda_{ab} \exp(i\varphi_{ab}) Z_a$. (U.19)

Measure:

$\varepsilon_{ab} = Z_b - \lambda_{ab} \exp(i\varphi_{ab}) Z_a$. (U.20)

Study:

- parameter stability;
- loop consistency;
- residual sources;
- gate differences;
- institutional asymmetry.

This stage tests whether phase is useful for reconciliation rather than only for one-model sensitivity.

U.9 Stage 8 — Gate-event models

Select explicit financial gates:

- margin call;
- exercise;
- covenant breach;
- impairment;

- default;
- capital threshold.

Model:

$$\Pr(\text{Gate}_{t+1}=1) = F(Q_t,\theta_t,\Delta\theta_t,\varepsilon_t,\text{controls}). \quad (U.21)$$

The framework gains practical significance if phase and residual measures improve:

- lead time;
- calibration;
- operational decision quality.

U.10 Stage 9 — Derivative composite-state experiments

For an underlying and derivative:

$$\Psi_{UD} = \Psi(Z_U,Z_D,C). \quad (U.22)$$

Research may compare:

- local underlying readout;
- local derivative readout;
- joint hedge state;
- collateral ledger;
- exercise gate.

Questions include:

- Does phase decomposition reveal residual hedge risk?
- Can joint Q-exposure predict hedge backreaction?
- Does gate structure explain abrupt contractual transitions?
- Can local states reconstruct the joint system?

This stage should remain within classical derivative finance before any quantum analogy is introduced.

U.11 Stage 10 — Quadrature comparison

Compare:

$$Q_{CAPM}(t) \quad (U.23)$$

with:

$$Q_H(t) = \mathscr{H}R. \quad (U.24)$$

And:

$\theta_{CAPM}(t)$ (U.25)

with:

$\varphi_{signal}(t)$. (U.26)

Possible research questions include:

- Do the phases converge in stable regimes?
- Does disagreement identify transitions?
- Is the signal phase robust to causal filtering?
- Does CAPM phase provide a useful fundamental anchor?

The comparison must control for look-ahead bias and boundary effects.

U.12 Stage 11 — Alternative metrics

Test a generalized norm:

$A^2 = x^T G x$. (U.27)

Where:

$x = [R,Q]^T$. (U.28)

Possible G matrices may be estimated from:

- empirical covariance;
- liquidity weights;
- risk-capital weights;
- information geometry.

The research question is whether a non-Euclidean metric improves:

- stability;
- prediction;
- frame reconciliation;
- residual control.

The Euclidean model should remain the benchmark.

U.13 Stage 12 — Higher-dimensional pressure channels

Extend the state to:

$\mathbf{X}$ = (R,Q_market,Q_credit,Q_liquidity,Q_funding,Q_model). (U.29)

A generalized magnitude may be:

$A^2 = R^2 + \mathbf{Q}^T G \mathbf{Q}$. (U.30)

Research challenges include:

- identifying orthogonal channels;
- avoiding double counting;
- defining compatible operators;
- preserving auditability;
- testing dimensional reduction.

A single complex plane may remain useful as a local projection of this larger state.

U.14 Stage 13 — Noncommuting institutional operators

Define operators such as:

$\mathcal{O}$_margin. (U.31)

$\mathcal{O}$_funding. (U.32)

$\mathcal{O}$_liquidation. (U.33)

$\mathcal{O}$_accounting. (U.34)

Test:

$[\mathcal{O}_a,\mathcal{O}_b] = \mathcal{O}_a\mathcal{O}_b - \mathcal{O}_b\mathcal{O}_a$. (U.35)

A nonzero result indicates order dependence.

The research must determine whether the effect arises from:

- thresholds;
- liquidity depletion;
- legal timing;
- behavioural feedback;
- data latency.

This is a classical institutional operator programme unless specifically quantum evidence emerges.

U.15 Stage 14 — Ledger-time measurement

Define:

$d\tau_L = G|d\tilde{\Phi}|/\Omega$. (U.36)

Questions include:

- Does ledger time better organize crises than calendar time?
- Does residual phase depth predict future gate events?
- Are different ledgers synchronized or desynchronized?
- Does backreaction accelerate subsequent ledger time?

This programme connects valuation geometry with event-time finance.

U.16 Stage 15 — Observer-bound world comparison

Represent a financial world as:

$\mathcal{W} = (\mathcal{S}, \square, \mathcal{G}, \mathcal{L}, \mathcal{E}, \mathcal{U}, \mathcal{B})$. (U.37)

Compare worlds such as:

- market world;
- accounting world;
- regulatory world;
- derivative-collateral world;
- algorithmic trading world.

The objective is to identify:

- shared invariants;
- incompatible gates;
- transport residuals;
- intervention asymmetries;
- world-to-world translation failure.

U.17 Stage 16 — Quantum-subtraction analysis

Only after the classical financial structures are operationally established should the research compare them with quantum measurement.

The procedure is:

1. identify a quantum-like feature;
2. reconstruct the feature in mature finance;
3. specify the exact formal mapping;

4. identify what the financial model does not reproduce;
5. define the remaining quantum residue.

The central discipline is:

Functional Homology ≠ Physical Identity. (U.38)

U.18 Success criteria for the full programme

The framework should be regarded as successful if it produces at least one of the following:

- improved risk attribution;
- improved hedge design;
- better protocol reconciliation;
- earlier gate-event detection;
- better residual governance;
- clearer distinction between exposure and realization;
- a testable subtraction of generic from specifically quantum structure.

A beautiful geometry without one of these gains remains mathematically interesting but operationally optional.

U.19 Failure criteria

The programme should be reduced if:

- results depend entirely on arbitrary baselines;
- Q adds no information beyond simple transformations;
- phase scenarios have no financial interpretation;
- residuals remain uncontrolled;
- the terminology confuses practitioners;
- cross-domain analogies dominate empirical work;
- higher-dimensional extensions become unauditable.

The strongest framework is not the one with the most concepts.

It is the one that preserves the necessary distinctions with the least unsupported structure.

Appendix V — Editorial and Publication Architecture

V.1 Recommended publication form

Because the article contains both a finance-first theorem and a broader conceptual programme, it may be published in one of three forms.

Option A — One long monograph article

Part I and Part II remain together with all appendices.

Best for:

- OSF;
- Zenodo;
- working-paper publication;
- conceptual completeness.

Option B — Two-paper series

Paper I

From Discounted Value to Conjugate Risk

Focus:

- CAPM completion;
- Q identity;
- Phase Delta;
- measurement cycle;
- duration comparison;
- empirical tests.

Paper II

When Conjugate Risk Becomes a Measurement World

Focus:

- relative frames;
- gates;
- ledgers;
- derivative composite states;
- observer-bound worlds;
- quantum subtraction.

This option is likely more accessible to traditional finance readers.

Option C — Main paper plus technical supplement

The main paper contains Sections 0–24.

Part II and the appendices become a separate technical and conceptual supplement.

V.2 Recommended finance-journal version

A conventional finance submission should emphasize:

- CAPM projection;
- Q as phase sensitivity;
- duration equivalence;
- multi-period exposure;
- empirical falsification.

It should minimize:

- quantum language;
- observer-world terminology;
- gauge analogies;
- internal time.

The finance-journal title could be:

The Conjugate Risk Structure of CAPM Valuation

A Complex-Plane Reconstruction of Discounted Value and Required-Return Sensitivity

---

V.3 Recommended interdisciplinary version

The interdisciplinary version may retain:

From Discounted Value to Conjugate Risk

CAPM Phase Geometry, the Financial Meaning of Q, and the Measurement Structure of Observer-Bound Valuation Worlds

This version should preserve Part II.

V.4 Recommended abstract discipline

The abstract should state explicitly:

1. Q is derived rather than independently observed;
2. the principal theorem is mathematical;
3. ordinary CAPM sensitivity is preserved;
4. Q is exposure rather than loss;
5. empirical value remains to be tested;
6. quantum comparison is methodological rather than ontological.

This prevents the strongest predictable misunderstandings.

V.5 Recommended figures for publication

Figure 1

CAPM scalar valuation and complex completion.

Figure 2

R, Q, A, and θ on the complex plane.

Figure 3

Q as tangent exposure:

$\partial R/\partial \theta = -Q.$ (V.1)

Figure 4

The measurement cycle:

$R \to -Q \to -R \to Q \to R.$ (V.2)

Figure 5

Required-return P&L versus phase P&L.

Figure 6

Haircut H versus marginal pressure Q.

Figure 7

Multi-period Q-term structure.

Figure 8

Measurement, movement, gate, ledger, and residual.

Figure 9

What finance reproduces versus what remains quantum.

V.6 Recommended tables

Table 1

Symbol dictionary.

Table 2

What Q is and is not.

Table 3

Comparison with duration and option Greeks.

Table 4

Long–short measurement structure.

Table 5

Static identities versus falsifiable hypotheses.

Table 6

Finance-reproducible structures versus quantum residue.

Table 7

Research-stage roadmap.

V.7 Final editorial principle

The article should lead with the strongest precise result:

$\partial R/\partial \theta = -Q.$ (V.3)

It should delay the most speculative implications until after:

- the construction;
- the financial interpretation;
- the limitations;
- the empirical burden;

have all been made explicit.

The preferred rhetorical sequence is:

Mature Finance
→ Exact Reconstruction
→ Operational Interpretation
→ Limitations
→ Empirical Programme
→ Broader Measurement Implications. (V.4)

This ordering allows the complex valuation framework to be judged first as finance, and only afterward as a possible contribution to a wider theory of observer-bound worlds.

Appendix W — Reproducible Data Schema and Model-Audit Protocol

W.1 Purpose

The complex valuation framework is unusually sensitive to protocol declaration.

A reported value of Q is not meaningful unless the reader can reconstruct:

- the cash-flow input;
- the baseline valuation;
- the CAPM filter;
- the phase convention;
- the measurement orientation;
- the domain assumptions.

A reproducible implementation should therefore treat every calculated state as a structured record rather than as an isolated number.

W.2 Minimal single-cash-flow record

For each valued cash flow, store:

Field	Description
Claim_ID	unique identifier
Valuation_Date	date of calculation
Currency	valuation currency
CF_t	future cash flow
t	horizon
r_base	baseline discount rate
β	beta
ERP	equity risk premium
r_CAPM	required return

Field	Description
A_t	baseline amplitude
R_t	admitted value
Q_t	conjugate coordinate
θ_t	valuation phase
H_t	scalar haircut
Phase_Delta	−Q_t
Rate_Exposure	tR_t/(1 + r_CAPM)
Metric_ID	declared norm
Orientation_ID	sign convention
Protocol_Version	model version
Domain_Status	valid, warning, or failed

The core fields satisfy:

r_CAPM = r_base + βERP. (W.1)

A_t = CF_t/(1 + r_base)^t. (W.2)

R_t = CF_t/(1 + r_CAPM)^t. (W.3)

Q_t = √(A_t² − R_t²). (W.4)

θ_t = arccos(R_t/A_t). (W.5)

W.3 Protocol identifier

Define a protocol hash:

Protocol_ID = Hash(CF_Model,r_base_Rule,CAPM_Specification,Metric,Orientation,Gate_Rules). (W.6)

Two Q values should be compared directly only when their protocol identifiers are compatible.

If:

Protocol_ID,a ≠ Protocol_ID,b, (W.7)

then a reconciliation step is required before interpreting:

Q_a − Q_b. (W.8)

W.4 Input provenance

Every input should record:

- source;
- timestamp;
- revision status;

- estimation window;
- confidence level;
- transformation rule.

For beta, store:

β = Cov(r_i,r_m)/Var(r_m). (W.9)

But also report:

- market index;
- return frequency;
- observation window;
- leverage adjustment;
- shrinkage method;
- treatment of outliers.

A phase state is only as credible as the inputs producing it.

W.5 Cash-flow provenance

The cash-flow field should distinguish:

CF_t,forecast, (W.10)

CF_t,contractual, (W.11)

CF_t,scenario, (W.12)

and:

CF_t,market-implied. (W.13)

Combining these without labels can create false precision.

A contractual bond coupon and an analyst's expected equity cash flow do not possess the same uncertainty status.

W.6 Baseline taxonomy

The baseline field should identify whether r_base is:

- risk-free rate;
- sovereign curve;
- collateral rate;
- funding rate;

- inflation-adjusted rate;
- internal hurdle baseline;
- scenario reference rate.

Write:

r_base = r_base^(type). (W.14)

Then:

A_t = A_t^(type). (W.15)

And:

Q_t = Q_t^(type). (W.16)

This makes baseline dependence visible in notation and data.

W.7 Metric declaration

The standard model uses:

$A^2 = R^2 + Q^2$. (W.17)

Store:

Metric_ID = Euclidean_2D. (W.18)

A generalized model may use:

$A^2 = x^T G x$. (W.19)

Then store:

Metric_ID = General_G. (W.20)

And preserve the matrix G with the calculation.

Without G, a reported orthogonal coordinate cannot be independently reconstructed.

W.8 Orientation declaration

The article adopts:

$\mathcal{J} = [0\ -1;\ 1\ 0]$. (W.21)

Therefore:

$R \to -Q$. (W.22)

Store:

Orientation_ID = Counterclockwise_Positive_θ. (W.23)

An implementation using the opposite sign must not report Q without indicating that its measurement cycle is:

$$R \to Q \to -R \to -Q \to R. \quad (W.24)$$

W.9 Domain status

Define:

$$c = R/A. \quad (W.25)$$

The principal real domain is:

$$0 \le c \le 1. \quad (W.26)$$

A domain status may be:

Valid

$$0 \le c \le 1. \quad (W.27)$$

Numerical Boundary Warning

$$|1 - c| < \varepsilon_{num}. \quad (W.28)$$

Baseline Failure

$$c > 1 + \varepsilon_{num}. \quad (W.29)$$

Signed-Claim Extension Required

$$c < -\varepsilon_{num}. \quad (W.30)$$

The software should never silently convert a domain failure into a valid Q.

W.10 Derived-field consistency checks

For each record, calculate the residuals:

$$\varepsilon_{norm} = A^2 - R^2 - Q^2. \quad (W.31)$$

$$\varepsilon_{projection} = R - A\cos\theta. \quad (W.32)$$

$$\varepsilon_{quadrature} = Q - A\sin\theta. \quad (W.33)$$

$$\varepsilon_{phase_delta} = \partial R/\partial\theta + Q. \quad (W.34)$$

A valid numerical implementation should satisfy:

$$|\varepsilon_{norm}| < \varepsilon_{tol}. \quad (W.35)$$

$$|\varepsilon_{projection}| < \varepsilon_{tol}. \quad (W.36)$$

$$|\varepsilon_{quadrature}| < \varepsilon_{tol}. \quad (W.37)$$

|ε_phase_delta| < ε_tol. (W.38)

W.11 Bump-and-revalue audit

For each risk factor x, store:

x_+, (W.39)

x_−, (W.40)

R_+, (W.41)

R_−, (W.42)

θ_+, (W.43)

θ_−. (W.44)

The numerical phase exposure is:

Q_est = −(R_+ − R_−)/(θ_+ − θ_−). (W.45)

Define the estimation error:

ε_Q = Q_est − Q_analytic. (W.46)

Report:

Relative_Error_Q = ε_Q/Q_analytic. (W.47)

when:

Q_analytic ≠ 0. (W.48)

W.12 Multi-period data structure

For each horizon t, store one row:

Claim_ID t CF_t A_t R_t Q_t θ_t

Aggregate:

Z_total = Σ_t(R_t + iQ_t). (W.49)

R_total = Σ_t R_t. (W.50)

Q_total = Σ_t Q_t. (W.51)

A_effective = √(R_total² + Q_total²). (W.52)

Φ_effective = arg(Z_total). (W.53)

The report must distinguish:

A_effective (W.54)

from:

$A_{sum} = \Sigma_t A_t$. (W.55)

In general:

$A_{effective} \le A_{sum}$. (W.56)

W.13 Historical state table

A dynamic implementation should store:

$X_k = (A_k, R_k, Q_k, \theta_k, L_k, \varepsilon_k, P_k)$. (W.57)

The transition table should contain:

Event k Pre-State Shock Movement Gate Ledger Update Residual Post-State

This makes it possible to distinguish:

- movement in A;
- movement in θ;
- gate recognition;
- residual accumulation;
- protocol revision.

W.14 Version control

A valuation result should identify:

Model_Version. (W.58)

Data_Version. (W.59)

Protocol_Version. (W.60)

Code_Commit. (W.61)

A phase state calculated under one version should not be merged with another without reconciliation.

Define:

$Version_Residual = Z_{new_version} - Z_{old_version}$. (W.62)

This residual records model-change effects separately from economic movement.

W.15 Reproducibility bundle

A publishable analysis should provide:

1. raw or licensed-source references;
2. input extraction code;
3. model configuration;
4. calculation code;
5. unit tests;
6. output tables;
7. failure logs;
8. protocol documentation.

The minimum reproducibility principle is:

Same Inputs + Same Protocol + Same Version → Same Z. (W.63)

W.16 Audit hierarchy

The audit should proceed through four gates.

Gate 1 — Arithmetic integrity

Do the equations close?

Gate 2 — Financial construction integrity

Are A and R derived from coherent protocols?

Gate 3 — Empirical integrity

Are tests independent of defining identities?

Gate 4 — Interpretive integrity

Are exposure, loss, gate, ledger, and quantum analogy kept separate?

A model can pass arithmetic validation while failing all higher gates.

Appendix X — Extended Worked Example: A Three-Cash-Flow Asset

X.1 Declared cash flows

Consider an asset with three positive future cash flows:

CF_1 = 40.00. (X.1)

CF_2 = 50.00. (X.2)

CF_3 = 70.00. (X.3)

Let:

r_base = 0.03. (X.4)

β = 1.10. (X.5)

ERP = 0.05. (X.6)

Therefore:

r = 0.03 + 1.10(0.05). (X.7)

Hence:

r = 0.085. (X.8)

X.2 Horizon-one cash flow

The baseline amplitude is:

A_1 = 40/(1.03). (X.9)

Therefore:

A_1 ≈ 38.8350. (X.10)

The admitted value is:

R_1 = 40/(1.085). (X.11)

Therefore:

R_1 ≈ 36.8664. (X.12)

The ratio is:

c_1 = R_1/A_1. (X.13)

Therefore:

c_1 ≈ 0.9493. (X.14)

The phase is:

θ_1 = arccos(c_1). (X.15)

Therefore:

θ_1 ≈ 0.3195 radians. (X.16)

The conjugate coordinate is:

Q_1 = √(A_1² − R_1²). (X.17)

Therefore:

Q_1 ≈ 12.2037. (X.18)

X.3 Horizon-two cash flow

The baseline amplitude is:

A_2 = 50/(1.03)². (X.19)

Therefore:

A_2 ≈ 47.1298. (X.20)

The admitted value is:

R_2 = 50/(1.085)². (X.21)

Therefore:

R_2 ≈ 42.4660. (X.22)

The ratio is:

c_2 = R_2/A_2. (X.23)

Therefore:

c_2 ≈ 0.9010. (X.24)

The phase is:

θ_2 = arccos(c_2). (X.25)

Therefore:

θ_2 ≈ 0.4484 radians. (X.26)

The conjugate coordinate is:

Q_2 = √(A_2² − R_2²). (X.27)

Therefore:

Q_2 ≈ 20.4430. (X.28)

X.4 Horizon-three cash flow

The baseline amplitude is:

A_3 = 70/(1.03)³. (X.29)

Therefore:

A_3 ≈ 64.0451. (X.30)

The admitted value is:

$R_3 = 70/(1.085)^3$. (X.31)

Therefore:

$R_3 \approx 54.8247$. (X.32)

The ratio is:

$c_3 = R_3/A_3$. (X.33)

Therefore:

$c_3 \approx 0.8560$. (X.34)

The phase is:

$\theta_3 = \arccos(c_3)$. (X.35)

Therefore:

$\theta_3 \approx 0.5438$ radians. (X.36)

The conjugate coordinate is:

$Q_3 = \sqrt{(A_3^2 - R_3^2)}$. (X.37)

Therefore:

$Q_3 \approx 33.1340$. (X.38)

X.5 Cash-flow-state table

t	CF_t	A_t	R_t	Q_t	θ_t
1	40.0000	38.8350	36.8664	12.2037	0.3195
2	50.0000	47.1298	42.4660	20.4430	0.4484
3	70.0000	64.0451	54.8247	33.1340	0.5438

The phase grows with horizon because the difference between the baseline and CAPM discount protocols compounds through time.

X.6 Aggregate state

The aggregate admitted value is:

$R_{total} = R_1 + R_2 + R_3$. (X.39)

Therefore:

$R_{total} \approx 134.1571$. (X.40)

The aggregate conjugate coordinate is:

$Q_{total} = Q_1 + Q_2 + Q_3$. (X.41)

Therefore:

$Q_{total} \approx 65.7807$. (X.42)

Thus:

$Z_{total} \approx 134.1571 + i65.7807$. (X.43)

X.7 Effective aggregate amplitude

The magnitude of the aggregate state is:

$A_{effective} = \sqrt{(R_{total}^2 + Q_{total}^2)}$. (X.44)

Therefore:

$A_{effective} \approx 149.4140$. (X.45)

The sum of individual amplitudes is:

$A_{sum} = A_1 + A_2 + A_3$. (X.46)

Therefore:

$A_{sum} \approx 150.0099$. (X.47)

Hence:

$A_{effective} < A_{sum}$. (X.48)

The difference is:

$Alignment_Gap = A_{sum} - A_{effective}$. (X.49)

Therefore:

$Alignment_Gap \approx 0.5959$. (X.50)

The gap arises because the three cash-flow states have different phases.

X.8 Effective aggregate phase

The effective aggregate phase is:

$\Phi = \arctan(Q_{total}/R_{total})$. (X.51)

Therefore:

$\Phi \approx 0.4557$ radians. (X.52)

Equivalently:

$\Phi \approx 26.11^\circ$. (X.53)

The aggregate state can therefore be written:

$$Z_{total} = A_{effective} \exp(i\Phi). \quad (X.54)$$

X.9 Common-phase exposure

Under a common phase shock $d\varphi$:

$$dR_{total} = -Q_{total} d\varphi. \quad (X.55)$$

For:

$$d\varphi = 0.01, \quad (X.56)$$

the first-order P&L is:

$$dR_{total} \approx -65.7807(0.01). \quad (X.57)$$

Therefore:

$$dR_{total} \approx -0.6578. \quad (X.58)$$

X.10 Exact common-phase P&L

For:

$$\Delta\varphi = 0.10, \quad (X.59)$$

the exact P&L is:

$$\Delta R = R_{total}(\cos 0.10 - 1) - Q_{total} \sin 0.10. \quad (X.60)$$

Using:

$$\cos 0.10 \approx 0.995004, \quad (X.61)$$

and:

$$\sin 0.10 \approx 0.099833, \quad (X.62)$$

we obtain:

$$\Delta R \approx 134.1571(-0.004996) - 65.7807(0.099833). \quad (X.63)$$

Therefore:

$$\Delta R \approx -0.6702 - 6.5671. \quad (X.64)$$

Hence:

$$\Delta R \approx -7.2373. \quad (X.65)$$

The first-order approximation is:

$$\Delta R_1 = -Q_{total} \Delta\varphi. \quad (X.66)$$

Therefore:

$\Delta R_1 \approx -6.5781$. (X.67)

The second-order approximation is:

$\Delta R_2 = -Q_{total}\Delta\varphi - (R_{total}/2)(\Delta\varphi)^2$. (X.68)

Therefore:

$\Delta R_2 \approx -6.5781 - 0.6708$. (X.69)

Hence:

$\Delta R_2 \approx -7.2489$. (X.70)

The second-order approximation is close to the exact result.

X.11 Term-specific phase shock

Suppose:

$\Delta\theta_1 = 0.01$. (X.71)

$\Delta\theta_2 = 0.02$. (X.72)

$\Delta\theta_3 = 0.04$. (X.73)

The first-order P&L is:

$\Delta R \approx -Q_1\Delta\theta_1 - Q_2\Delta\theta_2 - Q_3\Delta\theta_3$. (X.74)

Therefore:

$\Delta R \approx -12.2037(0.01) - 20.4430(0.02) - 33.1340(0.04)$. (X.75)

Hence:

$\Delta R \approx -0.1220 - 0.4089 - 1.3254$. (X.76)

Therefore:

$\Delta R \approx -1.8563$. (X.77)

The longest cash flow contributes most of the phase P&L.

X.12 Q-weighted horizon

The Q-weighted horizon is:

$T_Q = [1Q_1 + 2Q_2 + 3Q_3]/[Q_1 + Q_2 + Q_3]$. (X.78)

Substitute:

$T_Q = [12.2037 + 40.8860 + 99.4020]/65.7807$. (X.79)

Therefore:

T_Q ≈ 2.3185. (X.80)

The aggregate phase exposure is concentrated beyond the midpoint of the three-year horizon.

X.13 Scalar haircuts

For each cash flow:

H_t = A_t − R_t. (X.81)

Therefore:

H_1 ≈ 1.9686. (X.82)

H_2 ≈ 4.6638. (X.83)

H_3 ≈ 9.2204. (X.84)

The total scalar haircut is:

H_total = H_1 + H_2 + H_3. (X.85)

Therefore:

H_total ≈ 15.8528. (X.86)

The total Q is:

Q_total ≈ 65.7807. (X.87)

Again:

Q_total ≠ H_total. (X.88)

Q_total measures common-phase exposure.

H_total measures accumulated same-axis discount haircuts.

X.14 Required-return exposure

For each cash flow:

D_r,t = tR_t/(1 + r). (X.89)

Therefore:

D_r,1 ≈ 1(36.8664)/1.085. (X.90)

D_r,1 ≈ 33.9782. (X.91)

D_r,2 ≈ 2(42.4660)/1.085. (X.92)

D_r,2 ≈ 78.2783. (X.93)

$D_{r,3} \approx 3(54.8247)/1.085$. (X.94)

$D_{r,3} \approx 151.5890$. (X.95)

The aggregate required-return exposure is:

$D_{r,\text{total}} \approx 263.8455$. (X.96)

For a one-percentage-point increase:

$\Delta r = 0.01$, (X.97)

the first-order rate P&L is:

$\Delta R \approx -2.6385$. (X.98)

X.15 Corresponding maturity-specific phase movements

For each horizon:

$d\theta_t/dr = tR_t/[(1 + r)Q_t]$. (X.99)

Therefore:

$d\theta_1/dr \approx 2.7842$. (X.100)

$d\theta_2/dr \approx 3.8291$. (X.101)

$d\theta_3/dr \approx 4.5754$. (X.102)

For:

$dr = 0.01$, (X.103)

the phase movements are:

$d\theta_1 \approx 0.027842$. (X.104)

$d\theta_2 \approx 0.038291$. (X.105)

$d\theta_3 \approx 0.045754$. (X.106)

The phase P&L is:

$dR = -\Sigma_t Q_t d\theta_t$. (X.107)

Substitution reproduces:

$dR \approx -2.6385$. (X.108)

This confirms that one parallel rate shock becomes a non-parallel phase movement across horizons.

X.16 Lessons from the example

The example illustrates six distinctions.

First:

$R_{total} = \Sigma_t R_t$. (X.109)

Second:

$Q_{total} = \Sigma_t Q_t$ (X.110)

under a common orientation.

Third:

$A_{effective} \neq \Sigma_t A_t$ (X.111)

when phases differ.

Fourth:

Q_{total} is exposure to a common phase rotation.

Fifth:

a parallel rate shock produces term-specific phase movements.

Sixth:

the scalar haircut and conjugate exposure remain fundamentally different quantities.

Appendix Y — Extended Glossary of Core Terms

Y.1 Admitted value

The value surviving the declared financial filter.

In the CAPM implementation:

$R = CF_t/(1 + r_{CAPM})^t$. (Y.1)

Admitted does not mean universally true.

It means accepted under a declared protocol.

Y.2 Amplitude

The magnitude A used to complete the valuation state.

In the baseline CAPM construction:

$A = CF_t/(1 + r_{base})^t$. (Y.2)

Amplitude is protocol-dependent.

It should not automatically be called intrinsic value.

Y.3 Baseline

The declared reference valuation rule against which the admitted value is compared.

The baseline determines A and therefore influences θ and Q.

Y.4 CAPM filter

The map:

$\beta ERP \rightarrow r_CAPM \rightarrow R.$ (Y.3)

It converts market-risk exposure into a required return and then into admitted present value.

Y.5 CAPM Phase Delta

The signed sensitivity:

$\Delta_\theta = \partial R/\partial\theta = -Q.$ (Y.4)

It measures currency change in admitted value per unit positive movement in valuation phase.

Y.6 Complex completion

The representation:

$Z = R + iQ = A \exp(i\theta).$ (Y.5)

It embeds the admitted scalar value inside a norm-preserving two-coordinate state.

Y.7 Complex structure

The operator $\mathcal{J}$ satisfying:

$\mathcal{J}^2 = -I.$ (Y.6)

In the valuation plane:

$\mathcal{J}[R,Q]^T = [-Q,R]^T.$ (Y.7)

Y.8 Conjugate exposure

The coordinate paired with R by the complex-structure operator.

In this article:

$-Q = \partial R/\partial\theta.$ (Y.8)

The word conjugate does not imply canonical Hamiltonian conjugacy.

Y.9 Contextual readout

A result that depends on both the state and the measurement protocol:

Observed Result = M_P(State). (Y.9)

Contextual here is operational, not automatically quantum-contextual in the technical sense.

Y.10 Economic movement

An actual change in A, θ, or another state variable.

Economic movement can generate P&L.

It is distinct from measurement rotation.

Y.11 Effective amplitude

For an aggregate complex state:

A_effective = |Σ_t Z_t|. (Y.10)

It may be smaller than the sum of individual amplitudes when phases differ.

Y.12 Effective phase

For:

Z_total ≠ 0, (Y.11)

the effective aggregate phase is:

Φ = arg(Z_total). (Y.12)

It summarizes the orientation of the aggregate vector.

Y.13 Exposure

A coefficient connecting a risk-factor movement to value change.

For phase:

Exposure = −Q. (Y.13)

Exposure is not itself a realized loss.

Y.14 Filter haircut

The same-axis difference:

$$H = A - R. \quad (Y.14)$$

It measures the scalar reduction from baseline value to admitted value.

Y.15 Frame

A declared valuation or measurement protocol defining how a claim is represented and read.

Examples include:

- CAPM frame;
- funding frame;
- accounting frame;
- regulatory frame.

Y.16 Frame residual

The unexplained difference after translating between two valuation frames:

$$\varepsilon_{ab} = Z_b - \lambda_{ab} \exp(i\varphi_{ab}) Z_a. \quad (Y.15)$$

Y.17 Gate

A rule determining whether an economic consequence is:

- admitted;
- deferred;
- partially admitted;
- rejected.

A gate is not multiplication by i.

Y.18 Geometric residual

The orthogonal coordinate Q generated by the Euclidean completion.

It should not be confused with model error or unrecognized P&L.

Y.19 Ledger

A persistent record of admitted financial consequences.

A ledger can alter future rights, constraints, and behaviour.

Y.20 Ledger time

An ordering generated by committed events rather than uniform calendar duration.

A simplified gate-weighted form is:

$d\tau_L = G|d\tilde{\Phi}|/\Omega.$ (Y.16)

Y.21 Mark

The ordinary admitted-value readout:

$M_0(Z) = R.$ (Y.17)

Y.22 Measurement orientation

The angle φ defining which real projection of Z is read:

$M_\varphi(Z) = R\cos\varphi - Q\sin\varphi.$ (Y.18)

Y.23 Measurement rotation

A change in the readout basis while the state may remain fixed.

It produces no economic P&L by itself.

Y.24 Model residual

The difference:

$\varepsilon_{model} = \text{Observed Value} - \text{Model Value}.$ (Y.19)

This is distinct from Q.

Y.25 Opportunity cost

The difference between the best feasible foregone alternative and the chosen alternative.

The equality:

$\text{Opportunity Cost} = A - R$ (Y.20)

requires A to represent that best alternative.

Y.26 Orthogonal completion

The process of defining Q so that:

$A^2 = R^2 + Q^2$. (Y.21)

Y.27 Passive rotation

A change in measurement basis without a change in the underlying state.

Y.28 Phase

The angular coordinate:

$\theta = \arccos(R/A)$. (Y.22)

It is protocol-relative and dimensionless.

Y.29 Phase coherence

The degree of directional alignment among multiple complex cash-flow states.

One measure is:

$\rho = |\Sigma_t A_t \exp(i\theta_t)|/\Sigma_t A_t$. (Y.23)

Y.30 Phase exposure term structure

The horizon-indexed vector:

$\mathbf{Q} = (Q_1,Q_2,\ldots,Q_T)$. (Y.24)

It measures where phase sensitivity is concentrated through time.

Y.31 Phase movement

An actual change:

$\Delta\theta \neq 0$. (Y.25)

At fixed amplitude, it generates:

$\Delta R \approx -Q\Delta\theta$. (Y.26)

Y.32 Projection

The extraction of one coordinate or readout from a larger state.

For the real axis:

$R = \mathrm{Re}(Z)$. (Y.27)

Projection does not necessarily destroy the remaining coordinate.

Y.33 Protocol

The declared set of rules producing and interpreting the valuation state.

A protocol includes:

- boundary;
- cash flow;
- baseline;
- filter;
- metric;
- orientation;
- gate;
- ledger.

Y.34 Q-coordinate

The positive first-quadrant coordinate:

$Q = \sqrt{(A^2 - R^2)}$. (Y.28)

It is the magnitude of the negative phase exposure:

$-Q = \partial R/\partial\theta$. (Y.29)

Y.35 Quadrature

A 90° phase relation between two real channels.

CAPM Q has quadrature-like structure but is not generally the Hilbert transform of R.

Y.36 Radial movement

A change in amplitude A.

Its first-order effect on R is:

$dR_{radial} = (R/A)dA$. (Y.30)

Y.37 Recognition

The institutional admission of an economic consequence into a ledger.

Recognition may lag economic movement.

Y.38 Residual

A general term for structure not exhausted by a selected model, projection, gate, or frame transformation.

Every use of residual should specify its type.

Y.39 Signed orientation

The direction distinguishing:

R (Y.31)

from:

−R. (Y.32)

And:

−Q (Y.33)

from:

Q. (Y.34)

Y.40 State evolution

A change in the underlying valuation state:

$Z \to Z'$. (Y.35)

It may involve radial, angular, or residual movement.

Y.41 Trace

A persistent record produced after gate admission.

A trace is more than a temporary calculation.

Y.42 Valuation world

An operational system containing:

- states;
- measurements;
- transition laws;
- gates;
- ledgers;
- residuals;
- interventions;
- backreaction.

World-like is a structural description, not a claim of physical universality.

Appendix Z — Final Claims, Qualifications, and Publication Checklist

Z.1 Claims established by construction

The following claims are exact under the declared Euclidean CAPM completion.

Claim 1

$R = A \cos \theta$. (Z.1)

Claim 2

$Q = A \sin \theta$. (Z.2)

Claim 3

$A^2 = R^2 + Q^2$. (Z.3)

Claim 4

$\partial R/\partial \theta = -Q$. (Z.4)

Claim 5

$\partial Q/\partial \theta = R$. (Z.5)

Claim 6

$\mathcal{J}^2 = -I$. (Z.6)

Claim 7

$\mathcal{J}^4 = I$. (Z.7)

Claim 8

$R \to -Q \to -R \to Q \to R$. (Z.8)

Claim 9

$dR = -Q d\theta$ (Z.9)

at fixed A.

Claim 10

$dR = (R/A)dA - Q d\theta$ (Z.10)

when A also changes.

Z.2 Claims established through CAPM translation

For:

$r = r_base + \beta ERP$, (Z.11)

the required-return sensitivity is:

$\partial R/\partial r = -tR/(1 + r)$. (Z.12)

The phase sensitivity is:

$\partial \theta/\partial r = tR/[(1 + r)Q]$. (Z.13)

Therefore:

$-Q d\theta = -[tR/(1 + r)]dr$. (Z.14)

The phase and required-return representations give the same first-order P&L.

Z.3 Claims requiring additional conditions

The following statements are not universally valid.

Conditional Claim 1

Q is marginal opportunity cost.

Required condition:

A = best feasible foregone alternative value. (Z.15)

Conditional Claim 2

−R is the counterparty's value.

Required conditions include:

- exact opposite claim;
- same protocol;
- linear signed position;
- no asymmetric adjustment.

Conditional Claim 3

Q values aggregate across assets.

Required conditions include:

- compatible baseline;
- common metric;
- common orientation;
- compatible phase scenario.

Conditional Claim 4

A common phase shock has market meaning.

Required condition:

an empirically or institutionally identifiable common reorientation factor.

Z.4 Claims remaining empirical

The following require independent testing.

1. Q improves risk communication.
2. Q improves P&L attribution.
3. Q predicts gate events.
4. phase coherence improves term-structure analysis.
5. phase-neutral hedging improves performance under declared scenarios.
6. frame translation reduces valuation disagreement.
7. residual phase measures identify regime shifts.

No construction identity proves these claims.

Z.5 Claims explicitly rejected

The article rejects the following unrestricted statements.

Q = Hidden Loss. (Z.16)

Q = Realized Loss. (Z.17)

Q = A − R. (Z.18)

Q = Volatility. (Z.19)

Q = Beta. (Z.20)

Q = VaR. (Z.21)

Q = Opportunity Cost under every baseline. (Z.22)

i = Financial Gate. (Z.23)

i^2 = Two Losses. (Z.24)

Derivative Coupling = Bell Entanglement. (Z.25)

Complex Finance = Quantum Finance. (Z.26)

Z.6 Publication checklist

Before publication, verify that the manuscript:

- defines A before Q;
- defines the protocol dependence of A;
- states the domain $0 \le R/A \le 1$;
- distinguishes Q from A − R;
- derives $\partial R/\partial \theta = -Q$;
- translates the result back into r, β, and ERP;
- separates exposure from movement;
- separates movement from recognition;
- separates recognition from total resolution;
- explains long–short limitations;
- states static non-independence;
- includes falsification conditions;
- limits the quantum comparison;
- preserves consistent sign conventions;
- uses one notation for option Theta and another for valuation phase;
- provides reproducible numerical examples.

Z.7 Recommended final boxed theorem

CAPM Conjugate Risk Theorem

Given:

$A_t = CF_t/(1 + r_{base})^t$, (Z.27)

$R_t = CF_t/(1 + r_{base} + \beta ERP)^t$, (Z.28)

and:

$\theta_t = \arccos(R_t/A_t)$, (Z.29)

define:

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (Z.30)

Then, at fixed A_t:

$\partial R_t/\partial \theta_t = -Q_t$. (Z.31)

Moreover:

$\partial Q_t/\partial \theta_t = R_t$. (Z.32)

Therefore the completed valuation state:

$Z_t = R_t + iQ_t$ (Z.33)

supports the closed measurement cycle:

$R_t \to -Q_t \to -R_t \to Q_t \to R_t$. (Z.34)

Z.8 Recommended final boxed interpretation

> **R is the admitted financial mark. Q is the magnitude of its conjugate exposure to valuation-phase movement. The imaginary unit rotates the measurement between these channels; it does not by itself create economic loss or ledgered history.**

Z.9 Recommended final boxed runtime

Measurement:

$M_{\pi/2}(Z) = -Q$. (Z.35)

Movement:

$\Delta\theta \neq 0$. (Z.36)

Economic consequence:

$\Delta R = R(\cos \Delta\theta - 1) - Q \sin \Delta\theta$. (Z.37)

Gate:

$G_P(\Delta R, X, L)$. (Z.38)

Ledger:

$L_k \to L_{k+1}$. (Z.39)

Residual:

$\varepsilon_{gate} = \Delta R_{economic} - \Delta R_{ledger}$. (Z.40)

Backreaction:

$X_{k+1} = \mathscr{B}(X_k, L_{k+1}, \varepsilon_{gate})$. (Z.41)

Z.10 Final authorial statement

The complex valuation framework should be judged in the following order:

1. Is the mathematics internally correct?
2. Is the financial construction explicit?
3. Is Q interpreted as an exposure rather than a hidden loss?
4. Does the coordinate change improve any real financial task?
5. Are residuals and failure conditions auditable?
6. Do broader analogies remain within their justified limits?

The framework's strongest achievement is not that it makes finance look exotic.

Its strongest achievement is that it takes an apparently decorative imaginary coordinate and derives a precise mature-finance identity:

$Q = -\partial R/\partial \theta$. (Z.42)

This converts Q from an unexplained geometric remainder into a conjugate risk exposure.

The deeper architecture then follows:

Mark
→ Conjugate Exposure
→ State Movement
→ Economic P&L
→ Commitment Gate
→ Ledger Trace
→ Residual
→ Backreaction. (Z.43)

That architecture is broad enough to support future work in:

- valuation;
- risk management;

- derivative systems;
- institutional accounting;
- regulatory finance;
- observer-bound modelling;
- cross-domain measurement theory.

Yet it remains disciplined by one final rule:

> **A richer representation is justified only when it preserves distinctions that matter and produces consequences that can be tested.**

The article and its appendices are now complete.

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.